

Hydraulic cylinders Mill type

Series CDH3, CGH3, CSH3



- Component series 3X
- Maximum operating pressure 350 bar

Features

- 6 types of mounting
- Piston Ø (ØAL): 40 ... 320 mm
- Piston rod Ø (ØMM): 28 ... 220 mm
- Stroke lengths up to 6 m
- Self-adjusting and adjustable end position damping
- 8 position measurement system interfaces

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Ordering codes: Series CDH3

01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16
CD	H3		/		/		/		A	3X	/				

01	Differential cylinder	CD
02	Series	H3

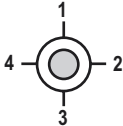
Types of mounting

03	Swivel eye at base	MP3 ¹⁾
	Self-aligning clevis at base	MP5
	Round flange at head	MF3
	Round flange at base	MF4
	Trunnion mounting	MT4 ²⁾
	Foot mounting	MS2
04	Piston Ø (ØAL) 40 ... 320 mm	
05	Piston rod Ø (ØMM) 28 ... 220 mm	
06	Stroke length in mm ³⁾	
07	Design principle: Head and base flanged	A
08	Component series 30 ... 39 (30 ... 39: unchanged installation and connection dimensions)	3X

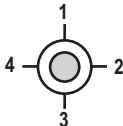
Line connection / version

09	According to ISO 1179-1 (pipe thread ISO 228-1)	B
	According to ISO 9974-1 (metric thread ISO 261)	M
	Flange hole pattern according to ISO 6162-2 tab. 2 type 1 (corresponds to SAE 6000 PSI)	D ^{4); 5)}
	Flange hole pattern according to ISO 6164 tab. 2	H ⁴⁾
	According to ISO 1179-1 (pipe thread ISO 228-1) with flat pipe flange	C ⁶⁾
	With mounted control block	Y ⁷⁾
	On/off and proportional directional control valves	
	Subplate NG6	P ^{4); 8); 9)}
	Subplate NG10	T ^{4); 9); 10)}
	Subplate NG16	U ^{4); 9); 11)}
	Subplate NG25	V ^{4); 9); 11)}
	Isolation valves type SL and SV	
	Subplate NG6	A ^{4); 8); 9); 12)}
	Subplate NG10	E ^{4); 9); 10); 12)}
	Subplate NG20	L ^{4); 9); 11); 12)}
	Subplate NG30	N ^{4); 9); 11); 12)}

Line connection / Head bearing

10	View to piston rod ¹³⁾		1
			2
			3
			4

Line connection/position at base

11	View to piston rod ¹³⁾		1
			2 ¹⁴⁾
			3
			4 ¹⁴⁾

Ordering codes: Series CDH3

01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16
CD	H3		/		/		/		A	3X	/				

Piston rod design

12	Hard chromium-plated	C
	Hardened and hard chromium-plated	H ¹⁵⁾
	Nickel-plated and hard chromium-plated	N ⁵⁾

Piston rod end (for mounting elements, see data sheet R. 17042)

13	Thread for swivel head CGAS	A
	Thread for swivel head CGA, CGAK, plain clevis CSA	G ¹⁶⁾
	With mounted swivel head CGAS	S
	With mounted swivel head CGA	L ¹⁶⁾
	With mounted swivel head CGAK	M ¹⁶⁾
	With mounted plain clevis CSA	N ¹⁾

End position damping

14	Without	U
	Both sides, self-adjusting	D ¹⁾
	Both sides, adjustable	E
	Base side, self-adjusting	K ^{1; 7)}
	Head side, self-adjusting	S ^{1; 7)}

Seal design - (Observe selection criteria for seals, see page 71)

15	Mineral oil HL, HLP and HFA	
	Standard seal system	M
	Standard seal system with guide rings	L
	Reduced friction, heavy industry	R
	Mineral oil HL, HLP, HFA and water glycol HFC	
	Standard seal system HFC	G ^{9); 17)}
	Servo quality / reduced friction	T
	Chevron seal kits	A
	HDFR phosphate ester and HFDU polyol ester	
	Servo quality / reduced friction	S
	Standard seal system FKM	V ^{9); 17)}
	Chevron seal kits	B

Option

16	Without additional options, do not fill fields for additional options	W
	Additional options, fill fields for additional options	Z

¹⁾ Only piston Ø 40 ... 200 mm.

²⁾ Trunnion mounting position freely selectable. Always indicate dimension "XV" in mm in the plain text with orders.

³⁾ Observe the maximum available stroke length page 14 and admissible stroke length (acc. to buckling calculation) page 63 ... 67.

⁴⁾ Not for type of mounting MF4.

⁵⁾ From piston Ø 63 mm.

⁶⁾ For type of mounting "MS2" only position 11.

⁷⁾ Upon request.

⁸⁾ Piston Ø 40 ... 80 mm, only position 11, subplates only possible in combination with line connection "B" at the head.

⁹⁾ Maximum operating pressure 315 bar.

¹⁰⁾ Piston Ø 63 ... 200 mm, only position 11, subplates only possible in combination with line connection "B" at the head.

¹¹⁾ Piston Ø 125 ... 200 mm, only position 11, subplates only possible in combination with line connection "B" at the head.

¹²⁾ Subplates for SL and SV isolation valves;

Seal designs "T", "G", "L", "R", "S" and "V" are not designed for the static holding function.

¹³⁾ All graphical pictures in the data sheet show position 1.

¹⁴⁾ Not for type of mounting "MF4" and "B", "M" or "C" line connection.

¹⁵⁾ Only piston rods Ø 28 ... 140 mm.

¹⁶⁾ Only piston Ø 40 ... 250 mm.

¹⁷⁾ Not for piston Ø 320 mm.

Ordering codes: Series CDH3 (fields for additional options)

01	02	03	04	05	06	07	08

01	Without inductive proximity switches	W
	Inductive proximity switches without mating connector - separate order, see page 16, 20, 52, 53	E ¹⁾
02	Without additional guide rings	W
	Additional guide rings	F ^{2); 3)}
03	Without measuring coupling	W
	Measuring coupling, on both sides	A
	Measuring coupling, on both sides, stainless steel version	E ⁴⁾
04	Standard conical grease nipples, DIN 71412 form A	W
	Spherical bearing, maintenance-free	A ^{5); 6)}
	Flat type grease nipples, DIN 3404 form A	B
05	Without piston rod extension	W
	Specify piston rod extension "LY" in mm in plain text	Y
06	Priming class CP3	W
	Painting class CP4	B ⁷⁾
	Painting class CP5	L ⁷⁾
	Painting class CP6	U ⁷⁾
	Painting class CP7	E ⁷⁾
07	Without oil filling	W
	With corrosion protection oil VG 68	F
08	Without test certificate	W
	With certificate of compliance 2.1 based on EN 10204	B
	With acceptance test certificate 3.1 based on EN 10204	C

Order examples:

Without additional options: CDH3MP5/100/56/300A3X/B11CADMW

With additional options: CDH3MP5/100/56/300A3X/B11CADMZEWABWWWW

- ¹⁾ Minimum stroke length = 20 mm.
²⁾ Not for seal design "A", "B";
Piston Ø 220 ... 320 mm standard.
³⁾ Standard with seal design "L".
⁴⁾ Upon request.
⁵⁾ Not for plain clevis "N".
⁶⁾ Not for type of mounting "MP3".
⁷⁾ Specify RAL color in the plain text.

Ordering codes: Series CGH3

01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16
CG	H3		/		/		/		A	3X	/				

01	Double-acting cylinder	CG ¹⁾
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02	Series	H3
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Types of mounting

03	Round flange at head	MF3
	Trunnion mounting	MT4 ²⁾
	Foot mounting	MS2

04	Piston Ø (ØAL) 40 ... 320 mm	
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05	Piston rod Ø (ØMM) 28 ... 220 mm	
----	----------------------------------	--

06	Stroke length in mm ³⁾	
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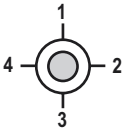
07	Design principle: Head and base flanged	A
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08	Component series 30 ... 39 (30 ... 39: unchanged installation and connection dimensions)	3X
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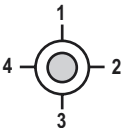
Line connection / version

09	According to ISO 1179-1 (pipe thread ISO 228-1)	B
	According to ISO 9974-1 (metric thread ISO 261)	M
	Flange hole pattern according to ISO 6162-2 tab. 2 type 1 (≠SAE 6000 PSI)	D ⁴⁾
	Flange hole pattern according to ISO 6164 tab. 2	H
	According to ISO 1179-1 (pipe thread ISO 228-1) with flat pipe flange	C ⁵⁾
	With mounted control block	Y ⁶⁾

Line connection / Head bearing

10	View to piston rod ⁷⁾		1
			2
			3
			4

Line connection/position at base

11	View to piston rod ⁷⁾		1
			2
			3
			4

 **Notice:** Standard units are contained in the EPS (standard price list).

Ordering codes: Series CGH3

01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16
CG	H3		/		/		/		A	3X	/				

Piston rod design

12	Hard chromium-plated	C
	Hardened and hard chromium-plated	H ⁸⁾

Piston rod end (for mounting elements, see data sheet R. 17042)

13	Thread for swivel head CGAS	A
	Thread for swivel head CGA, CGAK, plain clevis CSA	G ⁹⁾
	With mounted swivel head CGAS	S ^{9); 10)}
	With mounted swivel head CGA	L ^{9); 10)}
	With mounted swivel head CGAK	M ^{9); 10)}
	With mounted plain clevis CSA	N ^{11); 10)}

End position damping

14	Without	U
	Both sides, self-adjusting	D ¹¹⁾
	Both sides, adjustable	E

Seal design - (Observe selection criteria for seals, see page 71)

15	Mineral oil HL, HLP and HFA	
	Standard seal system	M
	Standard seal system with guide rings	L
	Reduced friction, heavy industry	R
	Mineral oil HL, HLP, HFA and water glycol HFC	
	Standard seal system HFC	G ^{12); 13)}
	Servo quality / reduced friction	T
	Chevron seal kits	A
	HDFR phosphate ester and HFDU polyol ester	
	Servo quality / reduced friction	S
	Standard seal system FKM	V ^{12); 13)}
	Chevron seal kits	B

Option

16	Without additional options, do not fill fields for additional options	W
	Additional options, fill fields for additional options	Z

- 1) Not standardized.
- 2) Trunnion mounting position freely selectable. Always indicate dimension "XV" in mm in the plain text with orders.
- 3) Observe the maximum available stroke length page 14 and admissible stroke length (acc. to buckling calculation) page 63 ... 67.
- 4) From piston Ø 63 mm.
- 5) For type of mounting "MS2" only position 11.
- 6) Upon request.
- 7) All graphical pictures in the data sheet show position 1.
- 8) Only piston rods Ø 28 ... 140 mm.
- 9) Only piston Ø 40 ... 250 mm.
- 10) Only one swivel head / plain clevis mounted, left piston rod side (alignment: figures in the data sheet).
- 11) Only piston Ø 40 ... 200 mm.
- 12) Maximum operating pressure 315 bar.
- 13) Not for piston Ø 320 mm.

Ordering codes: Series CGH3 (fields for additional options)

	01	02	03	04	05	06	07	08

01	Without inductive proximity switches	W
	Inductive proximity switches without mating connector - separate order, see page 16, 20, 52, 53	E ¹⁾
02	Without additional guide rings	W
	Additional guide rings	F ^{2); 3)}
03	Without measuring coupling	W
	Measuring coupling, on both sides	A
	Measuring coupling, on both sides, stainless steel version	E ⁴⁾
04	Standard conical grease nipples, DIN 71412 form A	W
	Spherical bearing, maintenance-free	A ⁵⁾
	Flat type grease nipples, DIN 3404 form A	B
05	Without piston rod extension	W
	Specify piston rod extension "LY" in mm in plain text	Y ⁶⁾
06	Priming class CP3	W
	Painting class CP4	B ⁷⁾
	Painting class CP5	L ⁷⁾
	Painting class CP6	U ⁷⁾
	Painting class CP7	E ⁷⁾
07	Without oil filling	W
	With corrosion protection oil VG 68	F
08	Without test certificate	W
	With certificate of compliance 2.1 based on EN 10204	B
	With acceptance test certificate 3.1 based on EN 10204	C

Order examples:

Without additional options: CGH3MF3/100/56/300A3X/B11CADMW

With additional options: CGH3MF3/100/56/300A3X/B11CADMZEWABWWWW

¹⁾ Minimum stroke length = 20 mm.²⁾ Not for seal design "A", "B";
Piston Ø 220 ... 320 mm standard.³⁾ Standard with seal design "L".⁴⁾ Upon request.⁵⁾ Not for plain clevis "N".⁶⁾ Only at left piston rod side (Orientation: Figures in data sheet).⁷⁾ Specify RAL color in the plain text.

Ordering codes: Series CSH3

01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16
CS	H3		/		/		/	A	3X	/					

01	Differential cylinder with position measurement system	CS ¹⁾
02	Series	H3

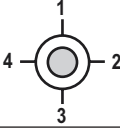
Types of mounting

03	Swivel eye at base	MP3 ²⁾
	Self-aligning clevis at base	MP5
	Round flange at head	MF3
	Round flange at base	MF4
	Trunnion mounting	MT4 ³⁾
	Foot mounting	MS2
04	Piston Ø (ØAL) 40 ... 320 mm	
05	Piston rod Ø (ØMM) 28 ... 220 mm	
06	Stroke length in mm ⁴⁾	
07	Design principle: Head and base flanged	A
08	Component series 30 ... 39 (30 ... 39: unchanged installation and connection dimensions)	3X

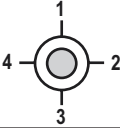
Line connection / version

09	According to ISO 1179-1 (pipe thread ISO 228-1)	B
	According to ISO 9974-1 (metric thread ISO 261)	M
	Flange hole pattern according to ISO 6162-2 tab. 2 type 1 (corresponds to SAE 6000 PSI)	D ^{5); 6)}
	Flange hole pattern according to ISO 6164 tab. 2	H ⁵⁾
	According to ISO 1179-1 (pipe thread ISO 228-1) with flat pipe flange	C ⁷⁾
	With mounted control block	Y ⁸⁾
	On/off and proportional directional control valves	
	Subplate NG6	P ^{5); 9); 10)}
	Subplate NG10	T ^{5); 10); 11)}
	Subplate NG16	U ^{5); 10); 12)}
	Subplate NG25	V ^{5); 10); 12)}
	Isolation valves type SL and SV	
	Subplate NG6	A ^{5); 9); 10); 13)}
	Subplate NG10	E ^{5); 10); 11); 13)}
	Subplate NG20	L ^{5); 10); 12); 13)}
	Subplate NG30	N ^{5); 10); 12); 13)}

Line connection / Head bearing

10	View to piston rod ¹⁴⁾		1
			2
			3
			4

Line connection/position at base

11	View to piston rod ¹⁴⁾		1
			2 ¹⁵⁾
			3
			4 ¹⁵⁾

Ordering codes: Series CSH3

01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16
CS	H3		/	/	/	A	3X	/							Z

Piston rod design

12	Hard chromium-plated	C
	Hardened and hard chromium-plated	H ¹⁶⁾

Piston rod end (for mounting elements, see data sheet R. 17042)

13	Thread for swivel head CGAS	A
	Thread for swivel head CGA, CGAK, plain clevis CSA	G ¹⁷⁾
	With mounted swivel head CGAS	S
	With mounted swivel head CGA	L ¹⁷⁾
	With mounted swivel head CGAK	M ¹⁷⁾
	With mounted plain clevis CSA	N ²⁾

End position damping

14	Without	U
	Both sides, adjustable	E ¹⁸⁾

Seal design - (Observe selection criteria for seals, see page 71)

15	Mineral oil HL, HLP and HFA	
	Standard seal system	M ¹⁹⁾
	Standard seal system with guide rings	L
	Reduced friction, heavy industry	R ¹⁹⁾
	Mineral oil HL, HLP, HFA and water glycol HFC	
	Standard seal system HFC	G ^{10); 19); 20)}
	Servo quality / reduced friction	T ¹⁹⁾
	HDFR phosphate ester and HFDU polyol ester	
	Servo quality / reduced friction	S ¹⁹⁾
	Standard seal system FKM	V ^{10); 19); 20)}

Option

16	Additional options, fill fields for additional options	Z
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- 1) Not standardized.
- 2) Only piston Ø 40 ... 200 mm.
- 3) Trunnion mounting position freely selectable. Always indicate dimension "XV" in mm in the plain text with orders.
- 4) Observe the maximum available stroke length page 14 and admissible stroke length (acc. to buckling calculation) page 63 ... 67.
- 5) Not for type of mounting "MF4".
- 6) From piston Ø 63 mm.
- 7) For type of mounting "MS2" only position 11.
- 8) Upon request.
- 9) Piston Ø 40 ... 80 mm, only position 11, subplates only possible in combination with line connection "B" at the head.
- 10) Maximum operating pressure 315 bar.
- 11) Piston Ø 63 ... 200 mm, only position 11, subplates only possible in combination with line connection "B" at the head.
- 12) Piston Ø 125 ... 200 mm, only position 11, subplates only possible in combination with line connection "B" at the head.
- 13) Subplates for SL and SV isolation valves.
Seal designs "T", "G", "L", "R", "S" and "V" are not designed for the static holding function.
- 14) All graphical pictures in the data sheet show position 1.
- 15) Not for type of mounting "MF4" and "B", "M" or "C" line connection
- 16) Only piston rods Ø 28 ... 140 mm.
- 17) Only piston Ø 40 ... 250 mm.
- 18) From piston rod Ø 45 mm.
- 19) With the CSH series with guide belts as standard.
- 20) Not for piston Ø 320 mm.

Ordering codes: Series CSH3 (fields for additional options)

01	02	03	04	05	06	07	08
T							

01	Position measurement system (magnetostrictive) without mating connector – separate order, see page 16 ... 22	T
02	Analog output 4 ... 20 mA	C
	Analog output 0 ... 10 V	F
	Digital output SSI (resolution 5 µm, asynchronous forward)	D
	Digital output SSI (resolution 1 µm, synchronous forward)	S
	Profibus D63 (integrated supply)	N
	Profibus D53 (separate supply line)	P
	IO-Link	L ^{1); 2)}
	Profinet RT and IRT with encoder profile	R
03	Without measuring coupling	W
	Measuring coupling, on both sides	A
	Measuring coupling, on both sides, stainless steel version	E ³⁾
04	Standard conical grease nipples, DIN 71412 form A	W
	Spherical bearing, maintenance-free	A ^{4); 5)}
	Flat type grease nipples, DIN 3404 form A	B
05	Without piston rod extension	W
	Specify piston rod extension "LY" in mm in plain text	Y
06	Priming class CP3	W
	Painting class CP4	B ⁶⁾
	Painting class CP5	L ⁶⁾
	Painting class CP6	U ⁶⁾
	Painting class CP7	E ⁶⁾
07	Without oil filling	W
	With corrosion protection oil VG 68	F
08	Without test certificate	W
	With certificate of compliance 2.1 based on EN 10204	B
	With acceptance test certificate 3.1 based on EN 10204	C

Order example:

With additional options: CSH3MP5/100/70/300A3X/B11 CAUMZTFAWWWWW

- 1) Not for type of mounting "MP3" and "MP5".
2) Minimum stroke length 50 mm; maximum stroke length 2540 mm.
3) Upon request.
4) Not for plain clevis "N".
5) Not for type of mounting "MP3".
6) Specify RAL color in the plain text.

Technical data

(For applications outside these values, please consult us.)

General		
Installation position		Any
Ambient temperature range	°C	See page 71
Priming		See page 81
Painting		See page 81
Weight		See page 13
Hydraulic		
Maximum operating pressure ¹⁾	bar	350
Minimum operating pressure ²⁾ (without load)	bar	10
Static test pressure	bar	525
Hydraulic fluid		See table below
Hydraulic fluid temperature range	°C	See page 71
Optimum viscosity range	mm ² /s	20 to 100 mm ² /s
Viscosity range	mm ² /s	12 to 380 mm ² /s
Maximum admissible degree of contamination of the hydraulic fluid; cleanliness class according to ISO 4406 (c)		Class 20/18/15
Bleeding		By default secured against screwing out
Hydraulic fluid	Classification	Data sheet
Mineral oils	HL, HLP	90220
Phosphate ester	HFDR	90222
Polyol ester	HFDU	90222
Oil-in-water emulsion	HFA	90223
Water glycol	HFC	90223

¹⁾ With extreme shock loads, mounting elements and threaded piston rod connections must be designed for durability.

²⁾ Depending on the application, application conditions and technical design, a certain minimum pressure is required in order to guarantee technically unobjectionable functioning of the hydraulic cylinder.

Technical data
 (For applications outside these values, please consult us.)

Stroke velocity

Piston Ø in mm	Line connection	Maximum stroke velocity in m/s
40	G 1/2	0.31
50	G 1/2	0.20
63	G 3/4	0.28
80	G 3/4	0.18
100	G 1	0.20
125	G 1 1/4	0.20
140	G 1 1/4	0.16
160	G 1 1/2	0.18
180	G 1 1/2	0.14
200	G 1 1/2	0.11
220	G 1 1/2	0.09
250	G 1 1/2	0.07
280	G 1 1/2	0.06
320	G 1 1/2	0.04



Notice:

Please observe the guideline on maximum stroke velocities (with recommended flow velocity of 5 m/s in the line connection) in the table. Higher stroke velocity on request. If the extension velocity is considerably higher than the retraction velocity of the piston rod, drag-out losses of the hydraulic fluid may result. If necessary, please consult us.

Technical data

(For applications outside these values, please consult us.)

Weight (in kg)

Piston ØAL in mm	Piston rod ØMM in mm	CD/CS cylinder with 0 mm stroke length					per 100 mm stroke length in kg	CG cylinder with 0 mm stroke length			per 100 mm stroke length in kg
		"MP3" ¹⁾ "MP5" ¹⁾ in kg	"MP3" ²⁾ "MP5" ²⁾ in kg	"MF3" "MF4" in kg	"MT4" in kg	"MS2" in kg		"MF3" in kg	"MT4" in kg	"MS2" in kg	
40	28	8	14	11	11	9	1.2	12	12	10	1.6
50	36	12	20	17	15	15	1.6	19	17	17	2.4
63	45	26	41	32	30	32	2.6	37	35	36	3.8
80	56	33	44.5	43	40	42	4.2	49	46	48	6.1
100	70	58	74.5	72	71	73	5.7	80	79	81	8.8
125	90	120	150	148	145	149	11.1	170	166	171	16.1
140	100	167	203	205	202	206	13.0	236	233	236	19.1
160	110	229	284	276	276	275	16.3	316	316	315	23.8
180	125	317	383	387	386	404	19.5	456	455	473	29.1
200	140	425	500	506	504	531	24.4	562	560	587	36.5
220	160	514	623	653	570	590	37.8	753	671	690	53.6
250	180	777	959	939	854	829	46.2	1057	972	948	66.2
280	200	915	1147	1073	1028	984	59.7	1224	1179	1135	84.3
320	220	1200	1479	1274	1211	1211	68.3	1431	1369	1369	98.1

ØAL = piston Ø

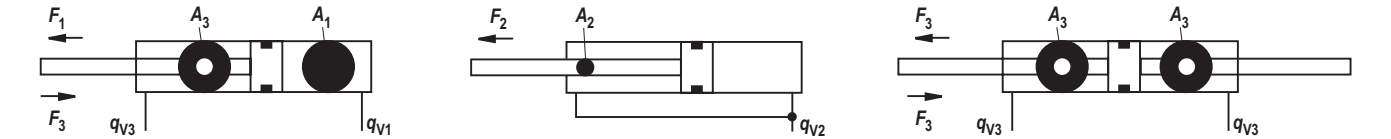
ØMM = piston rod Ø

¹⁾ CD cylinder.²⁾ CS cylinder.

Technical data
(For applications outside these values, please consult us.)

Areas, forces, flow

Piston	Piston rod	Area ratio	Areas			Force at 350 bar ¹⁾			Flow at 0.1 m/s ²⁾			max. available stroke length in mm
			Piston	Rod	Ring	Pressure	Diff.	Pulling	OFF	Diff.	ON	
ØAL in mm	ØMM in mm	φ A ₁ /A ₃	A ₁ in cm ²	A ₂ in cm ²	A ₃ in cm ²	F ₁ in kN	F ₂ in kN	F ₃ in kN	q _{v1} in l/min	q _{v2} in l/min	q _{v3} in l/min	
40	28	1.96	12.56	6.16	6.40	43.96	21.56	22.40	7.5	3.7	3.8	2000
50	36	2.08	19.63	10.18	9.45	68.71	35.63	33.08	11.8	6.1	5.7	2000
63	45	2.04	31.17	15.90	15.27	109.10	55.65	53.45	18.7	9.5	9.2	2000
80	56	1.96	50.26	24.63	25.63	175.91	86.21	89.71	30.2	14.8	15.4	2000
100	70	1.96	78.54	38.48	40.06	274.89	134.68	140.21	47.1	23.1	24.0	3000
125	90	2.08	122.72	63.62	59.10	429.52	222.67	206.85	73.6	38.2	35.4	3000
140	100	2.04	153.94	78.54	75.40	538.79	274.89	263.90	92.4	47.1	45.3	3000
160	110	1.90	201.06	95.06	106.00	703.71	332.71	371.00	120.6	57.0	63.6	3000
180	125	1.93	254.47	122.72	131.75	890.65	429.52	461.13	152.7	73.6	79.1	3000
200	140	1.96	314.16	153.96	160.20	1099.56	538.86	560.70	188.5	92.4	96.1	3000
220	160	2.12	380.1	201.0	179.1	1330.5	703.7	626.8	228.1	120.7	107.4	6000
250	180	2.08	490.8	254.4	236.4	1718.1	890.6	827.4	294.5	152.7	141.8	6000
280	200	2.04	615.7	314.1	301.6	2155.1	1099.6	1055.6	369.4	188.5	180.9	6000
320	220	1.90	804.2	380.1	424.2	2814.9	1330.5	1484.4	482.5	228.1	254.4	6000



¹⁾ Theoretical static cylinder force (without consideration of the efficiency and admissible load for attachment parts such as swivel heads, plates or valves, etc.)
²⁾ Stroke velocity.

Technical data

(For applications outside these values, please consult us.)

Tolerances according to DIN ISO 6022

Installation dimensions	WC	XC ¹⁾	XO ¹⁾	XS ¹⁾	XV ¹⁾	ZP ¹⁾	Stroke tolerances ²⁾
Type of mounting	MF3	MP3	MP5	MS2	MT4	MF4	
Stroke length	Tolerances						
≤ 1250	±2	±1.5	±1.5	±2	±2	±1.5	+2
> 1250 – ≤ 3150	±4	±3	±3	±4	±4	±3	+5
> 3150 – ≤ 6000	±8	±5	±5	±8	±8	±5	+8

¹⁾ Including stroke length.²⁾ Stroke tolerances must not be added to the tolerances listed in this table.

Technical data
 (For applications outside these values, please consult us.)

Proximity switch

Electrical			
Function type			PNP normally open contact
Admissible pressure		bar	500
Supply voltage	► Nominal value range	V DC	10 ... 30
	► Including residual ripple	%	≤ 15
Voltage drop		V	≤ 15
Rated operating voltage		V DC	24
Rated operating current		mA	200
Idle current		mA	≤ 8
Residual current		µA	≤ 10
Repetition accuracy		%	≤ 5
Hysteresis		%	≤ 15
Ambient temperature range		°C	-25 ... +80
Temperature drift		%	≤ 10
Switching frequency		Hz	1000
Protection class according to EN 60529	► Active area		IP68
	► Proximity switch		IP67
Housing material			Material no. 1.4104



Notice:

Inductive proximity switches are used as reliable end position control for hydraulic cylinders. They are an important element for the safe and exact monitoring of safety equipment, lockings and/or other machine functions in their end position by means of the output of signals.

The proximity switch which is high-pressure-resistant up to 500 bar works in a contactless manner. Consequently, it is wear-free. The proximity switch has been set at the factory. The switching distance must not be adjusted. The lock nut of the proximity switch is marked at the factory using sealing wax. On versions with proximity switch, the cylinders are provided with proximity switches on both sides.

Position measurement system

General	
Protection class according to EN 60529	IP67 (connector professionally mounted) IP68 at cable outlet
EMC test	Electro-magnetic interference emission according to EN 61000-6-3 Electro-magnetic interference resistance according to EN 61000-6-2 The sensor corresponds to EC directives and bears the CE marking



Notice:

The magnetostrictive position measurement system is contactless and absolute.

Technical data

(For applications outside these values, please consult us.)

Analog output

Version "F"; "C"			
Type of connection			1 x M16 connector (6-pole)
Power consumption		W	<3.25
Ambient temperature range		°C	-40 ... +85
Tightening torque M_A	► Measurement length $\leq$ 1500 mm	Nm	50 $\pm$ 4 %
	► Measurement length > 1500mm	Nm	65 $\pm$ 4%
Wrench size SW		mm	46
Voltage "F"	► Voltage range	V	0 ... 10
	► Input resistance control system	k Ω	>5
	► Resolution		16 bit (internal resolution 0.1 μ m)
Current "C"	► Current range	mA	4 ... 20
	► Minimum load	Ω	0
	► Maximum load	Ω	500
	► Resolution		16 bit (internal resolution 0.1 μ m)

Digital output

Version "D"; "S"			
Type of connection			1 x M16 connector (7-pole)
Power consumption		W	1.2 typical
Ambient temperature range		°C	-40 ... +85
Tightening torque M_A	► Measurement length $\leq$ 1500 mm	Nm	50 $\pm$ 4 %
	► Measurement length > 1500mm	Nm	65 $\pm$ 4%
Wrench size SW		mm	46
SSI "D"	► Interface		SSI 24 bit (RS-485/RS-422)
	► Resolution	μ m	5
	► Direction of measurement		Asynchronous forward
	► Data format		Gray
SSI "S"	► Interface		SSI 24 bit (RS-485/RS-422)
	► Resolution	μ m	1
	► Direction of measurement		Synchronous forward
	► Data format		Gray

Technical data

(For applications outside these values, please consult us.)

Profibus

Version "N"; "P"		Profibus 63 / Profibus 53	
Type of connection		2 x M16 connector (6-pole), alternatively 2 x M12 connector (5-pole) and 1 x M8 connector (4-pole)	
Current consumption		mA	90, typical
Ambient temperature range		°C	-40 ... +75
Tightening torque M_A		Nm	65 ±4%
Wrench size SW		mm	46
Output	► Interface		IEC 61158 CPF3 PROFIBUS
	► Data record		PROFIBUS-DP slave
	► Transmission rate	MBit/s	Max. 12
Measurement accuracy	► Travel resolution	µm	1 ... 1000 selectable as parameter
	► Velocity (5 µm travel resolution ¹⁾)		
	– up to 500 mm measurement length	mm/s	0.64
	– up to 2000 mm measurement length	mm/s	0.43
	– up to 4500 mm measurement length	mm/s	0.21
	– up to 7600 mm measurement length	mm/s	0.14
	► Linearity		< ± 0.01% Full Scale (Minimum +/- 50 µm)
	► Repeatability		< ± 0.001% Full Scale (Minimum +/- 2.5 µm)
	► Temperature coefficient	ppm/°C	< 15
	► Hysteresis	µm	< 4

¹⁾ 2 µm travel resolution corresponds to 2.5 times smaller values.**IO-Link**

Version "L"		IO-Link V1.1	
Type of connection		1 x M12 connector (4-pole)	
Operating voltage		VDC	+24 (± 25%) / residual ripple ≤ 0.28 Vpp
Current consumption		mA	< 50
Ambient temperature ranges		°C	-40 ... +75
Tightening torque M_A		Nm	50 ±4 %
Wrench size SW		mm	34
Output	► Interface		Digital
	► Transmission record		IO-Link V1.1
	► Data format	Bit	32 signed (position in µm)
	► Data transmission rate		COM3 (230.4 kBaud)
	► Process data		
	– Device – Master	Bytes	4
	– Master – Device	Bytes	0
Measured values	► Measured variable	MBit/s	Position
	► Resolution ²⁾	µm	5; 10; 20; 50 or 100
	► Cycle time:	ms	Minimum 1 (depending on master)
	► Linearity ³⁾	%	≤ ±0.02 F.S. (Minimum ±60 µm)
	► Measurement repetition accuracy	%	≤ ±0.005 F.S. (Minimum ±20 µm)

²⁾ Selectable via IO-Link Master.³⁾ Tested with position magnet 251 416-2.

Technical data

(For applications outside these values, please consult us.)

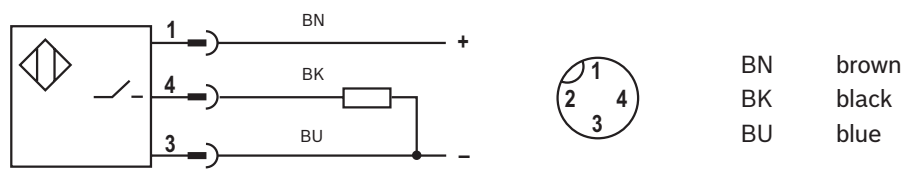
Profinet

Version "R"		Profinet RT / Profinet IRT version 2.3						
Electrical connection	► Type of connection		2 x M12 device socket (5-pole) 1 x M12 connector (4-pole)					
	► Operating voltage	VDC	+12 ... 30 (± 20%) (9.6 ... 36)					
	► Power consumption	W	< 4 typical					
Ambient temperature ranges		°C	-40 ... +85					
Tightening torque M _A		Nm	65 ±4%					
Wrench size SW		mm	46					
Output	► Interface		Profinet RT					
			Profinet IRT version 2.3					
	► Data record		MTS profile and encoder profile 4.1					
	► Data transmission rate	MBit/s	Max. 100					
Measured values	► Resolution	µm	0.5 ... 100 (selectable)					
	► Cycle time ¹⁾							
	– Measurement length	mm	≤ 50	≤ 715	≤ 2000	≤ 4675	≤ 7620	
	– Cycle time:	µs	250	500	1000	2000	4000	
	► Linearity deviation ²⁾ Measurement length		mm	≤ 500			> 500	
	► Linearity deviation		µm	≤ ± 50			< 0.01% F.S	
	► Measurement repetition accuracy		%	≤ ±0.001 F.S. (Minimum ±2.5 µm) typical				
	► Hysteresis		µm	< 4 typical				
	► Temperature coefficient		ppm/K	< 15 typical				

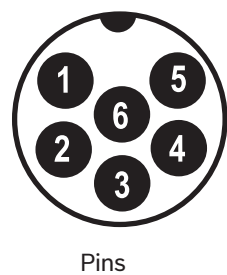
¹⁾ Sensor with standard settings.²⁾ Tested with position magnet 251 416-2.

Electrical connections and assignment

► Proximity switch

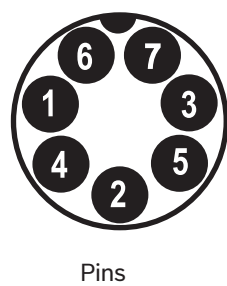


► Analog output



Pin	Cable	Function
1	gray	Position 1 (solenoid)
2	pink	DC ground
3	yellow	Not used
4	green	DC ground
5	brown	+12 ... 30 VDC ±20% (9.6 ... 36 VDC)
6	white	DC ground (0 V)

► Digital output

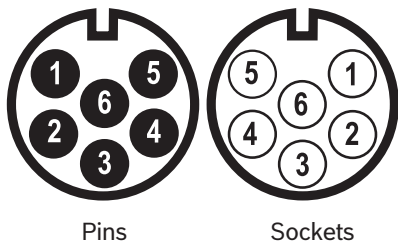


Pin	Cable	Function
1	gray	Data (-)
2	pink	Data (+)
3	yellow	Clock (+)
4	green	Clock (-)
5	brown	+12 ... 30 VDC ±20% (9.6 ... 36 VDC)
6	white	DC ground (0 V)
7	-	Not used

Notice:
Mating connectors, separate order, see page 82 and data sheet 08006.

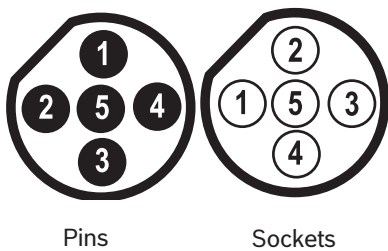
Electrical connections and assignment

► Profibus "D63"



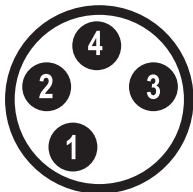
Pin	Cable	Function
1	green	RxD/TxD-N (bus)
2	red	RxD/TxD-P (bus)
3	--	DGND (terminating resistor) *
4	--	VP (terminating resistor) *
5	black	+24 V DC (-15 / +20%)
6	blue	DC ground (0 V)
--	yellow/green	Shield compensating line, is usually not to be connected

► Profibus "D53"



Pin	Cable	Function
1	--	VP+5 (terminating resistor) *
2	green	RxD/TxD-N (bus)
3	--	DGND (terminating resistor) *
4	red	RxD/TxD-P (bus)
5	Shield	Shield

► Supply



View connector side

Pin	Cable	Function
1	brown	+24 V DC (-15 / +20%)
2	white	Not used
3	blue	DC ground (0 V)
4	black	Not used

* only with sockets



Notice:

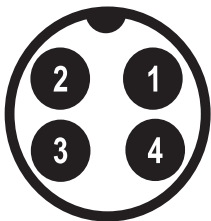
Mating connectors, separate order, see page 82 and data sheet 08006.

Electrical connections and assignment

► IO-Link

M12 connector (A-coded)

View to sensor

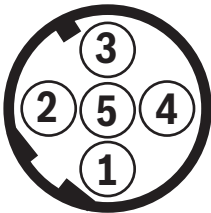


Pin	Function
1	+ 24 VDC (± 25%)
2	DI / DQ
3	DC ground (0 V)
4	C / Q

► Profinet

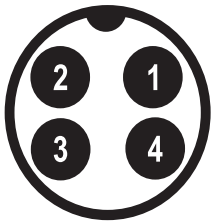
M12 device socket (D-coded)

Port 1 and 2 - signal
(View to sensor)



Pin	Function
1	Tx (+)
2	Rx (+)
3	Tx (-)
4	Rx (-)
5	Not used

M12 connector (A-coded)



Pin	Function
1	+12...30 VDC (±20%)
2	Not used
3	DC ground (0 V)
4	Not used



Notice:

Mating connectors, separate order, see page 82 and data sheet 08006.

Types of mounting: Series CDH1; CGH1

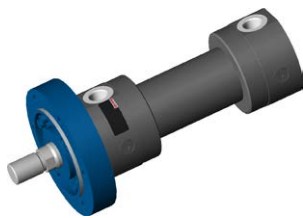
CDH3 "MP3"; see page 26, 27



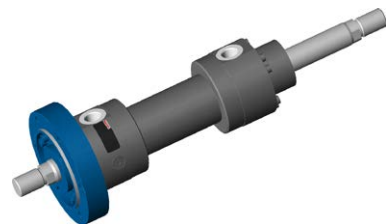
CDH3 "MP5"; see page 28, 29



CDH3 "MF3"; see page 30, 31



CGH3 "MF3"; see page 30, 31



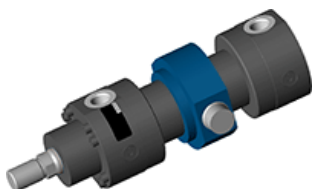
CDH3 "MF4"; see page 32, 33



CGH3 "MT4"; see page 36, 37



CDH3 "MT4"; see page 34, 35



CDH3 "MS2"; see page 38, 39



CGH3 "MS2"; see page 38, 39

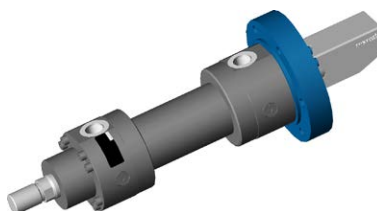


Types of mounting: Series CSH1

CSH3 "MP3"; see page 40, 41



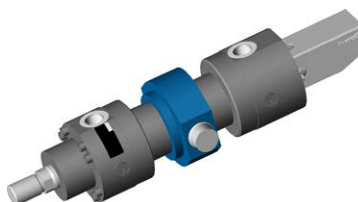
CSH3 "MF4"; see page 46, 47



CSH3 "MP5"; see page 42, 43



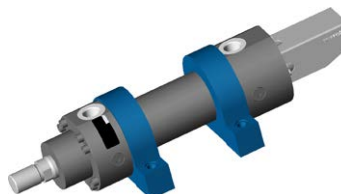
CSH3 "MT4"; see page 48, 49



CSH3 "MF3"; see page 44, 45



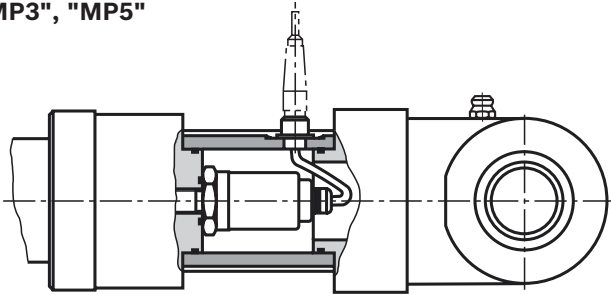
CSH3 "MS2"; see page 50, 51



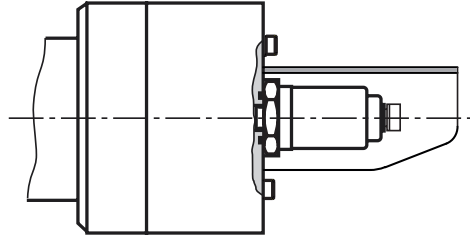
Types of mounting

► Analog and digital output

"MP3", "MP5"



"MF3", "MF4", "MT4", "MS2"

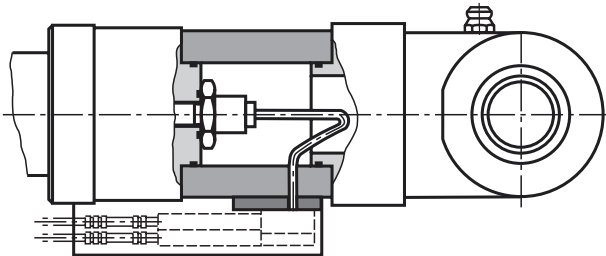


Notice:

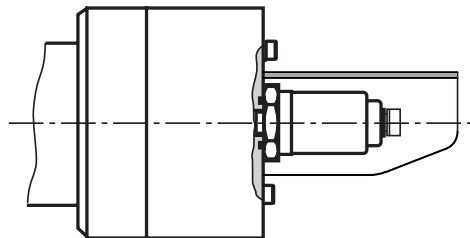
- For analog output: 6-pole amphenol mating connector, material no. R901511626 (to be ordered separately; see page 82 and data sheet 08006)
- For digital output: 7-pole amphenol mating connector, material no. R900079551 (to be ordered separately; see page 82 and data sheet 08006)

► Profinet and Profibus

"MP3", "MP5"



"MF3", "MF4", "MT4", "MS2"

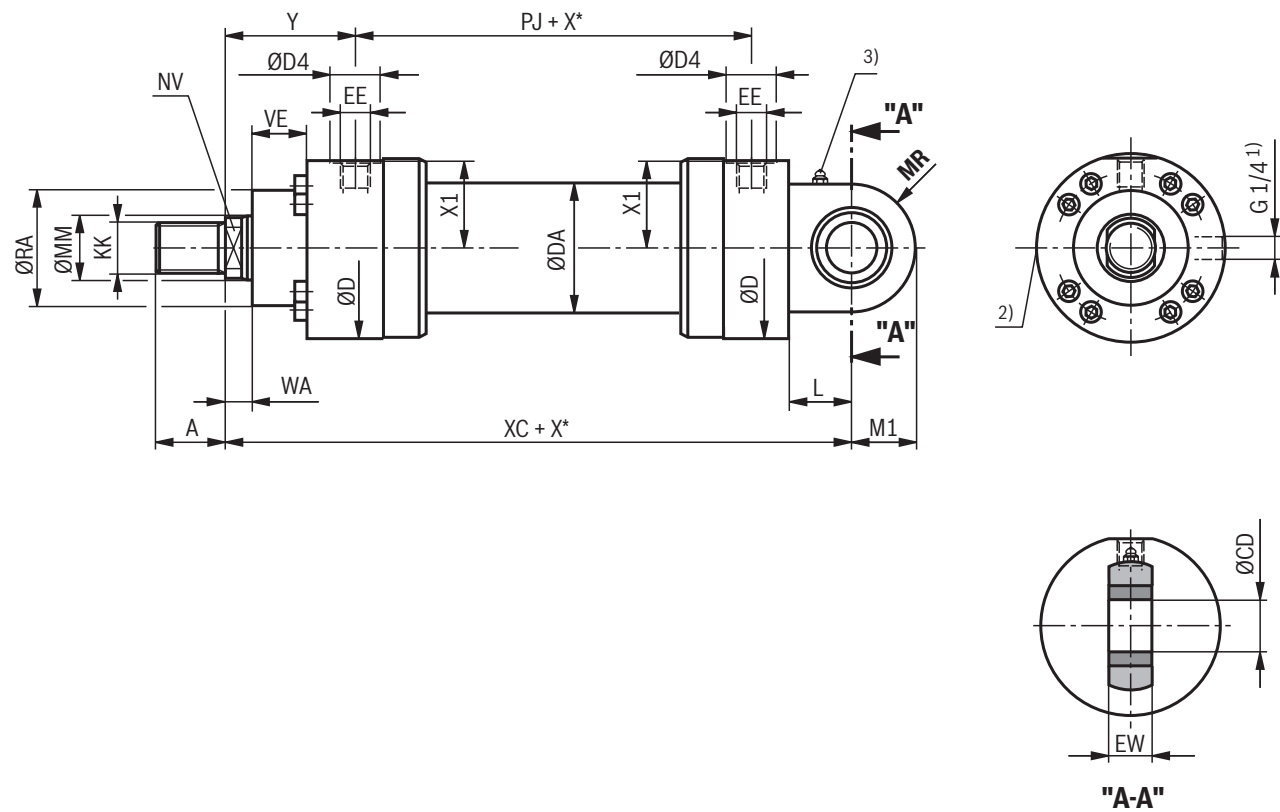


Notice:

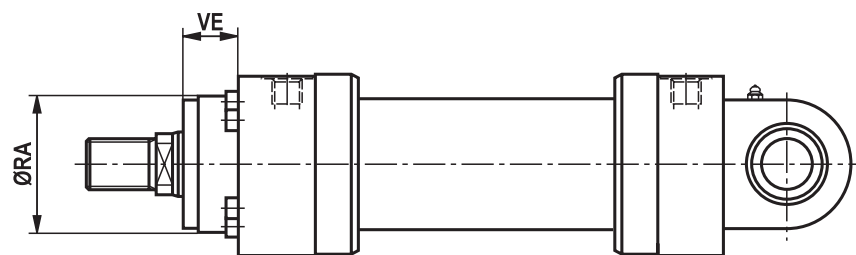
- The output of the position measurement system is by default always rotated by 180° to the selected position of the hydraulic connection in the cylinder base.

Dimensions: Swivel eye at base CDH3 – "MP3"
(Dimensions in mm)

CDH3 "MP3"



CDH3 "MP3": Seal design "A", "B" and $\varnothing AL$ 160 ... 200 mm



Dimensions: Swivel eye at base CDH3 – "MP3"

(Dimensions in mm)

ØAL	ØMM	KK ⁴⁾	A ⁴⁾	KK ⁵⁾	A ⁵⁾	NV	ØD	ØDA	ØD4 ⁶⁾	EE ^{7); 8)}	EE ^{7); 9)}	Y	PJ
40	28	M22 x 1.5	22	M24 x 2	35	22	92	52	34	G 1/2	M22 x 1.5	91	120
50	36	M28 x 1.5	28	M30 x 2	45	30	108	62	34	G 1/2	M22 x 1.5	90	120
63	45	M35 x 1.5	35	M39 x 3	55	36	140	78	42	G 3/4	M27 x 2	117	133
80	56	M45 x 1.5	45	M50 x 3	75	46	148	100	42	G 3/4	M27 x 2	124	146
100	70	M58 x 1.5	58	M64 x 3	95	60	186	125	47	G 1	M33 x 2	119	171
125	90	M65 x 1.5	65	M80 x 3	110	75	235	160	58	G 1 1/4	M42 x 2	170	205
140	100	M80 x 2	80	M90 x 3	120	85	258	175	58	G 1 1/4	M42 x 2	186	219
160	110	M100 x 2	100	M100 x 3	140	95	292	200	65	G 1 1/2	M48 x 2	210	240
180	125	M110 x 2	110	M110 x 4	150	110	325	220	65	G 1 1/2	M48 x 2	241	264
200	140	M120 x 3	120	M120 x 4	160	120	350	245	65	G 1 1/2	M48 x 2	262	278

ØAL	ØMM	X1	WA	XC	L	MR	M1	ØCD H11	EW h12	ØRA ¹⁰⁾	VE ¹⁰⁾	ØRA ¹¹⁾	VE ¹¹⁾
40	28	43	18	268	35	36	34	30	28	52	45	52	20
50	36	51.5	18	280	45	42	40	35	30	70	47	70	19
63	45	67	22	330	50	52	50	40	35	88	43	88	13
80	56	71.5	22	355	55	65	62.5	50	40	98	53	98	15
100	70	90.5	25	390	65	70	70	60	50	120	55	120	17
125	90	114	32	495	75	82	82	70	55	150	68	150	20
140	100	126	35	530	80	95	95	80	60	170	75	170	23
160	110	142.5	40	600	90	113	113	90	65	200	90	200	90
180	125	159.5	45	665	105	125	125	100	70	230	100	230	100
200	140	172.5	45	710	115	142.5	142.5	110	80	250	110	250	110

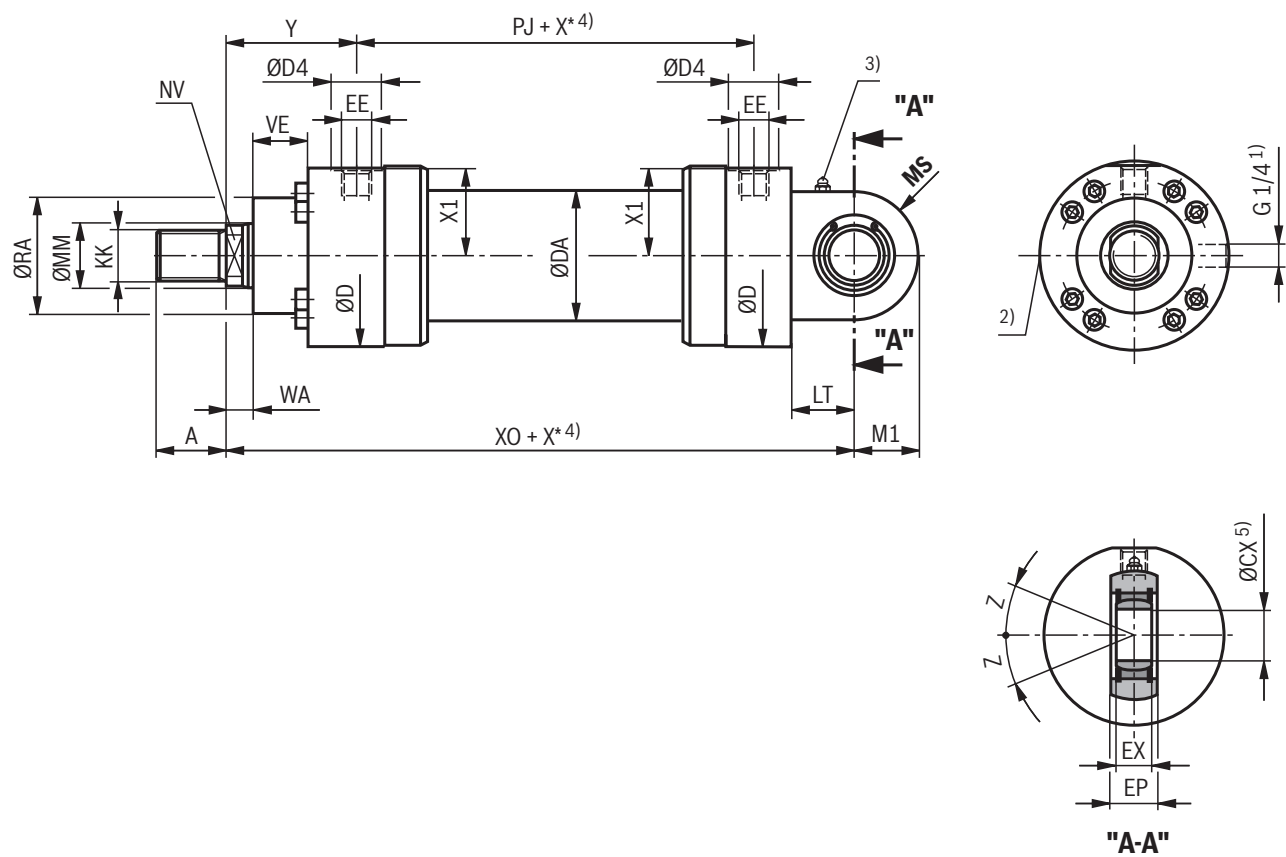
ØAL = piston Ø
 ØMM = piston rod Ø
 X* = stroke length

- 1) Bleeding: With view to the piston rod, the position is offset by 90° in relation to the line connection (clockwise)
- 2) Throttle valve only with end position damping "E" (180° for bleeding).
- 3) Standard version "W";
Grease nipple, cone head form A according to DIN 71412.
- 4) Thread design "G".

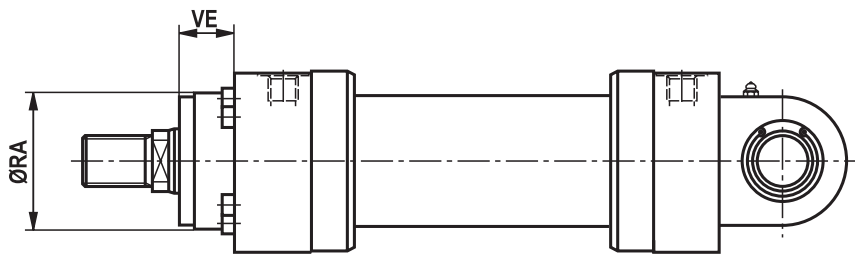
- 5) Thread design "A".
- 6) ØD4 maximum depth 0.5 mm.
- 7) Flange connections see separate table page 54 and 55.
- 8) Line connection "B" and "C".
- 9) Line connection "M".
- 10) Dimensions for cylinders with seal design "M", "T", "G", "L", "R", "S" and "V".
- 11) Dimensions for cylinders with seal design "A" and "B".

Dimensions: Self-aligning clevis at base CDH3 – "MP5"
(Dimensions in mm)

CDH3 "MP5"



CDH3 "MP5": Seal design "A", "B" and $\varnothing AL$ 160 ... 320 mm



Dimensions: Self-aligning clevis at base CDH3 – "MP5"

(Dimensions in mm)

ØAL	ØMM	KK ⁽⁶⁾	A ⁽⁶⁾	KK ⁽⁷⁾	A ⁽⁷⁾	NV	ØD	ØDA	ØD4 ⁽⁸⁾	EE ^(9); 10)	EE ^(9); 11)	Y	PJ	X1	C ₀ ⁽¹²⁾ kN
40	28	M22 x 1.5	22	M24 x 2	35	22	92	52	34	G 1/2	M22 x 1.5	91	120	43	106
50	36	M28 x 1.5	28	M30 x 2	45	30	108	62	34	G 1/2	M22 x 1.5	90	120	51.5	153
63	45	M35 x 1.5	35	M39 x 3	55	36	140	78	42	G 3/4	M27 x 2	117	133	67	250
80	56	M45 x 1.5	45	M50 x 3	75	46	148	100	42	G 3/4	M27 x 2	124	146	71.5	365
100	70	M58 x 1.5	58	M64 x 3	95	60	186	125	47	G 1	M33 x 2	119	171	90.5	400
125	90	M65 x 1.5	65	M80 x 3	110	75	235	160	58	G 1 1/4	M42 x 2	170	205	114	540
140	100	M80 x 2	80	M90 x 3	120	85	258	175	58	G 1 1/4	M42 x 2	186	219	126	670
160	110	M100 x 2	100	M100 x 3	140	95	292	200	65	G 1 1/2	M48 x 2	210	240	142.5	980
180	125	M110 x 2	110	M110 x 4	150	110	325	220	65	G 1 1/2	M48 x 2	241	264	159.5	1120
200	140	M120 x 3	120	M120 x 4	160	120	350	245	65	G 1 1/2	M48 x 2	262	278	172.5	1700
220	160	M120 x 3	120	M120 x 4	160	140	375	292	65	G 1 1/2	M48 x 2	262	326	185	1700
250	180	M130 x 3	130	M150 x 4	190	160	440	324	65	G 1 1/2	M48 x 2	272	336	218	2900
280	200	–	–	M160 x 4	200	180	460	368	65	G 1 1/2	M48 x 2	282	366	228	–
320	220	–	–	M180 x 4	220	200	490	406	65	G 1 1/2	M48 x 2	287	391	243	–

ØAL	ØMM	F _{adm} ⁽¹³⁾ kN	WA	XO	X _{min} [*]	LT	M1	MS	ØCX	EP -0.4	EX	Z	ØRA ⁽¹⁵⁾	VE ⁽¹⁵⁾	ØRA ⁽¹⁶⁾	VE ⁽¹⁶⁾
40	28	38.2	18	268	–	35	34	36	30 _{-0.010}	28	22 _{-0.12}	6 °	52	45	52	20
50	36	55.1	18	280	–	45	40	42	35 _{-0.012}	30	25 _{-0.12}	6 °	70	47	70	19
63	45	90.0	22	330	–	50	50	52	40 _{-0.012}	35	28 _{-0.12}	7 °	88	43	88	13
80	56	131.4	22	355	–	55	62.5	65	50 _{-0.012}	40	35 _{-0.12}	6 °	98	53	98	15
100	70	144.0	25	390	–	65	70	70	60 _{-0.015}	50	44 _{-0.15}	6 °	120	55	120	17
125	90	194.4	32	495	–	75	82	82	70 _{-0.015}	55	49 _{-0.15}	6 °	150	68	150	20
140	100	241.2	35	530	–	80	95	95	80 _{-0.015}	60	55 _{-0.15}	6 °	170	75	170	23
160	110	352.8	40	600	–	90	113	113	90 _{-0.020}	65	60 _{-0.20}	5 °	200	90	200	90
180	125	403.2	45	665	–	105	125	125	100 _{-0.020}	70	70 _{-0.20}	7 °	230	100	230	100
200	140	612.0	45	710	–	115	142.5	142.5	110 _{-0.020}	80	70 _{-0.20}	6 °	250	110	250	110
220	160	612.0	40	760	–	115	150 ⁽¹⁴⁾	140 ⁽¹⁴⁾	110 _{-0.020}	80	70 _{-0.20}	6 °	275	125	275	125
250	180	1044.0	40	825	20	140	188 ⁽¹⁴⁾	178 ⁽¹⁴⁾	120 _{-0.020}	90	85 _{-0.20}	6 °	320	135	320	135
280	200	–	40	895	–	170	210 ⁽¹⁴⁾	200 ⁽¹⁴⁾	140 _{-0.025}	100	90 _{-0.25}	7 °	335	150	335	150
320	220	–	40	965	340	200	260 ⁽¹⁴⁾	250 ⁽¹⁴⁾	160 _{-0.025}	110	105 _{-0.25}	8 °	350	165	350	165

ØAL = piston Ø

ØMM = piston rod Ø

X* = stroke length

X*_{min} = minimum stroke length

(With hydraulic cylinders with end position damping,
see notice on page 68.)

- 1) Bleeding: With view to the piston rod, the position is offset by 90° in relation to the line connection (clockwise)
- 2) Throttle valve only with end position damping "E" (180° for bleeding).
- 3) Standard version "W";
Grease nipple, cone head form A according to DIN 71412;
not applicable to spherical bearing, maintenance-free "A".
- 4) Observe minimum stroke length "X*_{min}".
- 5) Related bolt Ø m6;
Related bolt Ø j6 for maintenance-free spherical bearing.
- 6) Thread design "G".

7) Thread design "A".

8) ØD4 maximum depth 0.5 mm.

9) Flange connections see separate table page 54 and 55.

10) Line connection "B" and "C".

11) Line connection "M".

12) C₀ = static load rating of the swivel head

13) F_{adm} = Maximum admissible load of the swivel head with oscillatory or alternating loads

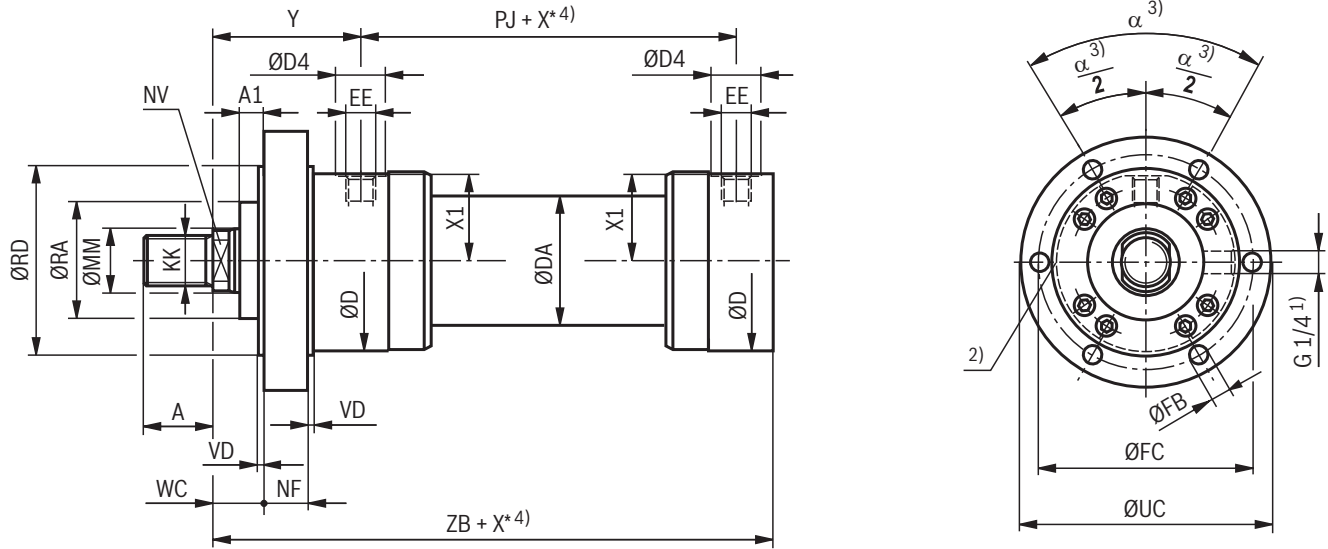
14) The specified dimensions are maximum values, tolerance classes 342 according to ISO 9013 Thermal cutting.

15) Dimensions for cylinders with seal design "M", "T", "G", "L", "R", "S" and "V".

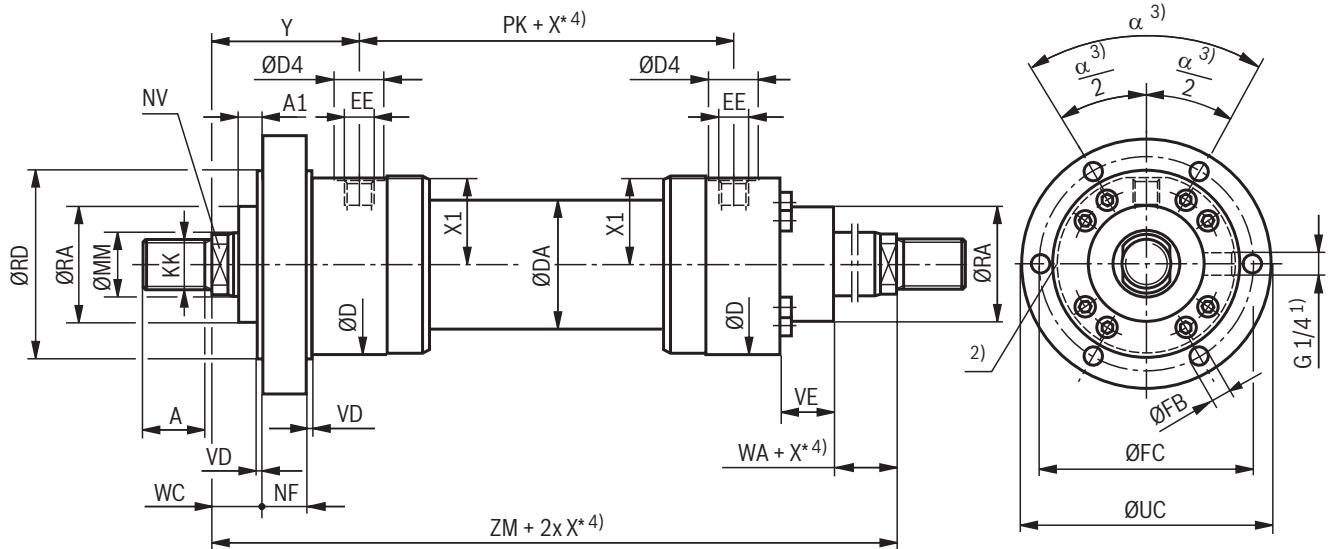
16) Dimensions for cylinders with seal design "A" and "B".

Dimensions: Round flange at head CDH3, CGH3 – "MF3"
(Dimensions in mm)

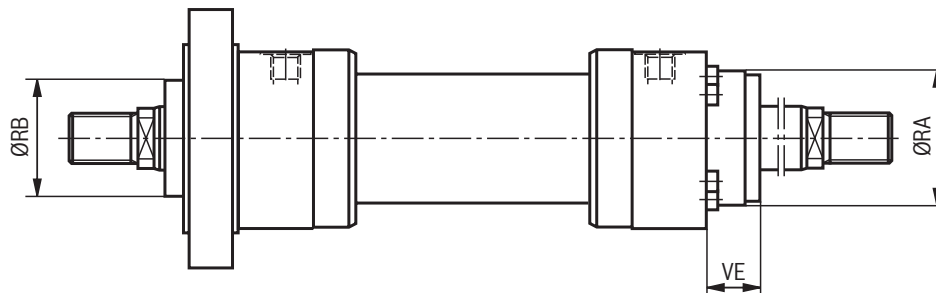
CDH3 "MF3"



CGH3 "MF3"



CGH3 "MF3": Seal design "A", "B" and ØAL 160 ... 320 mm



Dimensions: Round flange at head CDH3, CGH3 – "MF3"

(Dimensions in mm)

ØAL	ØMM	KK ⁵⁾	A ⁵⁾	KK ⁶⁾	A ⁶⁾	NV	ØD	ØDA	ØD4 ⁷⁾	EE ^{8); 9)}	EE ^{8); 10)}	Y	PJ	x 1
40	28	M22 x 1.5	22	M24 x 2	35	22	92	52	34	G 1/2	M22 x 1.5	91	120	43
50	36	M28 x 1.5	28	M30 x 2	45	30	108	62	34	G 1/2	M22 x 1.5	90	120	51.5
63	45	M35 x 1.5	35	M39 x 3	55	36	140	78	42	G 3/4	M27 x 2	117	133	67
80	56	M45 x 1.5	45	M50 x 3	75	46	148	100	42	G 3/4	M27 x 2	124	146	71.5
100	70	M58 x 1.5	58	M64 x 3	95	60	186	125	47	G 1	M33 x 2	119	171	90.5
125	90	M65 x 1.5	65	M80 x 3	110	75	235	160	58	G 1 1/4	M42 x 2	170	205	114
140	100	M80 x 2	80	M90 x 3	120	85	258	175	58	G 1 1/4	M42 x 2	186	219	126
160	110	M100 x 2	100	M100 x 3	140	95	292	200	65	G 1 1/2	M48 x 2	210	240	142.5
180	125	M110 x 2	110	M110 x 4	150	110	325	220	65	G 1 1/2	M48 x 2	241	264	159.5
200	140	M120 x 3	120	M120 x 4	160	120	350	245	65	G 1 1/2	M48 x 2	262	278	172.5
220	160	M120 x 3	120	M120 x 4	160	140	375	292	65	G 1 1/2	M48 x 2	262	326	185
250	180	M130 x 3	130	M150 x 4	190	160	440	324	65	G 1 1/2	M48 x 2	272	336	218
280	200	–	–	M160 x 4	200	180	460	368	65	G 1 1/2	M48 x 2	282	366	228
320	220	–	–	M180 x 4	220	200	490	406	65	G 1 1/2	M48 x 2	287	391	243

ØAL	ØMM	ØRD e8	WC	VD	NF	PK	A1	ZB	ZM	X _{min} [*]	ØFB H13	ØFC js13	ØUC -1	α	WA	ØRA ¹¹⁾	VE ¹¹⁾	ØRA ¹²⁾	VE ¹²⁾
40	28	95	23	5	35	120	0	238	302	–	13.5	120	145	60°	18	52	45	52	20
50	36	115	20	5	40	120	0	237	300	–	13.5	140	165	60°	18	70	47	70	19
63	45	150	20	5	40	133	0	285	367	–	17.5	180	210	60°	22	88	43	88	13
80	56	160	20	5	50	146	0	305	394	–	17.5	195	230	60°	22	98	53	98	15
100	70	200	20	5	55	171	0	330	409	–	22	230	270	60°	25	120	55	120	17
125	90	245	25	5	70	205	0	425	545	–	26	290	335	60°	32	150	68	150	20
140	100	280	30	10	70	219	0	457	591	–	30	330	380	60°	35	170	75	170	23
160	110	300	40	10	80	240	0	515	660	–	30	360	420	45°	40	200	90	200	90
180	125	335	40	10	95	264	0	565	746	–	36	400	470	45°	45	230	100	230	100
200	140	360	40	10	105	278	0	600	802	–	36	430	500	45°	45	250	110	250	110
220	160	400	40	10	115	326	0	655	850	–	39	475	550	45°	40	275	125	275	125
250	180	450	40	10	125	336	0	695	880	20	45	530	610	45°	40	320	135	320	135
280	200	470	50	10	130	366	0	735	930	–	45	550	630	45°	40	335	150	335	150
320	220	510	55	10	140	391	5	775	965	340	45	590	670	30°	40	350	165	350	165

ØAL = piston Ø

ØMM = piston rod Ø

X* = stroke length

X*_{min} = minimum stroke length

(With hydraulic cylinders with end position damping,
see notice on page 68.)

1) Bleeding: With view to the piston rod, the position is offset by 90° in
relation to the line connection (clockwise)

2) Throttle valve only with end position damping "E" (180° for bleeding).

3) With piston Ø 160 ... 280 mm 8 mounting bores.
With piston Ø 320 mm 12 mounting bores.

4) Observe minimum stroke length "X*_{min}".

5) Thread design "G".

6) Thread design "A".

7) Ø D4 maximum depth 0.5 mm.

8) Flange connections see separate table page 54 and 55.

9) Line connection "B" and "C".

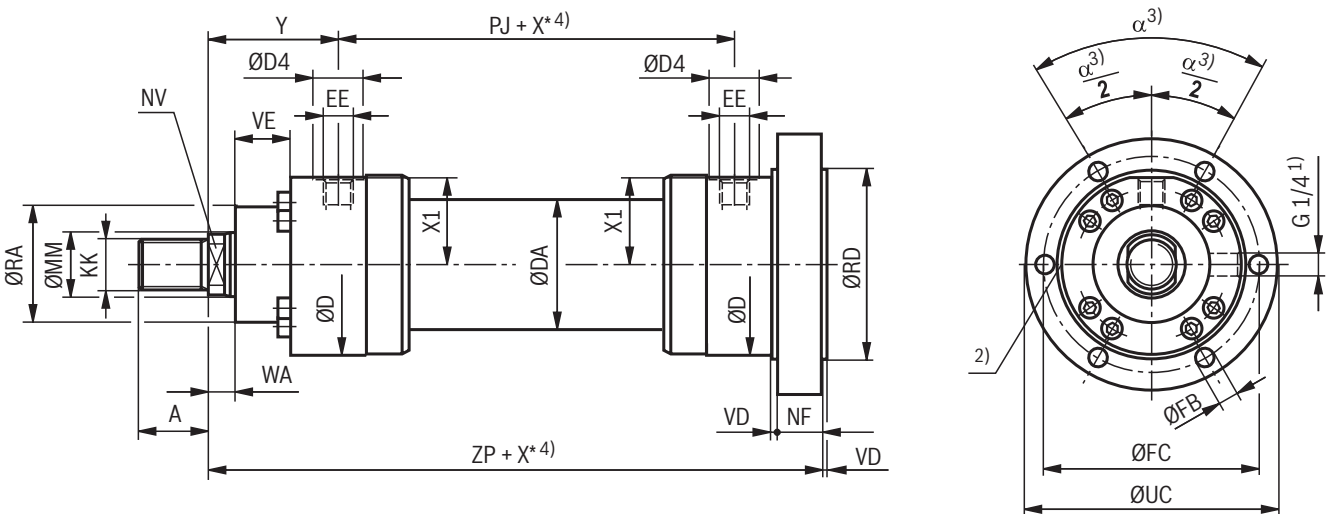
10) Line connection "M".

11) Dimensions for cylinders with seal design "M", "T", "G", "L", "R", "S"
and "V".

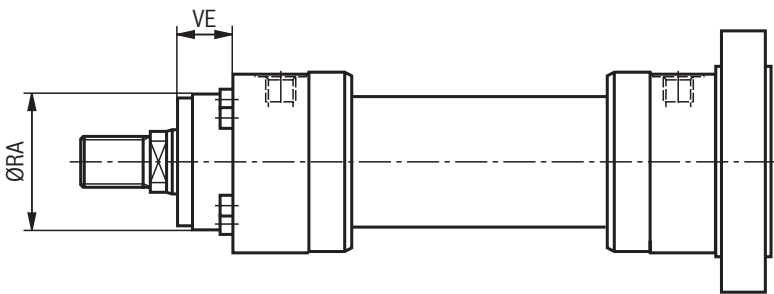
12) Dimensions for cylinders with seal design "A" and "B".

Dimensions: Round flange at base CDH3 – "MF4"
(Dimensions in mm)

CDH3 "MF4"



CDH3 "MF4": Seal design "A", "B" and $\varnothing AL$ 160 ... 320 mm



Dimensions: Round flange at base CDH3 – "MF4"

(Dimensions in mm)

ØAL	ØMM	KK ⁵⁾	A ⁵⁾	KK ⁶⁾	A ⁶⁾	NV	ØD	ØDA	ØD4 ⁷⁾	EE ^{8); 9)}	EE ^{8); 10)}	Y	PJ	X1
40	28	M22 x 1.5	22	M24 x 2	35	22	92	52	34	G 1/2	M22 x 1.5	91	120	43
50	36	M28 x 1.5	28	M30 x 2	45	30	108	62	34	G 1/2	M22 x 1.5	90	120	51.5
63	45	M35 x 1.5	35	M39 x 3	55	36	140	78	42	G 3/4	M27 x 2	117	133	67
80	56	M45 x 1.5	45	M50 x 3	75	46	148	100	42	G 3/4	M27 x 2	124	146	71.5
100	70	M58 x 1.5	58	M64 x 3	95	60	186	125	47	G 1	M33 x 2	119	171	90.5
125	90	M65 x 1.5	65	M80 x 3	110	75	235	160	58	G 1 1/4	M42 x 2	170	205	114
140	100	M80 x 2	80	M90 x 3	120	85	258	175	58	G 1 1/4	M42 x 2	186	219	126
160	110	M100 x 2	100	M100 x 3	140	95	292	200	65	G 1 1/2	M48 x 2	210	240	142.5
180	125	M110 x 2	110	M110 x 4	150	110	325	220	65	G 1 1/2	M48 x 2	241	264	159.5
200	140	M120 x 3	120	M120 x 4	160	120	350	245	65	G 1 1/2	M48 x 2	262	278	172.5
220	160	M120 x 3	120	M120 x 4	160	140	375	292	65	G 1 1/2	M48 x 2	262	326	185
250	180	M130 x 3	130	M150 x 4	190	160	440	324	65	G 1 1/2	M48 x 2	272	336	218
280	200	–	–	M160 x 4	200	180	460	368	65	G 1 1/2	M48 x 2	282	366	228
320	220	–	–	M180 x 4	220	200	490	406	65	G 1 1/2	M48 x 2	287	391	243

ØAL	ØMM	WA	ZP	X _{min} [*]	NF	VD	ØRD e8	ØFB H13	ØFC js13	ØUC -1	α	ØRA ¹¹⁾	VE ¹¹⁾	ØRA ¹²⁾	VE ¹²⁾
40	28	18	273	–	35	5	95	13.5	120	145	60°	52	45	52	20
50	36	18	277	–	40	5	115	13.5	140	165	60°	70	47	70	19
63	45	22	325	–	40	5	150	17.5	180	210	60°	88	43	88	13
80	56	22	355	–	50	5	160	17.5	195	230	60°	98	53	98	15
100	70	25	385	–	55	5	200	22	230	270	60°	120	55	120	17
125	90	32	495	–	70	5	245	26	290	335	60°	150	68	150	20
140	100	35	532	–	70	10	280	30	330	380	60°	170	75	170	23
160	110	40	600	–	80	10	300	30	360	420	45°	200	90	200	90
180	125	45	665	–	95	10	335	36	400	470	45°	230	100	230	100
200	140	45	710	–	105	10	360	36	430	500	45°	250	110	250	110
220	160	40	770	–	115	10	400	39	475	550	45°	275	125	275	125
250	180	40	820	20	125	10	450	45	530	610	45°	320	135	320	135
280	200	40	865	–	130	10	470	45	550	630	45°	335	150	335	150
320	220	40	915	340	140	10	510	45	590	670	30°	350	165	350	165

ØAL = piston Ø

ØMM = piston rod Ø

X* = stroke length

X_{min}^{*} = minimum stroke length

(With hydraulic cylinders with end position damping,
see notice on page 68.)

1) Bleeding: With view to the piston rod, the position is offset by 90° in relation to the line connection (clockwise)

2) Throttle valve only with end position damping "E" (180° for bleeding).

3) With piston Ø 160 ... 280 mm 8 mounting bores.

With piston Ø 320 mm 12 mounting bores.

4) Observe minimum stroke length "X_{min}^{*}".

5) Thread design "G".

6) Thread design "A".

7) ØD4 maximum depth 0.5 mm.

8) Flange connections see separate table page 54 and 55.

9) Line connection "B" and "C".

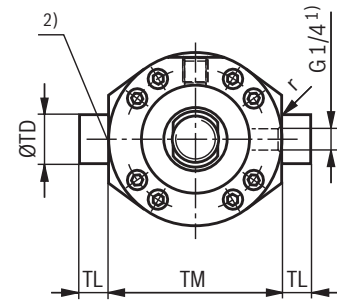
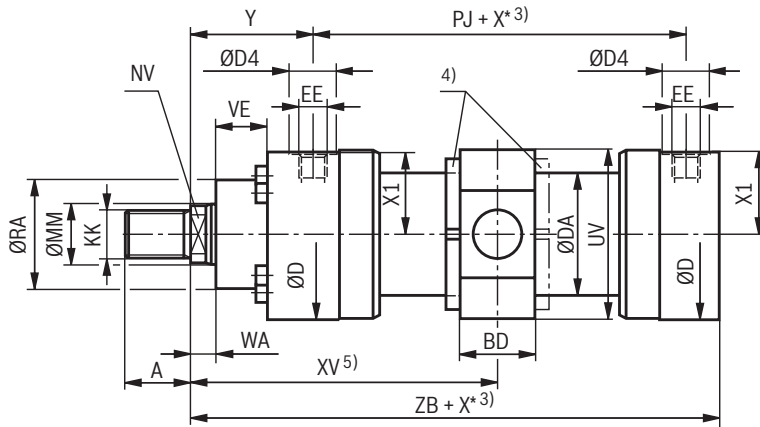
10) Line connection "M".

11) Dimensions for cylinders with seal design "M", "T", "G", "L", "R", "S" and "V".

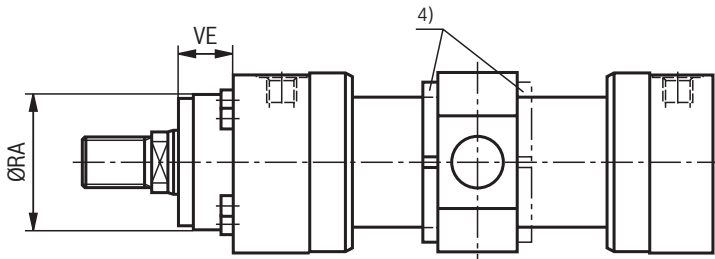
12) Dimensions for cylinders with seal design "A" and "B".

Dimensions: Trunnion mounting CDH3 – "MT4" (Dimensions in mm)

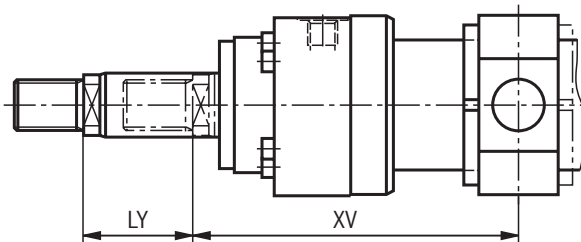
CDH3 "MT4"



CDH3 "MT4": Seal design "A", "B" and ΔAL 160 ... 320 mm



Dimensions for cylinder with piston rod extension "LY" in retracted condition



Notice:

During installation, it must be ensured that the trunnion bearings are installed up to the trunnion shoulders. Any variation may reduce the product's service life.

Dimensions: Trunnion mounting CDH3 – "MT4"

(Dimensions in mm)

ØAL	ØMM	KK ⁶⁾	A ⁶⁾	KK ⁷⁾	A ⁷⁾	NV	ØD	ØDA	ØD4 ⁸⁾	EE ⁹⁾ ; 10)	EE ⁹⁾ ; 11)	Y	PJ	X1	WA
40	28	M22 x 1.5	22	M24 x 2	35	22	92	52	34	G 1/2	M22 x 1.5	91	120	43	18
50	36	M28 x 1.5	28	M30 x 2	45	30	108	62	34	G 1/2	M22 x 1.5	90	120	51.5	18
63	45	M35 x 1.5	35	M39 x 3	55	36	140	78	42	G 3/4	M27 x 2	117	133	67	22
80	56	M45 x 1.5	45	M50 x 3	75	46	148	100	42	G 3/4	M27 x 2	124	146	71.5	22
100	70	M58 x 1.5	58	M64 x 3	95	60	186	125	47	G 1	M33 x 2	119	171	90.5	25
125	90	M65 x 1.5	65	M80 x 3	110	75	235	160	58	G 1 1/4	M42 x 2	170	205	114	32
140	100	M80 x 2	80	M90 x 3	120	85	258	175	58	G 1 1/4	M42 x 2	186	219	126	35
160	110	M100 x 2	100	M100 x 3	140	95	292	200	65	G 1 1/2	M48 x 2	210	240	142.5	40
180	125	M110 x 2	110	M110 x 4	150	110	325	220	65	G 1 1/2	M48 x 2	241	264	159.5	45
200	140	M120 x 3	120	M120 x 4	160	120	350	245	65	G 1 1/2	M48 x 2	262	278	172.5	45
220	160	M120 x 3	120	M120 x 4	160	140	375	292	65	G 1 1/2	M48 x 2	262	326	185	40
250	180	M130 x 3	130	M150 x 4	190	160	440	324	65	G 1 1/2	M48 x 2	272	336	218	40
280	200	–	–	M160 x 4	200	180	460	368	65	G 1 1/2	M48 x 2	282	366	228	40
320	220	–	–	M180 x 4	220	200	490	406	65	G 1 1/2	M48 x 2	287	391	243	40

ØAL	ØMM	ZB	X _{min} [*]	XV _{mid} ¹²⁾	XV _{min} ⁵⁾	XV _{max} ⁵⁾	BD	UV ¹³⁾	ØTD _{e8}	TL _{js16}	TM _{h12}	r	ØRA ¹⁴⁾	VE ¹⁴⁾	ØRA ¹⁵⁾	VE ¹⁵⁾
40	28	238	42	151+X*/2	172	138+X*	48	101	40	30	95	2	52	45	52	20
50	36	237	50	150+X*/2	175	134+X*	48	117	40	30	120	2	70	47	70	19
63	45	285	64	183.5+X*/2	215.5	163.5+X*	53	153	45	35	150	2	88	43	88	13
80	56	305	82	197+X*/2	238	168+X*	68	169	55	50	160	2	98	53	98	15
100	70	330	109	204.5+X*/2	259	165+X*	88	203	60	55	200	2	120	55	120	17
125	90	425	131	272.5+X*/2	338	207+X*	118	252	75	60	245	2.5	150	68	150	20
140	100	457	147	295.5+X*/2	369	222+X*	128	282	85	70	280	2.5	170	75	170	23
160	110	515	186	330+X*/2	423	237+X*	148	310	95	80	300	2.5	200	90	200	90
180	125	565	212	373+X*/2	479	267+X*	168	348	110	90	335	2.5	230	100	230	100
200	140	600	228	401+X*/2	515	287+X*	188	373	120	100	360	2.5	250	110	250	110
220	160	655	205	425+X*/2	527.5	322.5+X*	165	398	130	100	400	2.5	275	125	275	125
250	180	695	245	440+X*/2	562.5	317.5+X*	175	463	140	100	450	5	320	135	320	135
280	200	735	245	465+X*/2	587.5	342.5+X*	205	486	170	125	480	5	335	150	335	150
320	220	775	600	482.5+X*/2	782.5	182.5+X*	245	537	200	150	500	5	350	165	350	165

ØAL = piston Ø

ØMM = piston rod Ø

X* = stroke length

X*_{max} = maximum stroke length

X*_{min} = minimum stroke length

(With hydraulic cylinders with end position damping, see notice on page 68.)

1) Bleeding: With view to the piston rod, the position is offset by 90° in relation to the line connection (clockwise)

2) Throttle valve only with end position damping "E" (180° for bleeding).

3) Observe minimum stroke length "X*_{min}".

4) Trunnion nut for ØAL ≥ 125 mm depending on the position of the trunnion mounting (XV) either on the head or base side.

5) When ordering, always specify the "XV" dimension in the plain text.

Preferred dimension XV: Trunnion mounting position in cylinder center
Observe XV_{min} and XV_{max}

6) Thread design "G".

7) Thread design "A".

8) ØD4 maximum depth 0.5 mm.

9) Flange connections see separate table page 54 and 55.

10) Line connection "B" and "C".

11) Line connection "M".

12) XV_{mid} recommendation: Trunnion mounting position in cylinder center.

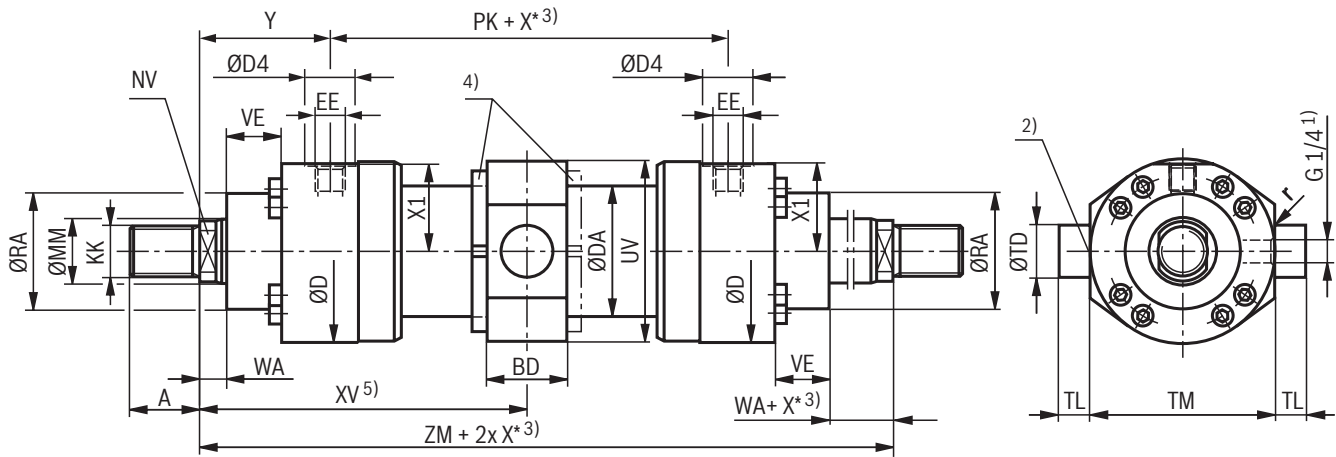
13) The dimensions given are maximum values, tolerance classes 342 according to ISO 9013 Thermal cutting.

14) Dimensions for cylinders with seal design "M", "T", "G", "L", "R", "S" and "V".

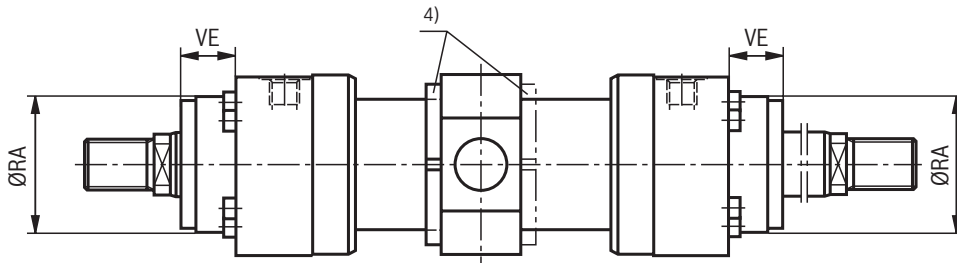
15) Dimensions for cylinders with seal design "A" and "B".

Dimensions: Trunnion mounting CGH3 – "MT4" (Dimensions in mm)

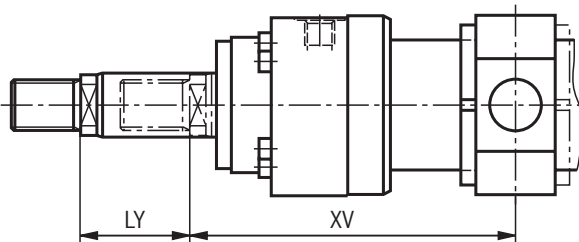
CGH3 "MT4"



CGH3 "MT4": Seal design "A", "B" and ØAL 160 ... 320 mm



Dimensions for cylinder with piston rod extension "LY" in retracted condition



Notice:

During installation, it must be ensured that the trunnion bearings are installed up to the trunnion shoulders. Any variation may reduce the product's service life.

Dimensions: Trunnion mounting CGH3 – "MT4"

(Dimensions in mm)

ØAL	ØMM	KK ⁶⁾	A ⁶⁾	KK ⁷⁾	A ⁷⁾	NV	ØD	ØDA	ØD4 ⁸⁾	EE ^{9); 10)}	EE ^{9); 11)}	Y	X1	WA
40	28	M22 x 1.5	22	M24 x 2	35	22	92	52	34	G 1/2	M22 x 1.5	91	43	18
50	36	M28 x 1.5	28	M30 x 2	45	30	108	62	34	G 1/2	M22 x 1.5	90	51.5	18
63	45	M35 x 1.5	35	M39 x 3	55	36	140	78	42	G 3/4	M27 x 2	117	67	22
80	56	M45 x 1.5	45	M50 x 3	75	46	148	100	42	G 3/4	M27 x 2	124	71.5	22
100	70	M58 x 1.5	58	M64 x 3	95	60	186	125	47	G 1	M33 x 2	119	90.5	25
125	90	M65 x 1.5	65	M80 x 3	110	75	235	160	58	G 1 1/4	M42 x 2	170	114	32
140	100	M80 x 2	80	M90 x 3	120	85	258	175	58	G 1 1/4	M42 x 2	186	126	35
160	110	M100 x 2	100	M100 x 3	140	95	292	200	65	G 1 1/2	M48 x 2	210	142.5	40
180	125	M110 x 2	110	M110 x 4	150	110	325	220	65	G 1 1/2	M48 x 2	241	159.5	45
200	140	M120 x 3	120	M120 x 4	160	120	350	245	65	G 1 1/2	M48 x 2	262	172.5	45
220	160	M120 x 3	120	M120 x 4	160	140	375	292	65	G 1 1/2	M48 x 2	262	185	40
250	180	M130 x 3	130	M150 x 4	190	160	440	324	65	G 1 1/2	M48 x 2	272	218	40
280	200	–	–	M160 x 4	200	180	460	368	65	G 1 1/2	M48 x 2	282	228	40
320	220	–	–	M180 x 4	220	200	490	406	65	G 1 1/2	M48 x 2	287	243	40

ØAL	ØMM	PK	ZM	X* _{min}	XV _{mid} ¹²⁾	XV _{min} ⁵⁾	XV _{max} ⁵⁾	BD	UV ¹³⁾	ØTD _{e8}	TL _{js16}	TM _{h12}	r	ØRA ₁₄₎	VE ¹⁴⁾	ØRA ₁₅₎	VE ¹⁵⁾
40	28	120	302	42	151+X*/2	172	138+X*	48	101	40	30	95	2	52	45	52	20
50	36	120	300	50	150+X*/2	175	134+X*	48	117	40	30	120	2	70	47	70	19
63	45	133	367	64	183.5+X*/2	215.5	163.5+X*	53	153	45	35	150	2	88	43	88	13
80	56	146	384	82	197+X*/2	238	168+X*	68	169	55	50	160	2	98	53	98	15
100	70	171	409	109	204.5+X*/2	259	165+X*	88	203	60	55	200	2	120	55	120	17
125	90	205	545	131	272.5+X*/2	338	207+X*	118	252	75	60	245	2.5	150	68	150	20
140	100	219	591	147	295.5+X*/2	369	222+X*	128	282	85	70	280	2.5	170	75	170	23
160	110	240	660	186	330+X*/2	423	237+X*	148	310	95	80	300	2.5	200	90	200	90
180	125	264	746	212	373+X*/2	479	267+X*	168	348	110	90	335	2.5	230	100	230	100
200	140	278	802	228	401+X*/2	515	287+X*	188	373	120	100	360	2.5	250	110	250	110
220	160	326	850	205	425+X*/2	527.5	322.5+X*	165	398	130	100	400	2.5	275	125	275	125
250	180	336	880	245	440+X*/2	562.5	317.5+X*	175	463	140	100	450	5	320	135	320	135
280	200	366	930	245	465+X*/2	587.5	342.5+X*	205	486	170	125	480	5	335	150	335	150
320	220	391	965	600	482.5+X*/2	782.5	182.5+X*	245	537	200	150	500	5	350	165	350	165

ØAL = piston Ø

ØMM = piston rod Ø

X* = stroke length

X*_{max} = maximum stroke length

X*_{min} = minimum stroke length

(With hydraulic cylinders with end position damping, see notice on page 68.)

1) Bleeding: With view to the piston rod, the position is offset by 90° in relation to the line connection (clockwise)

2) Throttle valve only with end position damping "E" (180° for bleeding).

3) Observe minimum stroke length "X*_{min}".

4) Trunnion nut for ØAL ≥ 125 mm depending on the position of the trunnion mounting (XV) either on the head or base side.

5) When ordering, always specify the "XV" dimension in the plain text. Preferred dimension XV: Observe trunnion mounting position in cylinder center XV_{min} and XV_{max}.

6) Thread design "G".

7) Thread design "A".

8) ØD4 maximum depth 0.5 mm.

9) Flange connections see separate table page 54 and 55.

10) Line connection "B" and "C".

11) Line connection "M".

12) XV mid recommendation: Trunnion mounting position in cylinder center.

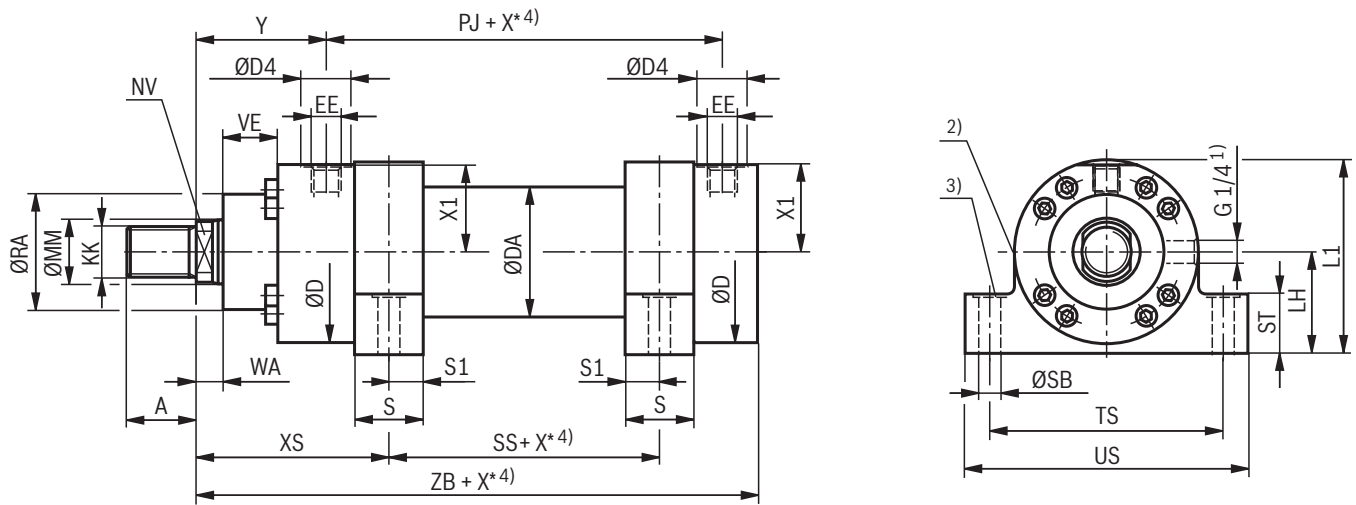
13) The dimensions given are maximum values, tolerance classes 342 according to ISO 9013 Thermal cutting.

14) Dimensions for cylinders with seal design "M", "T", "G", "L", "R", "S" and "V".

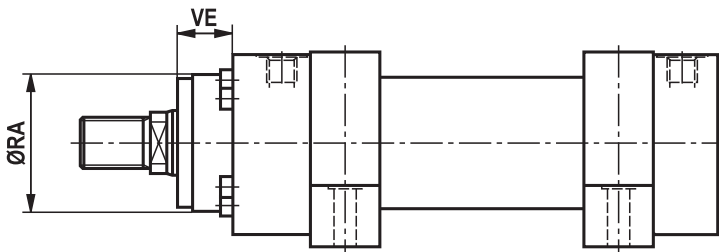
15) Dimensions for cylinders with seal design "A" and "B".

Dimensions: Foot mounting CDH3, CGH3 – "MS2"
(Dimensions in mm)

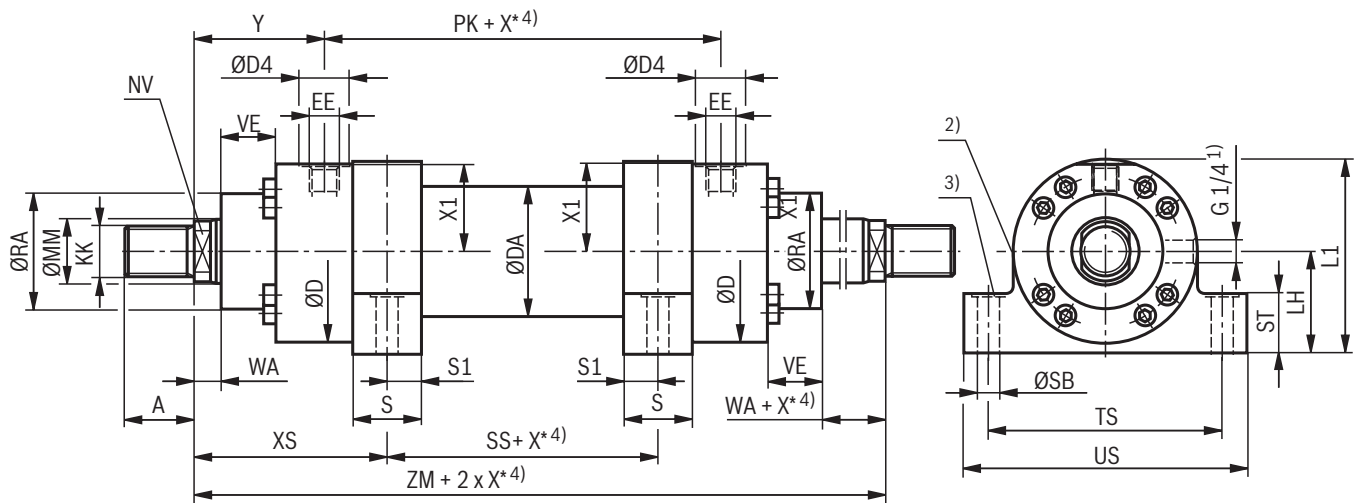
CDH3 "MS2"



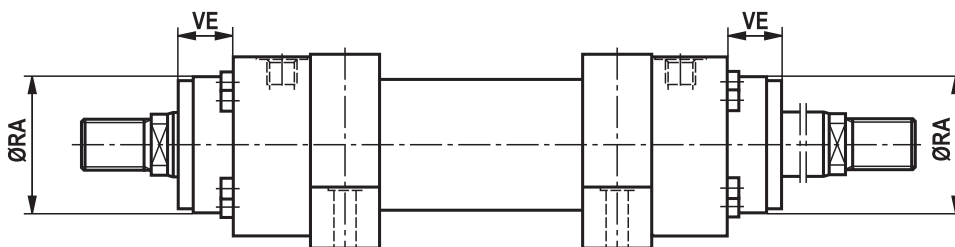
CDH3 "MS2": Seal design "A", "B" and ØAL 160 ... 320 mm



CGH3 "MS2"



CGH3 "MS2": Seal design "A", "B" and ØAL 160 ... 320 mm



Dimensions: Foot mounting CDH3, CGH3 – "MS2"

(Dimensions in mm)

ØAL	ØMM	KK ⁵⁾	A ⁵⁾	KK ⁶⁾	A ⁶⁾	NV	ØD	ØDA	ØD4 ⁷⁾	EE ^{8); 9)}	EE ^{8); 10)}	Y	PJ	X1	WA
40	28	M22 x 1.5	22	M24 x 2	35	22	92	52	34	G 1/2	M22 x 1.5	91	120	43	18
50	36	M28 x 1.5	28	M30 x 2	45	30	108	62	34	G 1/2	M22 x 1.5	90	120	51.5	18
63	45	M35 x 1.5	35	M39 x 3	55	36	140	78	42	G 3/4	M27 x 2	117	133	67	22
80	56	M45 x 1.5	45	M50 x 3	75	46	148	100	42	G 3/4	M27 x 2	124	146	71.5	22
100	70	M58 x 1.5	58	M64 x 3	95	60	186	125	47	G 1	M33 x 2	119	171	90.5	25
125	90	M65 x 1.5	65	M80 x 3	110	75	235	160	58	G 1 1/4	M42 x 2	170	205	114	32
140	100	M80 x 2	80	M90 x 3	120	85	258	175	58	G 1 1/4	M42 x 2	186	219	126	35
160	110	M100 x 2	100	M100 x 3	140	95	292	200	65	G 1 1/2	M48 x 2	210	240	142.5	40
180	125	M110 x 2	110	M110 x 4	150	110	325	220	65	G 1 1/2	M48 x 2	241	264	159.5	45
200	140	M120 x 3	120	M120 x 4	160	120	350	245	65	G 1 1/2	M48 x 2	262	278	172.5	45
220	160	M120 x 3	120	M120 x 4	160	140	375	292	65	G 1 1/2	M48 x 2	262	326	185	40
250	180	M130 x 3	130	M150 x 4	190	160	440	324	65	G 1 1/2	M48 x 2	272	336	218	40
280	200	–	–	M160 x 4	200	180	460	368	65	G 1 1/2	M48 x 2	282	366	228	40
320	220	–	–	M180 x 4	220	200	490	406	65	G 1 1/2	M48 x 2	287	391	243	40

ØAL	ØMM	PK	XS	ZB	ZM	SS	X* _{min}	S	S1	ØSB H13	ST	TS js13	US _{max}	LH	L1	ØRA (11)	VE ¹¹⁾ (12)	ØRA (12)	VE (12)
40	28	120	126	238	302	50	–	30	15	17.5	32	125	164	50	100	52	45	52	20
50	36	120	130	237	300	40	4	40	20	22	37	150	197	60	118	70	47	70	19
63	45	133	164	285	367	39	15	50	25	24	47	185	235	75	149	88	43	88	13
80	56	146	176	305	394	42	22	60	30	26	52	210	270	80	160	98	53	98	15
100	70	171	179	330	409	51	23	70	35	33	62	250	320	100	200	120	55	120	17
125	90	205	245	425	545	55	39	90	45	40	72	310	392	120	245	150	68	150	20
140	100	219	265.5	457	591	60	39	95	47.5	40	77	340	422	135	271	170	75	170	23
160	110	240	302.5	515	660	55	64	115	57.5	45	87	370	462	150	305	200	90	200	90
180	125	264	353.5	565	746	39	110	145	72.5	45	79	415	515	165	337	230	100	230	100
200	140	278	379.5	600	802	43	116	155	77.5	52	112	460	570	180	366	250	110	250	110
220	160	326	387.5	655	850	75	100	155	77.5	52	112	500	610	200	398	275	125	275	125
250	180	336	397.5	695	880	85	90	155	77.5	52	122	550	660	225	456	320	135	320	135
280	200	366	410	735	930	110	70	160	80	62	142	600	722	235	476	335	150	335	150
320	220	391	440	775	965	85	400	190	95	74	162	650	785	255	512	350	165	350	165

ØAL = piston Ø

ØMM = piston rod Ø

X* = stroke length

X*_{min} = minimum stroke length

(With hydraulic cylinders with end position damping,
see notice on page 68.)

- 1) Bleeding: With view to the piston rod, the position is offset by 90° in relation to the line connection (clockwise)
- 2) Throttle valve only with end position damping "E" (180° for bleeding).
- 3) Recess 2 mm deep for hexagon socket head cap screws; ISO 4762 (for spoolØ 320 mm DIN 931). The screws must not be subjected to shear force. Force distribution via additional external fitting strip.

4) Observe minimum stroke length "X*_{min}".

5) Thread design "G".

6) Thread design "A".

7) Ø D4 maximum depth 0.5 mm.

8) Flange connections see separate table page 54 and 55.

9) Line connection "B" and "C".

10) Line connection "M".

11) Dimensions for cylinders with seal design "M", "T", "G", "L", "R", "S" and "V".

12) Dimensions for cylinders with seal design "A" and "B".

Dimensions: Swivel eye at base CSH3 – "MP3"

(Dimensions in mm)

ØAL	ØMM	KK ⁴⁾	A ⁴⁾	KK ⁵⁾	A ⁵⁾	NV	ØD	ØDA	ØD4 ⁶⁾	EE ^{7); 8)}	EE ^{7); 9)}	Y	PJ	X* _{max}
40	28	M22 x 1.5	22	M24 x 2	35	22	92	52	34	G 1/2	M22 x 1.5	91	120	1000
50	36	M28 x 1.5	28	M30 x 2	45	30	108	62	34	G 1/2	M22 x 1.5	90	120	1000
63	45	M35 x 1.5	35	M39 x 3	55	36	140	78	42	G 3/4	M27 x 2	117	133	2000
80	56	M45 x 1.5	45	M50 x 3	75	46	148	100	42	G 3/4	M27 x 2	124	146	2000
100	70	M58 x 1.5	58	M64 x 3	95	60	186	125	47	G 1	M33 x 2	119	171	3000
125	90	M65 x 1.5	65	M80 x 3	110	75	235	160	58	G 1 1/4	M42 x 2	170	205	3000
140	100	M80 x 2	80	M90 x 3	120	85	258	175	58	G 1 1/4	M42 x 2	186	219	3000
160	110	M100 x 2	100	M100 x 3	140	95	292	200	65	G 1 1/2	M48 x 2	210	240	3000
180	125	M110 x 2	110	M110 x 4	150	110	325	220	65	G 1 1/2	M48 x 2	241	264	3000
200	140	M120 x 3	120	M120 x 4	160	120	350	245	65	G 1 1/2	M48 x 2	262	278	3000

ØAL	ØMM	X1	WA	XC	L	MR	M1	ØCD H11	EW h12	ØRA	VE	L2	H ¹⁰⁾	H ¹¹⁾	L7	L8	B
40	28	43	18	433	35	36	34	30	28	52	45	102	200	83	108	115	64
50	36	51.5	18	445	45	42	40	35	30	70	47	115	200	102	116	125	64
63	45	67	22	508	50	52	50	40	35	88	43	127	200	104	133	140	64
80	56	71.5	22	540	55	65	62.5	50	40	98	53	137	200	109	137	125	64
100	70	90.5	25	565	65	70	70	60	50	120	55	155	200	127	156	135	64
125	90	114	32	668	75	82	82	70	55	150	68	185	200	161	181	150	64
140	100	126	35	705	80	95	95	80	60	170	75	192	200	166	192	160	64
160	110	142.5	40	785	90	113	113	90	65	200	90	225	200	193	210	170	64
180	125	159.5	45	838	105	125	125	100	70	230	100	235	200	202	226	180	64
200	140	172.5	45	888	115	142.5	142.5	110	80	250	110	245	200	214	239	195	64

ØAL = piston Ø

ØMM = piston rod Ø

X* = stroke length

X*_{max} = maximum stroke length

1) Bleeding: With view to the piston rod, the position is offset by 90° in relation to the line connection (clockwise)

2) Throttle valve only with end position damping "E" (180° for bleeding).

3) Standard version "W";

Grease nipple, cone head form A according to DIN 71412.

4) Thread design "G".

5) Thread design "A".

6) ØD4 maximum depth 0.5 mm.

7) Flange connections see separate table page 54 and 55.

8) Line connection "B" and "C".

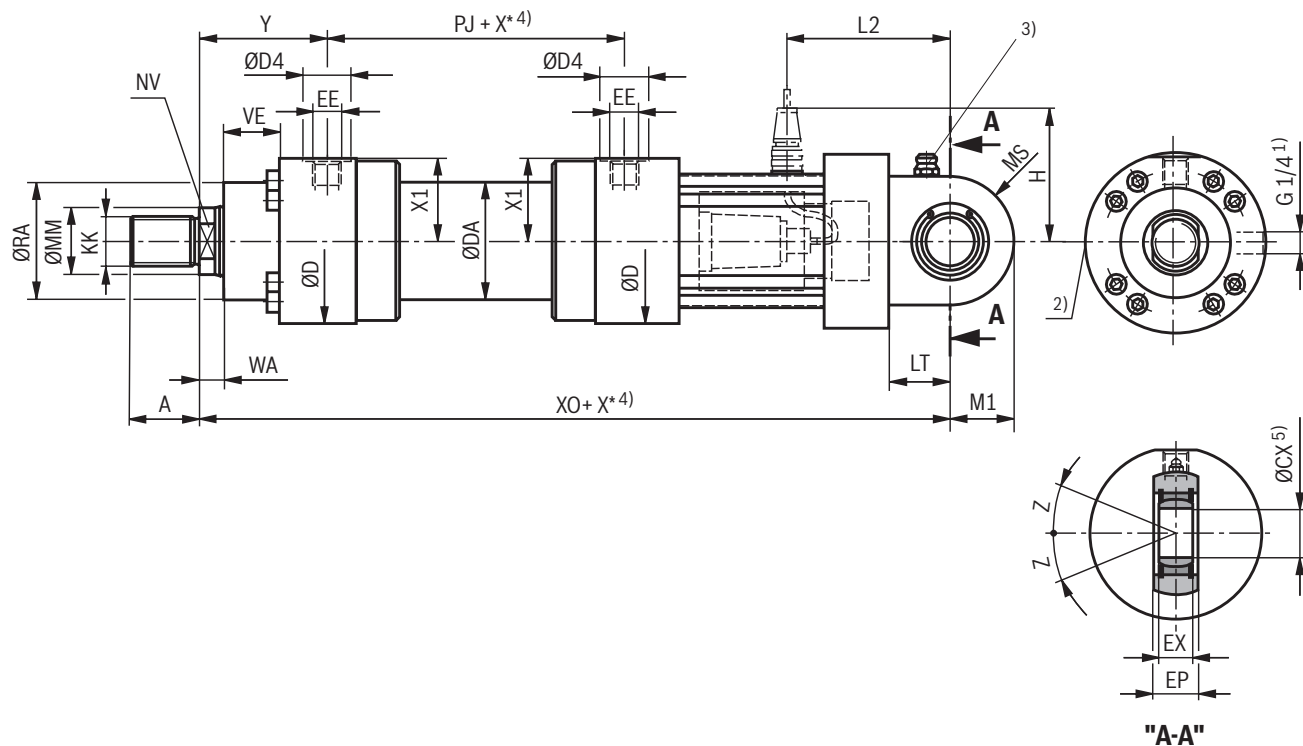
9) Line connection "M".

10) Dimensions for position measurement system output "C", "F", "D" and "S".

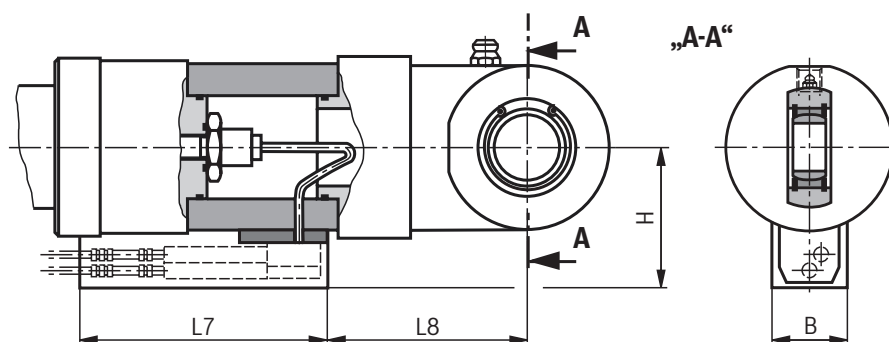
11) Dimensions for position measurement system output "N", "P" and "R".

Dimensions: Self-aligning clevis at base CSH3 – "MP5"
(Dimensions in mm)

CSH3 "MP5": position measurement system output "C", "F", "D" and "S"



CSH3 "MP5": position measurement system output "N", "P" and "R"



Dimensions: Self-aligning clevis at base CSH3 – "MP5"

(Dimensions in mm)

ØAL	ØMM	KK ⁽⁶⁾	A ⁽⁶⁾	KK ⁽⁷⁾	A ⁽⁷⁾	NV	ØD	ØDA	ØD4 ⁽⁸⁾	EE ^(9); 10)	EE ^(9); 11)	Y	PJ	X1	X* _{max}	C ₀ ⁽¹²⁾ kN
40	28	M22 x 1.5	22	M24 x 2	35	22	92	52	34	G 1/2	M22 x 1.5	91	120	43	1000	106
50	36	M28 x 1.5	28	M30 x 2	45	30	108	62	34	G 1/2	M22 x 1.5	90	120	51.5	1000	153
63	45	M35 x 1.5	35	M39 x 3	55	36	140	78	42	G 3/4	M27 x 2	117	133	67	2000	250
80	56	M45 x 1.5	45	M50 x 3	75	46	148	100	42	G 3/4	M27 x 2	124	146	71.5	2000	365
100	70	M58 x 1.5	58	M64 x 3	95	60	186	125	47	G 1	M33 x 2	119	171	90.5	3000	400
125	90	M65 x 1.5	65	M80 x 3	110	75	235	160	58	G 1 1/4	M42 x 2	170	205	114	3000	540
140	100	M80 x 2	80	M90 x 3	120	85	258	175	58	G 1 1/4	M42 x 2	186	219	126	3000	670
160	110	M100 x 2	100	M100 x 3	140	95	292	200	65	G 1 1/2	M48 x 2	210	240	142.5	3000	980
180	125	M110 x 2	110	M110 x 4	150	110	325	220	65	G 1 1/2	M48 x 2	241	264	159.5	3000	1120
200	140	M120 x 3	120	M120 x 4	160	120	350	245	65	G 1 1/2	M48 x 2	262	278	172.5	3000	1700
220	160	M120 x 3	120	M120 x 4	160	140	375	292	65	G 1 1/2	M48 x 2	262	326	185	3000	1700
250	180	M130 x 3	130	M150 x 4	190	160	440	324	65	G 1 1/2	M48 x 2	272	336	218	3000	2900
280	200	–	–	M160 x 4	200	180	460	368	65	G 1 1/2	M48 x 2	282	366	228	3000	–
320	220	–	–	M180 x 4	220	200	490	406	65	G 1 1/2	M48 x 2	287	391	243	3000	–

ØAL	ØMM	F _{adm} ⁽¹³⁾ kN	WA	XO	X* _{min}	LT	M1	MS	ØCX	EP -0.4	EX	Z	ØRA	VE	L2	L7	L8	H ⁽¹⁵⁾	H ⁽¹⁶⁾	B
40	28	38.2	18	433	–	35	34	36	30-0.010	28	22-0.12	6 °	52	45	102	200	83	108	115	64
50	36	55.1	18	445	–	45	40	42	35-0.012	30	25-0.12	6 °	70	47	115	200	102	116	125	64
63	45	90.0	22	508	–	50	50	52	40-0.012	35	28-0.12	7 °	88	43	127	200	104	133	140	64
80	56	131.4	22	540	–	55	62.5	65	50-0.012	40	35-0.12	6 °	98	53	137	200	109	137	125	64
100	70	144.0	25	565	–	65	70	70	60-0.015	50	44-0.15	6 °	120	55	155	200	127	156	135	64
125	90	194.4	32	668	–	75	82	82	70-0.015	55	49-0.15	6 °	150	68	185	200	161	181	150	64
140	100	241.2	35	705	–	80	95	95	80-0.015	60	55-0.15	6 °	170	75	192	200	166	192	160	64
160	110	352.8	40	785	–	90	113	113	90-0.020	65	60-0.20	5 °	200	90	225	200	193	210	170	64
180	125	403.2	45	838	–	105	125	125	100-0.020	70	70-0.20	7 °	230	100	235	200	202	226	180	64
200	140	612.0	45	888	–	115	142.5	142.5	110-0.020	80	70-0.20	6 °	250	110	245	200	214	239	195	64
220	160	612.0	40	970	–	115	150 ⁽¹⁴⁾	140 ⁽¹⁴⁾	110-0.020	80	70-0.20	6 °	275	125	270	200	238	254	215	64
250	180	1044.0	40	1055	20	140	188 ⁽¹⁴⁾	178 ⁽¹⁴⁾	120-0.020	90	85-0.20	6 °	320	135	320	200	283	284	235	64
280	200	–	40	1115	–	170	210 ⁽¹⁴⁾	200 ⁽¹⁴⁾	140-0.025	100	90-0.25	7 °	335	150	350	200	315	294	285	64
320	220	–	40	1195	340	200	260 ⁽¹⁴⁾	250 ⁽¹⁴⁾	160-0.025	110	105-0.25	8 °	350	165	400	200	400	309	300	64

ØAL = piston Ø

ØMM = piston rod Ø

X* = stroke length

X*_{max} = maximum stroke length

X*_{min} = minimum stroke length

(With hydraulic cylinders with end position damping, see notice on page 68.)

1) Bleeding: With view to the piston rod, the position is offset by 90° in relation to the line connection (clockwise)

2) Throttle valve only with end position damping "E" (180° for bleeding).

3) Standard version "W";

Grease nipple, cone head form A according to DIN 71412; not applicable to spherical bearing, maintenance-free "A".

4) Observe minimum stroke length "X*_{min}".

5) Related bolt Ø m6;

Related bolt Ø j6 for maintenance-free spherical bearing.

6) Thread design "G".

7) Thread design "A".

8) Ø D4 maximum depth 0.5 mm.

9) Flange connections see separate table page 54 and 55.

10) Line connection "B" and "C".

11) Line connection "M".

12) C₀ = static load rating of the swivel head.

13) F_{adm} = Maximum admissible load of the swivel head with oscillatory or alternating loads.

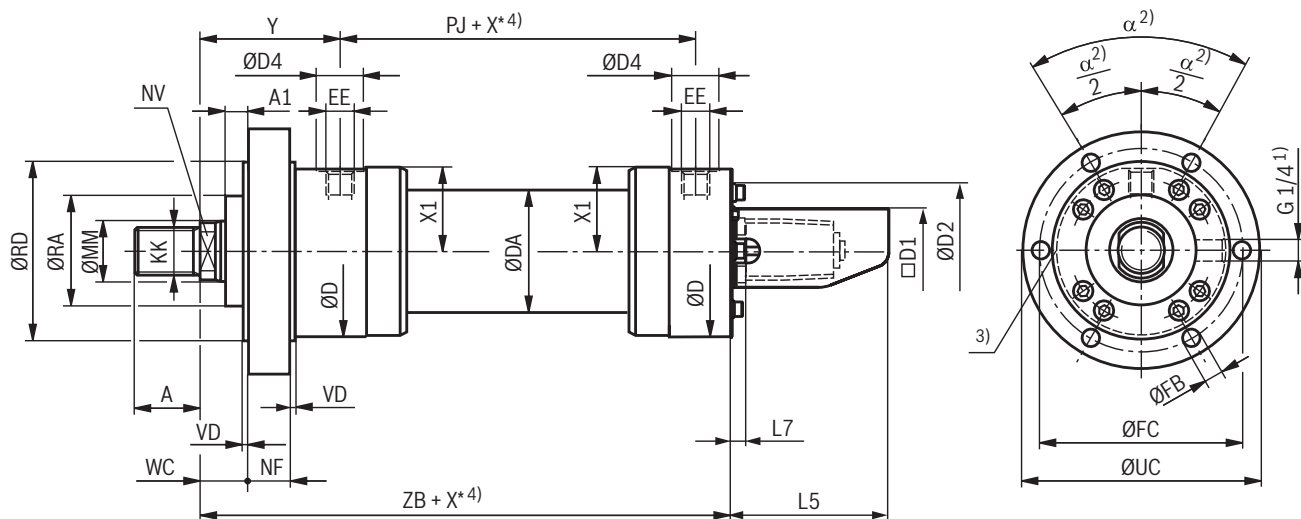
14) The specified dimensions are maximum values, tolerance classes 342 according to ISO 9013 Thermal cutting.

15) Dimensions for position measurement system output "C", "F", "D" and "S".

16) Dimensions for position measurement system output "N", "P" and "R".

Dimensions: Round flange at head CSH3 – "MF3"
(Dimensions in mm)

CSH3 "MF3"



Dimensions: Round flange at head CSH3 – "MF3"

(Dimensions in mm)

ØAL	ØMM	KK ⁵⁾	A ⁵⁾	KK ⁶⁾	A ⁶⁾	NV	ØD	ØDA	ØD4 ⁷⁾	EE ^{8); 9)}	EE ^{8); 10)}	Y	PJ	X1	L5	L7
40	28	M22 x 1.5	22	M24 x 2	35	22	92	52	34	G 1/2	M22 x 1.5	91	120	43	115	9.1
50	36	M28 x 1.5	28	M30 x 2	45	30	108	62	34	G 1/2	M22 x 1.5	90	120	51.5	115	9.1
63	45	M35 x 1.5	35	M39 x 3	55	36	140	78	42	G 3/4	M27 x 2	117	133	67	115	9.1
80	56	M45 x 1.5	45	M50 x 3	75	46	148	100	42	G 3/4	M27 x 2	124	146	71.5	115	9.1
100	70	M58 x 1.5	58	M64 x 3	95	60	186	125	47	G 1	M33 x 2	119	171	90.5	115	9.1
125	90	M65 x 1.5	65	M80 x 3	110	75	235	160	58	G 1 1/4	M42 x 2	170	205	114	115	9.1
140	100	M80 x 2	80	M90 x 3	120	85	258	175	58	G 1 1/4	M42 x 2	186	219	126	115	9.1
160	110	M100 x 2	100	M100 x 3	140	95	292	200	65	G 1 1/2	M48 x 2	210	240	142.5	115	9.1
180	125	M110 x 2	110	M110 x 4	150	110	325	220	65	G 1 1/2	M48 x 2	241	264	159.5	115	9.1
200	140	M120 x 3	120	M120 x 4	160	120	350	245	65	G 1 1/2	M48 x 2	262	278	172.5	115	9.1
220	160	M120 x 3	120	M120 x 4	160	140	375	292	65	G 1 1/2	M48 x 2	262	326	185	115	9.1
250	180	M130 x 3	130	M150 x 4	190	160	440	324	65	G 1 1/2	M48 x 2	272	336	218	115	9.1
280	200	–	–	M160 x 4	200	180	460	368	65	G 1 1/2	M48 x 2	282	366	228	115	9.1
320	220	–	–	M180 x 4	220	200	490	406	65	G 1 1/2	M48 x 2	287	391	243	115	9.1

ØAL	ØMM	X* _{max}	ØRD e8	WC	VD	NF	A1	ZB	X* _{min}	ØFB H13	ØFC js13	ØUC -1	α	ØRA	□D1	ØD2
40	28	1000	95	23	5	35	0	247	–	13.5	120	145	60°	52	58	93
50	36	1000	115	20	5	40	0	246	–	13.5	140	165	60°	70	58	93
63	45	2000	150	20	5	40	0	304	–	17.5	180	210	60°	88	58	93
80	56	2000	160	20	5	50	0	332	–	17.5	195	230	60°	98	58	93
100	70	3000	200	20	5	55	0	347	–	22	230	270	60°	120	58	93
125	90	3000	245	25	5	70	0	427	–	26	290	335	60°	150	58	93
140	100	3000	280	30	10	70	0	460	–	30	330	380	60°	170	58	93
160	110	3000	300	40	10	80	0	515	–	30	360	420	45°	200	58	93
180	125	3000	335	40	10	95	0	565	–	36	400	470	45°	230	58	93
200	140	3000	360	40	10	105	0	600	–	36	430	500	45°	250	58	93
220	160	3000	400	40	10	115	0	655	–	39	475	550	45°	275	58	93
250	180	3000	450	40	10	125	0	695	20	45	530	610	45°	320	58	93
280	200	3000	470	50	10	130	0	735	–	45	550	630	45°	335	58	93
320	220	3000	510	55	10	140	5	775	340	45	590	670	30°	350	58	93

ØAL = piston Ø

ØMM = piston rod Ø

X* = stroke length

X*_{max} = maximum stroke length

X*_{min} = minimum stroke length

(With hydraulic cylinders with end position damping,
see notice on page 68.)

1) Bleeding: With view to the piston rod, the position is offset by 90° in
relation to the line connection (clockwise)

2) With piston Ø 160 ... 280 mm 8 mounting bores;
With piston Ø 320 mm 12 mounting bores.

3) Throttle valve only with end position damping "E" (180° for bleeding).

4) Observe minimum stroke length "X*_{min}".

5) Thread design "G"

6) Thread design "A"

7) ØD4 maximum depth 0.5 mm.

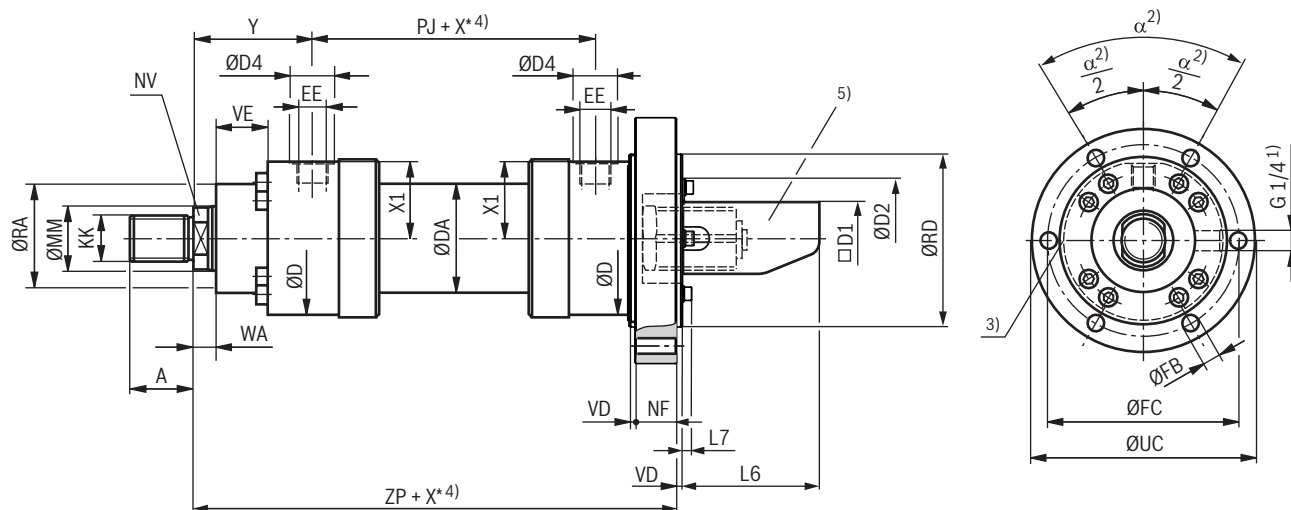
8) Flange connections see separate table page 54 and 55.

9) Line connection "B" and "C".

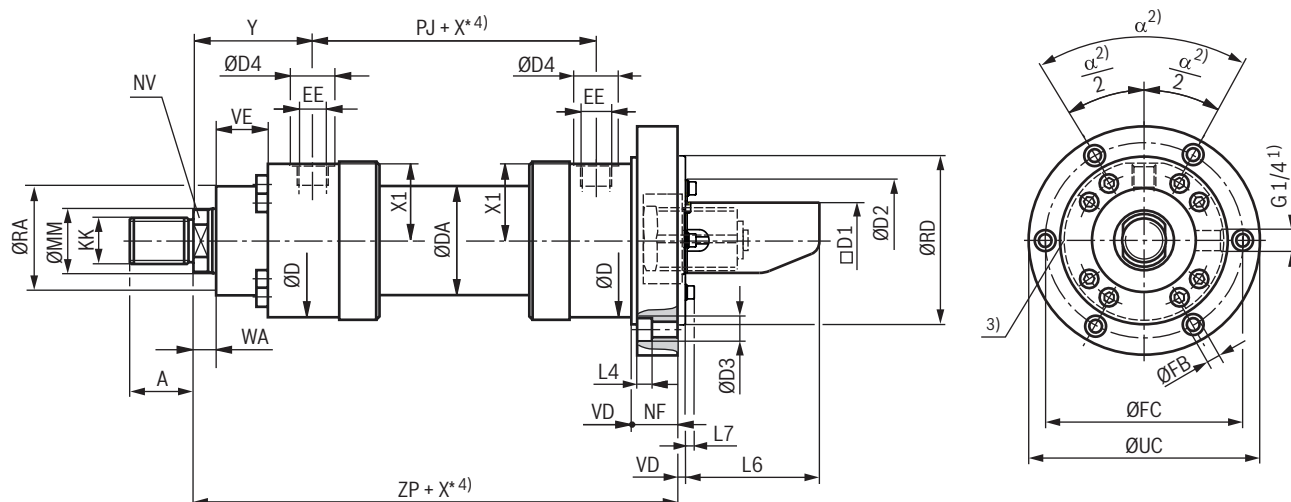
10) Line connection "M".

Dimensions: Round flange at base CSH3 – "MF4"
(Dimensions in mm)

CSH3 "MF4": ØAL 40 ... 100 mm



CSH3 "MF4": ØAL 125 ... 320 mm



Dimensions: Round flange at base CSH3 – "MF4"

(Dimensions in mm)

ØAL	ØMM	KK ⁶⁾	A ⁶⁾	KK ⁷⁾	A ⁷⁾	NV	ØD	ØDA	ØD4 ⁸⁾	EE ⁹⁾ ; 10)	EE ⁹⁾ ; 11)	Y	PJ	X1	L4	L7
40	28	M22 x 1.5	22	M24 x 2	35	22	92	52	34	G 1/2	M22 x 1.5	91	120	43	–	–
50	36	M28 x 1.5	28	M30 x 2	45	30	108	62	34	G 1/2	M22 x 1.5	90	120	51.5	–	–
63	45	M35 x 1.5	35	M39 x 3	55	36	140	78	42	G 3/4	M27 x 2	117	133	67	–	9.1
80	56	M45 x 1.5	45	M50 x 3	75	46	148	100	42	G 3/4	M27 x 2	124	146	71.5	–	9.1
100	70	M58 x 1.5	58	M64 x 3	95	60	186	125	47	G 1	M33 x 2	119	171	90.5	–	9.1
125	90	M65 x 1.5	65	M80 x 3	110	75	235	160	58	G 1 1/4	M42 x 2	170	205	114	25.5	9.1
140	100	M80 x 2	80	M90 x 3	120	85	258	175	58	G 1 1/4	M42 x 2	186	219	126	28.5	9.1
160	110	M100 x 2	100	M100 x 3	140	95	292	200	65	G 1 1/2	M48 x 2	210	240	142.5	28.5	9.1
180	125	M110 x 2	110	M110 x 4	150	110	325	220	65	G 1 1/2	M48 x 2	241	264	159.5	35	9.1
200	140	M120 x 3	120	M120 x 4	160	120	350	245	65	G 1 1/2	M48 x 2	262	278	172.5	35	9.1
220	160	M120 x 3	120	M120 x 4	160	140	375	292	65	G 1 1/2	M48 x 2	262	326	185	38	9.1
250	180	M130 x 3	130	M150 x 4	190	160	440	324	65	G 1 1/2	M48 x 2	272	336	218	44	9.1
280	200	–	–	M160 x 4	200	180	460	368	65	G 1 1/2	M48 x 2	282	366	228	44	9.1
320	220	–	–	M180 x 4	220	200	490	406	65	G 1 1/2	M48 x 2	287	391	243	44	–

ØAL	ØMM	WA	ZP	NF	VD	ØRD e8	ØFB H13	ØFC js13	ØUC -1	α	ØRA	VE	L6	□D1	ØD2	ØD3	X* _{min}	X* _{max}
40	28	18	282	35	5	95	13.5	120	145	60°	52	45	–	–	–	–	–	1000
50	36	18	285	40	5	115	13.5	140	165	60°	70	47	–	–	–	–	–	1000
63	45	22	340	40	5	150	17.5	180	210	60°	88	43	115	58	93	–	–	2000
80	56	22	370	50	5	160	17.5	195	230	60°	98	53	92	58	93	–	–	2000
100	70	25	402	55	5	200	22	230	270	60°	120	55	72	58	93	–	–	3000
125	90	32	495	70	5	245	26	290	335	60°	150	68	70	58	93	40	–	3000
140	100	35	532	70	10	280	30	330	380	60°	170	75	60	58	93	43	–	3000
160	110	40	600	80	10	300	30	360	420	45°	200	90	45	58	93	43	–	3000
180	125	45	665	95	10	335	36	400	470	45°	230	100	35	58	93	53	–	3000
200	140	45	710	105	10	360	36	430	500	45°	250	110	25	58	93	53	–	3000
220	160	40	770	115	10	400	39	475	550	45°	275	125	20	58	93	57	–	3000
250	180	40	820	125	10	450	45	530	610	45°	320	135	20	58	93	66	20	3000
280	200	40	865	130	10	470	45	550	630	45°	335	150	10	58	93	66	–	3000
320	220	40	915	140	10	510	45	590	670	30°	350	165	–	–	–	66	340	3000

ØAL = piston Ø

ØMM = piston rod Ø

X* = stroke length

X*_{max} = maximum stroke length

X*_{min} = minimum stroke length

(With hydraulic cylinders with end position damping,
see notice on page 68.)

1) Bleeding: With view to the piston rod, the position is offset by 90° in
relation to the line connection (clockwise)

2) With piston Ø 160 ... 280 mm 8 mounting bores.
With piston Ø 320 mm 12 mounting bores.

3) Throttle valve only with end position damping "E" (180° for bleeding).

4) Observe minimum stroke length "X*_{min}".

5) No protective housing for ØAL 40 mm and 50 mm.

6) Thread design "G".

7) Thread design "A".

8) ØD4 maximum depth 0.5 mm.

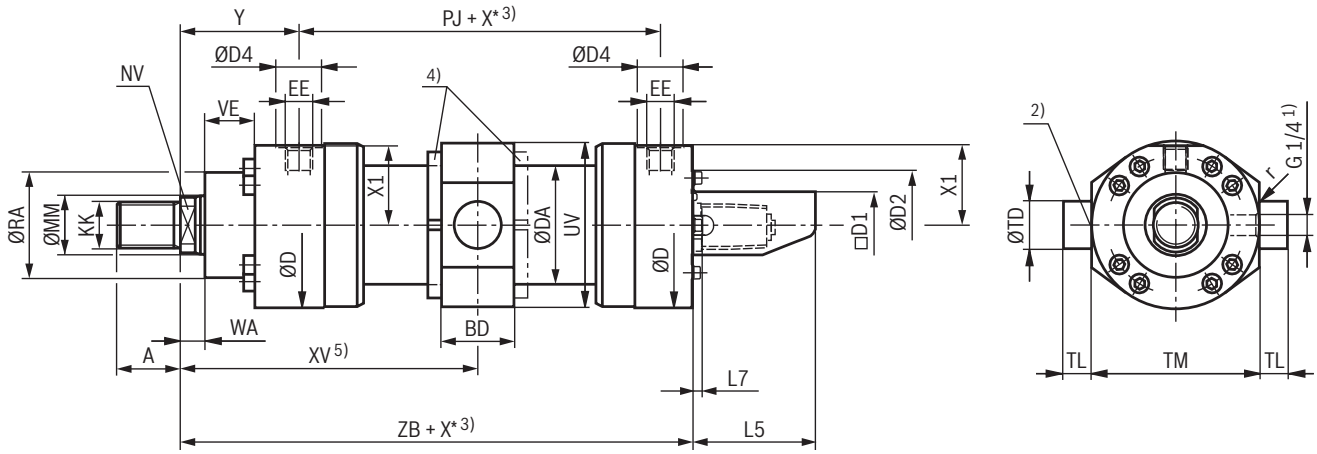
9) Flange connections see separate table page 54 and 55.

10) Line connection "B" and "C".

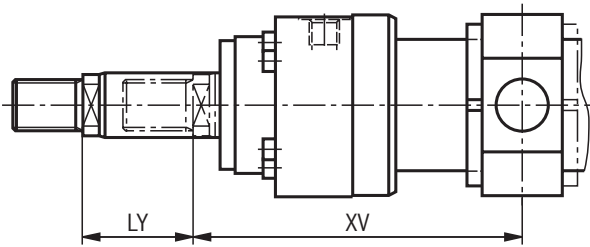
11) Line connection "M".

Dimensions: Trunnion mounting CSH3 – "MT4"
(Dimensions in mm)

CSH3 "MT4"



Dimensions for cylinder with piston rod extension "LY" in retracted condition



Notice:

During installation, it must be ensured that the trunnion bearings are installed up to the trunnion shoulders. Any variation may reduce the product's service life.

Dimensions: Trunnion mounting CSH3 – "MT4"

(Dimensions in mm)

ØAL	ØMM	KK ⁶⁾	A ⁶⁾	KK ⁷⁾	A ⁷⁾	NV	ØD	ØDA	ØD4 ⁸⁾	EE ^{9); 10)}	EE ^{9); 11)}	Y	PJ	X1	WA	L5	L7
40	28	M22 x 1.5	22	M24 x 2	35	22	92	52	34	G 1/2	M22 x 1.5	91	120	43	18	115	9.1
50	36	M28 x 1.5	28	M30 x 2	45	30	108	62	34	G 1/2	M22 x 1.5	90	120	51.5	18	115	9.1
63	45	M35 x 1.5	35	M39 x 3	55	36	140	78	42	G 3/4	M27 x 2	117	133	67	22	115	9.1
80	56	M45 x 1.5	45	M50 x 3	75	46	148	100	42	G 3/4	M27 x 2	124	146	71.5	22	115	9.1
100	70	M58 x 1.5	58	M64 x 3	95	60	186	125	47	G 1	M33 x 2	119	171	90.5	25	115	9.1
125	90	M65 x 1.5	65	M80 x 3	110	75	235	160	58	G 1 1/4	M42 x 2	170	205	114	32	115	9.1
140	100	M80 x 2	80	M90 x 3	120	85	258	175	58	G 1 1/4	M42 x 2	186	219	126	35	115	9.1
160	110	M100 x 2	100	M100 x 3	140	95	292	200	65	G 1 1/2	M48 x 2	210	240	142.5	40	115	9.1
180	125	M110 x 2	110	M110 x 4	150	110	325	220	65	G 1 1/2	M48 x 2	241	264	159.5	45	115	9.1
200	140	M120 x 3	120	M120 x 4	160	120	350	245	65	G 1 1/2	M48 x 2	262	278	172.5	45	115	9.1
220	160	M120 x 3	120	M120 x 4	160	140	375	292	65	G 1 1/2	M48 x 2	262	326	185	40	115	9.1
250	180	M130 x 3	130	M150 x 4	190	160	440	324	65	G 1 1/2	M48 x 2	272	336	218	40	115	9.1
280	200	–	–	M160 x 4	200	180	460	368	65	G 1 1/2	M48 x 2	282	366	228	40	115	9.1
320	220	–	–	M180 x 4	220	200	490	406	65	G 1 1/2	M48 x 2	287	391	243	40	115	9.1

ØAL	ØMM	ZB	X _{min} [*]	XV _{mid} ¹²⁾	XV _{min} ⁵⁾	XV _{max} ⁵⁾	BD	UV ¹³⁾	ØTD _{e8}	TL _{js16}	TM _{h12}	r	ØRA	VE	□D1	ØD2	X [*] _{max}
40	28	247	42	151+X*/2	172	138+X*	48	101	40	30	95	2	52	45	58	93	1000
50	36	246	50	150+X*/2	175	134+X*	48	117	40	30	120	2	70	47	58	93	1000
63	45	304	64	183.5+X*/2	215.5	163.5+X*	53	153	45	35	150	2	88	43	58	93	2000
80	56	332	82	197+X*/2	238	168+X*	68	169	55	50	160	2	98	53	58	93	2000
100	70	347	109	204.5+X*/2	259	165+X*	88	203	60	55	200	2	120	55	58	93	3000
125	90	427	131	272.5+X*/2	338	207+X*	118	252	75	60	245	2.5	150	68	58	93	3000
140	100	460	147	295.5+X*/2	369	222+X*	128	282	85	70	280	2.5	170	75	58	93	3000
160	110	515	186	330+X*/2	423	237+X*	148	310	95	80	300	2.5	200	90	58	93	3000
180	125	565	212	373+X*/2	479	267+X*	168	348	110	90	335	2.5	230	100	58	93	3000
200	140	600	228	401+X*/2	515	287+X*	188	373	120	100	360	2.5	250	110	58	93	3000
220	160	655	205	425+X*/2	527.5	322.5+X*	165	398	130	100	400	2.5	275	125	58	93	3000
250	180	695	245	440+X*/2	562.5	317.5+X*	175	463	140	100	450	5	320	135	58	93	3000
280	200	735	245	465+X*/2	587.5	342.5+X*	205	486	170	125	480	5	335	150	58	93	3000
320	220	775	600	482.5+X*/2	782.5	182.5+X*	245	537	200	150	500	5	350	165	58	93	3000

ØAL = piston Ø

ØMM = piston rod Ø

X* = stroke length

X*_{max} = maximum stroke length

X*_{min} = minimum stroke length

(With hydraulic cylinders with end position damping, see notice on page 68.)

- 1) Bleeding: With view to the piston rod, the position is offset by 90° in relation to the line connection (clockwise)
- 2) Throttle valve only with end position damping "E" (180° for bleeding).
- 3) Observe minimum stroke length "X*_{min}".
- 4) Trunnion nut with ØAL ≥ 125 mm either at the head or base side depending on the position of the trunnion mounting (XV).

- 5) When ordering, always specify the "XV" dimension in the clear text. Preferred dimension XV: Observe trunnion mounting position in cylinder center XV_{min.} and XV_{max.}

6) Thread design "G".

7) Thread design "A".

8) ØD4 maximum depth 0.5 mm.

9) Flange connections see separate table page 54 and 55.

10) Line connection "B" and "C".

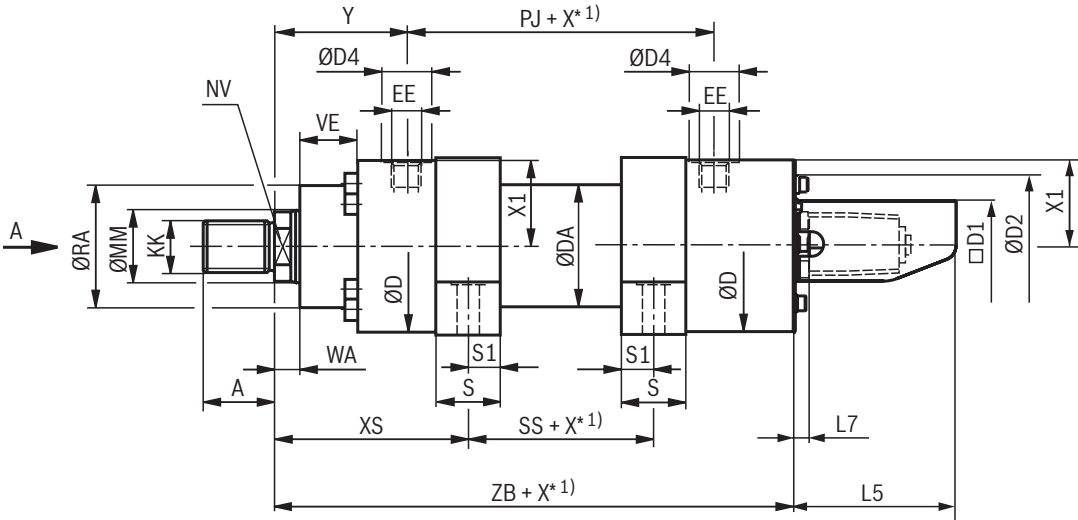
11) Line connection "M".

12) XV_{mid} recommendation: Trunnion mounting position in cylinder center.

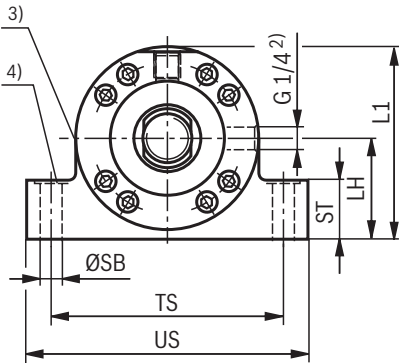
13) The specified dimensions are maximum values, tolerance classes 342 according to ISO 9013 Thermal cutting.

Dimensions: Foot mounting CSH3 – "MS2"
(Dimensions in mm)

CSH3 "MS2"



View A



Dimensions: Foot mounting CSH3 – "MS2"

(Dimensions in mm)

ØAL	ØMM	KK ⁵⁾	A ⁵⁾	KK ⁶⁾	A ⁶⁾	NV	ØD	ØDA	ØD4 ⁷⁾	EE ^{8); 9)}	EE ^{8); 10)}	Y	PJ	X1	WA	L5	L7
40	28	M22 x 1.5	22	M24 x 2	35	22	92	52	34	G 1/2	M22 x 1.5	91	120	43	18	115	9.1
50	36	M28 x 1.5	28	M30 x 2	45	30	108	62	34	G 1/2	M22 x 1.5	90	120	51.5	18	115	9.1
63	45	M35 x 1.5	35	M39 x 3	55	36	140	78	42	G 3/4	M27 x 2	117	133	67	22	115	9.1
80	56	M45 x 1.5	45	M50 x 3	75	46	148	100	42	G 3/4	M27 x 2	124	146	71.5	22	115	9.1
100	70	M58 x 1.5	58	M64 x 3	95	60	186	125	47	G 1	M33 x 2	119	171	90.5	25	115	9.1
125	90	M65 x 1.5	65	M80 x 3	110	75	235	160	58	G 1 1/4	M42 x 2	170	205	114	32	115	9.1
140	100	M80 x 2	80	M90 x 3	120	85	258	175	58	G 1 1/4	M42 x 2	186	219	126	35	115	9.1
160	110	M100 x 2	100	M100 x 3	140	95	292	200	65	G 1 1/2	M48 x 2	210	240	142.5	40	115	9.1
180	125	M110 x 2	110	M110 x 4	150	110	325	220	65	G 1 1/2	M48 x 2	241	264	159.5	45	115	9.1
200	140	M120 x 3	120	M120 x 4	160	120	350	245	65	G 1 1/2	M48 x 2	262	278	172.5	45	115	9.1
220	160	M120 x 3	120	M120 x 4	160	140	375	292	65	G 1 1/2	M48 x 2	262	326	185	40	115	9.1
250	180	M130 x 3	130	M150 x 4	190	160	440	324	65	G 1 1/2	M48 x 2	272	336	218	40	115	9.1
280	200	–	–	M160 x 4	200	180	460	368	65	G 1 1/2	M48 x 2	282	366	228	40	115	9.1
320	220	–	–	M180 x 4	220	200	490	406	65	G 1 1/2	M48 x 2	287	391	243	40	115	9.1

ØAL	ØMM	XS	ZB	SS	X* _{min}	S	S1	ØSB H13	ST	TS js13	US _{max}	LH	L1	ØRA	VE	□D1	ØD2	X* _{max}
40	28	126	247	50	–	30	15	17.5	32	125	164	50	100	52	45	58	93	1000
50	36	130	246	40	4	40	20	22	37	150	197	60	118	70	47	58	93	1000
63	45	164	304	39	15	50	25	24	47	185	235	75	149	88	43	58	93	2000
80	56	176	332	42	22	60	30	26	52	210	270	80	160	98	53	58	93	2000
100	70	179	347	51	23	70	35	33	62	250	320	100	200	120	55	58	93	3000
125	90	245	427	55	39	90	45	40	72	310	392	120	245	150	68	58	93	3000
140	100	265.5	460	60	39	95	47.5	40	77	340	422	135	271	170	75	58	93	3000
160	110	302.5	515	55	64	115	57.5	45	87	370	462	150	305	200	90	58	93	3000
180	125	353.5	565	39	110	145	72.5	45	79	415	515	165	337	230	100	58	93	3000
200	140	379.5	600	43	116	155	77.5	52	112	460	570	180	366	250	110	58	93	3000
220	160	387.5	655	75	100	155	77.5	52	112	500	610	200	398	275	125	58	93	3000
250	180	397.5	695	85	90	155	77.5	52	122	550	660	225	456	320	135	58	93	3000
280	200	410	735	110	70	160	80	62	142	600	722	235	476	335	150	58	93	3000
320	220	440	815	125	85	190	95	62	142	610	732	255	512	365	155	58	93	3000

ØAL = piston Ø

ØMM = piston rod Ø

X* = stroke length

X*_{max} = maximum stroke length

X*_{min} = minimum stroke length

(With hydraulic cylinders with end position damping, see notice on page 68.)

1) Observe minimum stroke length "X*_{min}".

2) Bleeding: With view to the piston rod, the position is offset by 90° in relation to the line connection (clockwise)

3) Throttle valve only with end position damping "E" (180° for bleeding).

4) Recess 2 mm deep, for hexagon socket head cap screws; ISO 4762 (for piston Ø 320 mm DIN 931). The screws must not be subjected to shear force. Force distribution via additional external fitting strip.

5) Thread design "G".

6) Thread design "A".

7) Ø D4 maximum depth 0.5 mm.

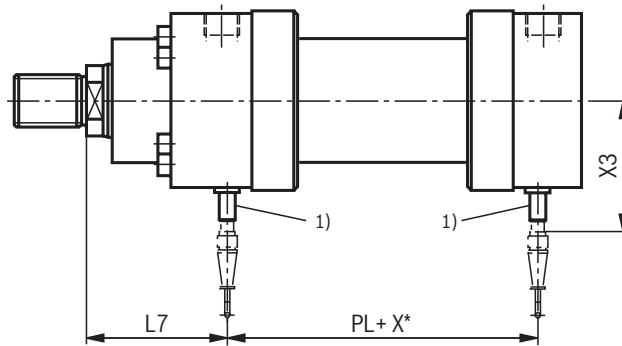
8) Flange connections see separate table page 54 and 55.

9) Line connection "B" and "C".

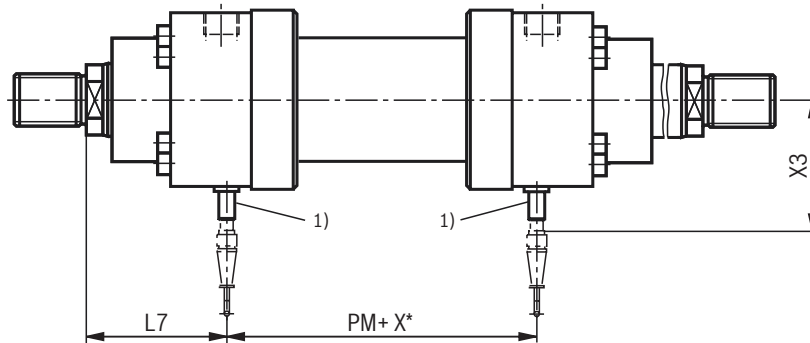
10) Line connection "M".

Dimensions: Proximity switch
(Dimensions in mm)

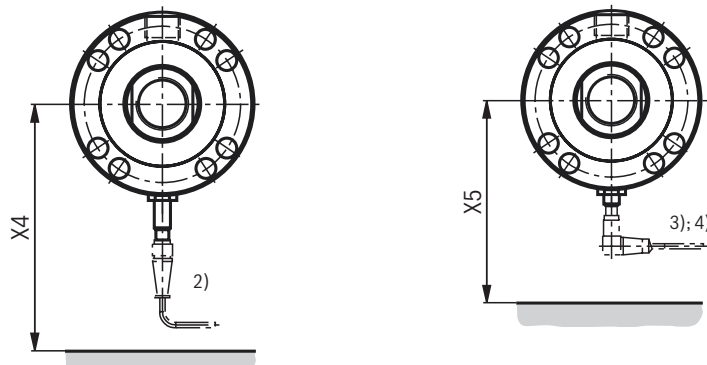
"CDH3"



"CGH3"



Installation space for mating connector



Notice:

Mating connectors, separate order, see page 82 and data sheet 08006.

Dimensions: Proximity switch

(Dimensions in mm)

ØAL	ØMM	PL	PM	L7	X3	X4	X5
40	28	112	112	95	94	170	125
50	36	110	110	95	98	175	130
63	45	125	125	121	103	180	135
80	56	138	138	128	108	185	140
100	70	161	161	124	116	195	150
125	90	189	189	178	126	205	160
140	100	209	209	191	146	225	180
160	110	228	228	216	151	230	185
180	125	254	254	246	159 ⁴⁾	235	190
200	140	264	264	269	166 ⁴⁾	245	200
220	160	310	310	270	177 ⁴⁾	255	– ⁵⁾
250	180	320	320	280	187 ⁴⁾	265	– ⁵⁾
280	200	360	360	285	199 ⁴⁾	275	– ⁵⁾
320	220	375	375	295	209 ⁴⁾	285	– ⁵⁾

For main dimensions, see page 26 ... 51

ØAL = piston Ø

ØMM = piston rod Ø

X* = stroke length

¹⁾ The proximity switch is always located opposite of the line connection.

²⁾ Mating connector with 5 m cable, material no. R913016852 (separate order, see page 82 and data sheet 08006).

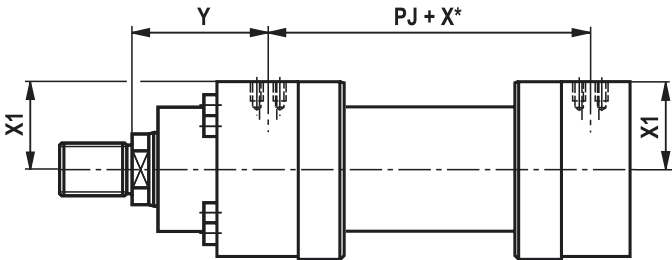
³⁾ Mating connector, angled (position of cable outlet cannot be defined), material no. R988064311 (separate order, see page 82 and data sheet 08006).

⁴⁾ Piston Ø 220 - 320 mm Proximity switch not protruding.

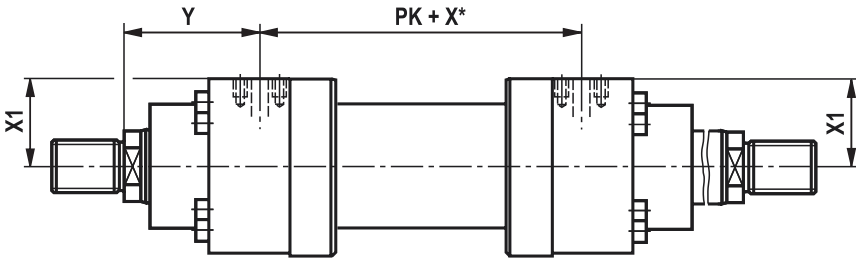
⁵⁾ Piston Ø 220 - 320 mm Angled mating connector not possible.

Dimensions: Flange ports
(Dimensions in mm)

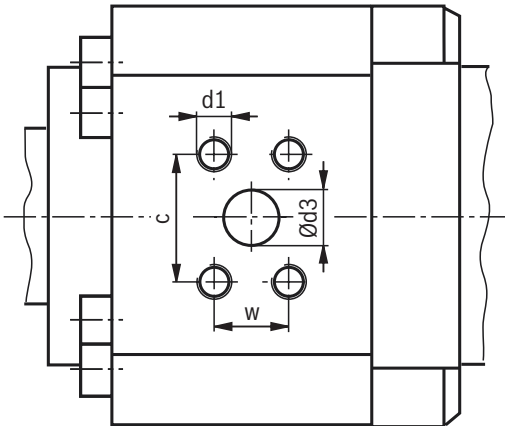
"CDH3", "CSH3"



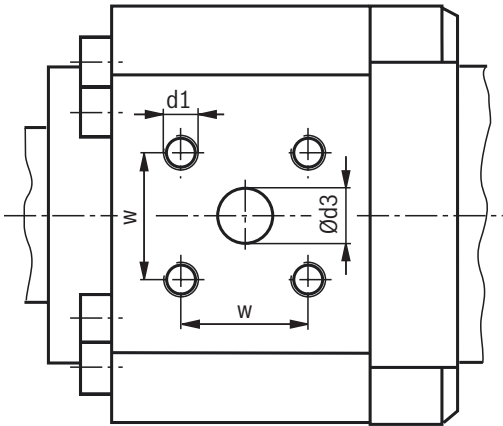
"CGH3"



Porting pattern for rectangular flange according to ISO 6162-2 table 2 type 1



Porting pattern for square flange according to ISO 6164 table 2



Dimensions: Flange ports (Dimensions in mm)

ØAL	Version "D" ISO 6162-2 tab.2 type1 (400 bar) (≅ SAE 6000 PSI)										Version "H" ISO 6164 tab.2 (400 bar)							
	Y	PJ PK	X1	Ød3	Ød3 ³⁾	c ±0.25	w ±0.25	d1	t1 ¹⁾	p ²⁾	Y	PJ PK	X1	Ød3	w ±0.25	d1	t1 ¹⁾	p ²⁾
40	–	–	–	–	–	–	–	–	–	–	90	122	42.5	10	24.7	M6	12.5	400
50	–	–	–	–	–	–	–	–	–	–	89	122	51	10	24.7	M6	12.5	400
63	113	141	65	13	1/2"	40.5	18.2	M8	16	400	113	141	66	19	35.4	M8	16	400
80	120	154	69	13	1/2"	40.5	18.2	M8	16	400	120	154	70	19	35.4	M8	16	400
100	114	181	87	19	3/4"	50.8	23.8	M10	20	400	118	173	89.5	19	35.4	M8	16	400
125	162.5	220	111.5	25	1"	57.2	27.8	M12	24	400	162.5	220	112.5	32	51.6	M12	24	400
140	179.5	232	121.5	32	1 1/4"	66.6	31.8	M14	26	400	179.5	232	124.5	32	51.6	M12	24	400
160	197.5	265	139.5	32	1 1/4"	66.6	31.8	M14	26	400	197.5	265	140.5	38	60.1	M16	30	400
180	233.5	279	156.5	32	1 1/4"	66.6	31.8	M14	26	400	233.5	279	156.5	38	60.1	M16	30	400
200	254.5	293	167.5	38	1 1/2"	79.3	36.5	M16	30	400	254.5	293	170.5	38	60.1	M16	30	400
220	262	326	178	38	1 1/2"	79.3	36.5	M16	30	400	262	326	182	38	60.1	M16	30	400
250	272	336	212	38	1 1/2"	79.3	36.5	M16	30	400	272	336	216	38	60.1	M16	30	400
280	282	366	222	38	1 1/2"	79.3	36.5	M16	30	400	282	366	226	38	60.1	M16	30	400
320	287	391	236	51	2"	96.8	44.5	M20	36	400	287	391	240	51	69.3	M16	30	400

For main dimensions, see page 26 ... 51

ØAL = piston Ø

X* = stroke length

¹⁾ Thread depth.

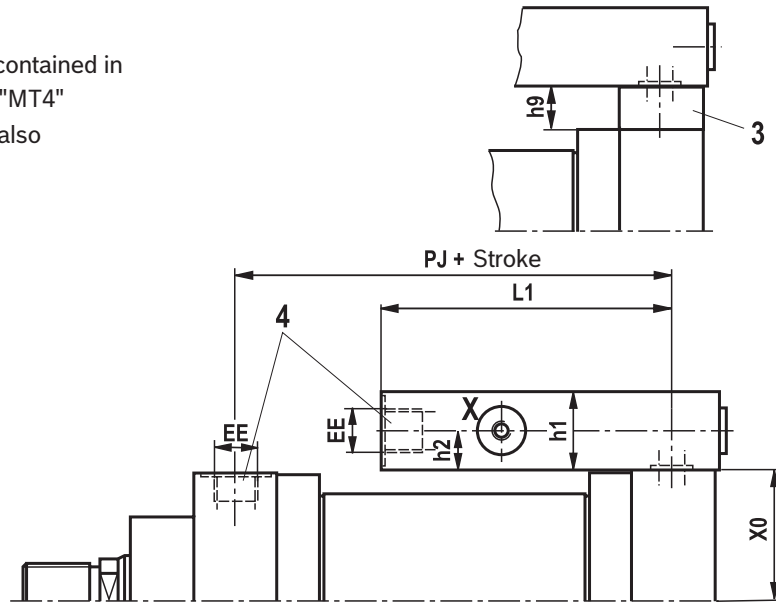
²⁾ Maximum operating pressure for related flanges in bar.

³⁾ Porting pattern according to ISO 6162-2 Table 2 Type 1 corresponds to porting pattern according to SAE 6000 PSI.

Dimensions: Subplates for valve mounting (check valves type SL and SV) (Dimensions in mm)

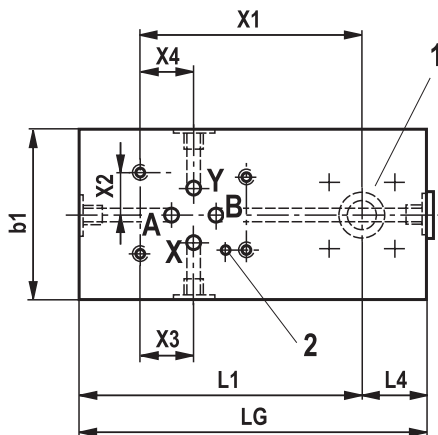
- 1 Port "B" to the piston side according to ISO 6164
- 2 Bore for locking pin
- 3 Adapter plate for type of mounting "MT4" contained in the scope of delivery for type of mounting "MT4"
- 4 Line connection "B" dimensions "EE" see also page 26 ... 51

Installation situation for type of mounting "MT4"



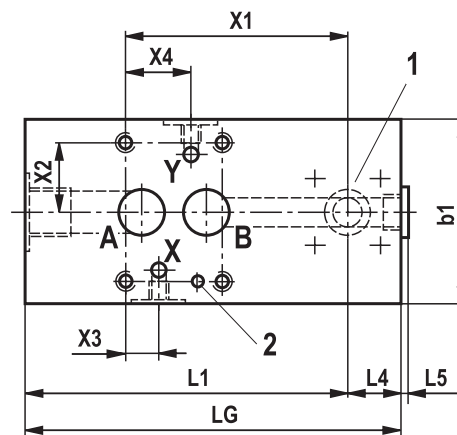
NG6

Porting pattern according to ISO 4401



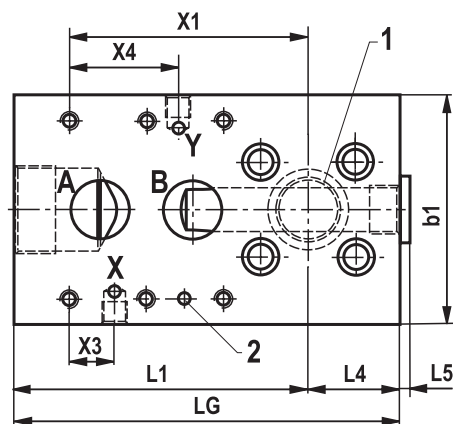
NG10 and 20

Porting pattern according to ISO 5781



NG30

Porting pattern according to ISO 5781



Notice:

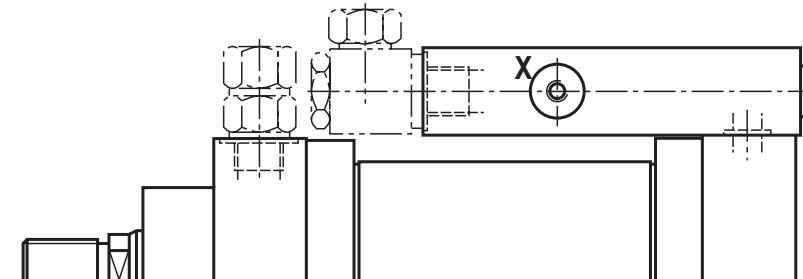
- Seal designs "T", "G", "L", "R", "S" and "V" are not designed for the static holding function.
- Valves, fittings and piping not included in the scope of delivery

Dimensions: Subplates for valve mounting (check valves type SL and SV) (Dimensions in mm)

ØAL	Valve size (NG)	PJ	EE	Stroke _{min} ¹⁾		Plate dimensions								Port size, Porting pattern						Position point Valve	
				2)	3)	X0	L1	L4	L5	LG	b1	h1	h9	h2	A	X	Y	X3	X4	X1	X2
40	6	121	G 1/2	50	50	42.5	90	20	4	110	55	40	10	20	G 1/2	G 1/4	G 1/4	21.5	21.5	65.5	15.5
50	6	121	G 1/2	50	50	51.0	90	20	4	110	55	40	10	20	G 1/2	G 1/4	G 1/4	21.5	21.5	65.5	15.5
63	6	137	G 3/4	64	64	66.0	105	30	5	135	60	45	20	22.5	G 3/4	G 1/4	G 1/4	21.5	21.5	75.5	15.5
	10	137	G 3/4	64	64	66.0	110	30	5	140	85	45	20	22.5	G 3/4	G 1/4	G 1/4	21.4	21.4	78	33.3
80	6	150	G 3/4	58	82	70.0	105	30	5	135	60	45	20	22.5	G 3/4	G 1/4	G 1/4	21.5	21.5	75.5	15.5
	10	150	G 3/4	58	82	70.0	110	30	5	140	85	45	20	22.5	G 3/4	G 1/4	G 1/4	21.4	21.4	78	33.3
100	10	172	G 1	50	109	89.5	102	28	5	130	85	50	20	25	G 1	G 1/4	G 1/4	21.4	21.4	70	33.3
125	10	212.5	G 1 1/4	80	131	112.5	120	40	5	160	85	60	30	30	G 1 1/4	G 1/4	G 1/4	21.4	21.4	90	33.3
	20	212.5	G 1 1/4	80	131	112.5	135	50	5	185	100	60	30	30	G 1 1/4	G 1/4	G 1/4	20.8	39.7	105	39.7
	30	212.5	G 1 1/4	80	131	112.5	160	50	5	210	125	60	30	30	G 1 1/4	G 1/4	G 1/4	24.6	59.6	130	48.4
140	10	225.5	G 1 1/4	60	147	124.5	120	40	5	160	85	60	30	30	G 1 1/4	G 1/4	G 1/4	21.4	21.4	90	33.3
	20	225.5	G 1 1/4	60	147	124.5	135	50	5	185	100	60	30	30	G 1 1/4	G 1/4	G 1/4	20.8	39.7	105	39.7
	30	225.5	G 1 1/4	60	147	124.5	160	50	5	210	125	60	30	30	G 1 1/4	G 1/4	G 1/4	24.6	59.6	130	48.4
160	10	252.5	G 1 1/2	60 ⁴⁾	186	140.5	130	45	5	175	95	70	20	35	G 1 1/2	G 1/4	G 1/4	21.4	21.4	100	33.3
	20	252.5	G 1 1/2	60 ⁴⁾	186	140.5	140	45	5	185	100	70	20	35	G 1 1/2	G 1/4	G 1/4	20.8	39.7	115	39.7
	30	252.5	G 1 1/2	60 ⁴⁾	186	140.5	165	45	5	210	125	70	20	35	G 1 1/2	G 1/4	G 1/4	24.6	59.6	140	48.4
180	10	271.5	G 1 1/2	50 ⁴⁾	212	156.5	130	45	5	175	95	70	20	35	G 1 1/2	G 1/4	G 1/4	21.4	21.4	100	33.3
	20	271.5	G 1 1/2	50 ⁴⁾	212	156.5	140	45	5	185	100	70	20	35	G 1 1/2	G 1/4	G 1/4	20.8	39.7	115	39.7
	30	271.5	G 1 1/2	50 ⁴⁾	212	156.5	165	45	5	210	125	70	20	35	G 1 1/2	G 1/4	G 1/4	24.6	59.6	140	48.4
200	10	285.5	G 1 1/2	30 ⁴⁾	228	170.5	130	45	5	175	95	70	20	35	G 1 1/2	G 1/4	G 1/4	21.4	21.4	100	33.3
	20	285.5	G 1 1/2	30 ⁴⁾	228	170.5	140	45	5	185	100	70	20	35	G 1 1/2	G 1/4	G 1/4	20.8	39.7	115	39.7
	30	285.5	G 1 1/2	30 ⁴⁾	228	170.5	165	45	5	210	125	70	20	35	G 1 1/2	G 1/4	G 1/4	24.6	59.6	140	48.4

ØAL = piston Ø

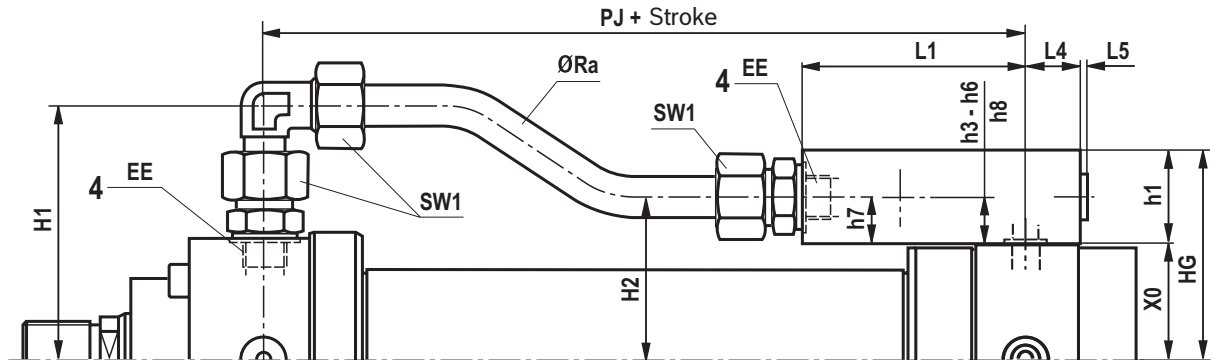
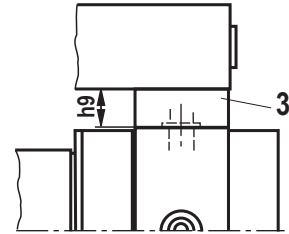
- 1) This applies only to the connection situation below.
- 2) Not for type of mounting "MT4".
- 3) Only for type of mounting "MT4".
- 4) With type of mounting "MS2", observe X*min on page 39 or 51.

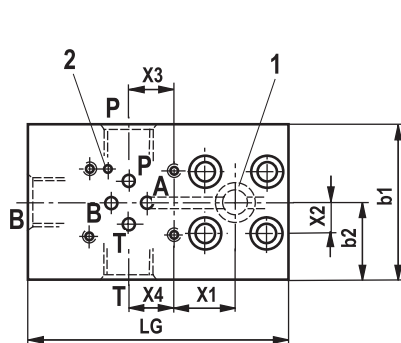
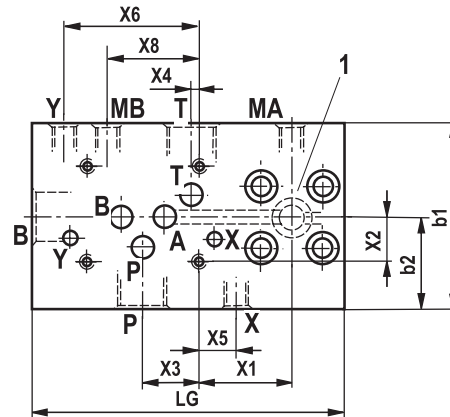
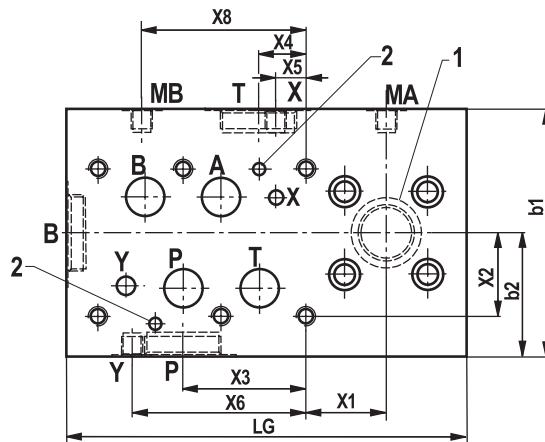
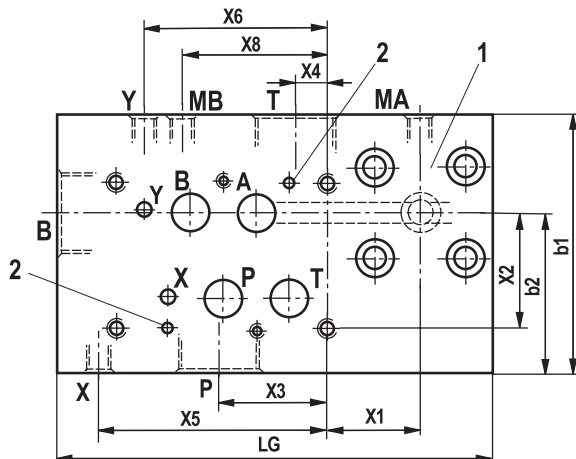


Dimensions: Subplates for valve mounting (on/off and proportional directional control valves) (Dimensions in mm)

Installation situation for type of mounting "MT4"

- 1 Port "A" to the piston side according to ISO 6164
- 2 Bore for locking pin
- 3 Adapter plate for type of mounting "MT4" contained in the scope of delivery for type of mounting "MT4"
- 4 Line connection "B" dimensions "EE" see also page 26 ... 51


NG6: Porting pattern according to ISO 4401

NG10: Porting pattern according to ISO 4401

NG16: Porting pattern according to ISO 4401

NG25: Porting pattern according to ISO 4401

Notice:

With larger stroke lengths and depending on the piston diameter, the pipeline is mounted at the cylinder pipe using pipe supports. A maximum of two sandwich plates is admissible.

Dimensions: Subplates for valve mounting (on/off and proportional directional control valves)
(Dimensions in mm)

ØAL	Valve size (NG)	PJ	EE	Stroke _{min}	Plate dimensions																
					L1	L4	L5 _{max}	H1	H2 ¹⁾	H2 ²⁾	SW1	ØRa	b1	h1	LG	HG ¹⁾	HG ²⁾	b2	X0	h7	h9
40	6	121	G 1/2	242	90	20	4	98.0	62.5	72.5	30	16.0x2.5	65	40	110	82.5	92.5	32.5	42.5	20	10
50	6	121	G 1/2	242	90	20	4	106.5	71.0	81.0	30	16.0x2.5	65	40	110	91.0	101.0	32.5	51.0	20	10
63	6	137	G 3/4	278	105	30	5	132.0	88.5	108.5	36	20.0x3.0	75	45	135	111.0	131.0	37.5	66.0	22.5	20
	10	137	G 3/4	303	130	30	5	132.0	89.0	109.0	36	20.0x3.0	90	70	160	136.0	156.0	45	66.0	23	20
80	6	150	G 3/4	265	105	30	5	136.5	92.5	112.5	36	20.0x3.0	75	45	135	115.0	135.0	37.5	70.0	22.5	20
	10	150	G 3/4	290	130	30	5	136.5	93.0	113.0	36	20.0x3.0	90	70	160	140.0	160.0	45	70.0	23	20
100	10	172	G 1	317	132	28	5	163.5	119.5	139.5	46	25.0x4.0	90	80	160	169.5	189.5	45	89.5	30	20
125	10	212.5	G 1 1/4	341	150	40	5	192.5	147.5	177.5	50	30.0x5.0	105	95	190	207.5	237.5	52.5	112.5	35	30
	16	212.5	G 1 1/4	371	180	40	5	192.5	162.5	192.5	50	30.0x5.0	125	105	220	217.5	247.5	62.5	112.5	50	30
	25	212.5	G 1 1/4	391	200	50	–	192.5	167.5	197.5	50	30.0x5.0	155	110	250	222.5	252.5	77.5	112.5	55	30
140	10	225.5	G 1 1/4	328	150	40	5	204.5	159.5	189.5	50	30.0x5.0	105	95	190	219.5	249.5	52.5	124.5	35	30
	16	225.5	G 1 1/4	358	180	40	5	204.5	174.5	204.5	50	30.0x5.0	125	105	220	229.5	259.5	62.5	124.5	50	30
	25	225.5	G 1 1/4	378	200	50	–	204.5	179.5	209.5	50	30.0x5.0	155	110	250	234.5	264.5	77.5	124.5	55	30
160	10	252.5	G 1 1/2	394	155	50	5	231.5	175.5	195.5	60	38.0x6.0	110	95	205	235.5	255.5	55	140.5	35	20
	16	252.5	G 1 1/2	429	190	50	5	231.5	190.5	210.5	60	38.0x6.0	125	105	240	245.5	265.5	62.5	140.5	50	20
	25	252.5	G 1 1/2	449	210	50	–	231.5	195.5	215.5	60	38.0x6.0	155	110	260	250.5	270.5	77.5	140.5	55	20
180	10	271.5	G 1 1/2	375	155	50	5	248.5	191.5	211.5	60	38.0x6.0	110	95	205	251.5	271.5	55	156.5	35	20
	16	271.5	G 1 1/2	248	190	50	5	248.5	206.5	226.5	60	38.0x6.0	125	105	240	261.5	281.5	62.5	156.5	50	20
	25	271.5	G 1 1/2	307	210	50	–	248.5	211.5	231.5	60	38.0x6.0	155	110	260	266.5	286.5	77.5	156.5	55	20
200	10	285.5	G 1 1/2	253	155	50	5	261.5	205.5	225.5	60	38.0x6.0	110	95	205	265.5	285.5	55	170.5	35	20
	16	285.5	G 1 1/2	234	190	50	5	261.5	220.5	240.5	60	38.0x6.0	125	105	240	275.5	295.5	62.5	170.5	50	20
	25	285.5	G 1 1/2	293	210	50	–	261.5	225.5	245.5	60	38.0x6.0	155	110	260	280.5	300.5	77.5	170.5	55	20

ØAL = piston Ø

¹⁾ Not for type of mounting "MT4".

²⁾ Only for type of mounting "MT4".

Dimensions: Subplates for valve mounting (on/off and proportional directional control valves)
(Dimensions in mm)

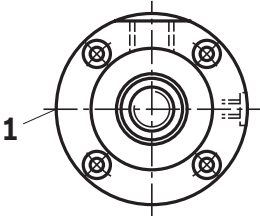
ØAL	Valve size (NG)	Port size, porting pattern																Position point Valve	
		P	X3	h3	T	X4	h4	X	X5	h5	Y	X6	h6	MA	MB	X8	h8	X1	X2
40	6	G 1/2	21.5	20	G 1/2	21.5	20	–	–	–	–	–	–	–	–	–	–	25	15.5
50	6	G 1/2	21.5	20	G 1/2	21.5	20	–	–	–	–	–	–	–	–	–	–	25	15.5
63	6	G 3/4	21.5	22.5	G 3/4	21.5	22.5	–	–	–	–	–	–	–	–	–	–	35	15.5
	10	G 3/4	27	33	G 3/4	3.5	33	G 1/4	18	57	G 1/4	64.0	57	G 1/4	G 1/4	50	17	50	21.4
80	6	G 3/4	21.5	22.5	G 3/4	21.5	22.5	–	–	–	–	–	–	–	–	–	–	35	15.5
	10	G 3/4	27	33	G 3/4	3.5	33	G1/4	18	57	G 1/4	64.0	57	G 1/4	G 1/4	50	17	50	21.4
100	10	G 1	27	30	G 1	3.5	40	G 1/4	18	57	G 1/4	65.0	57	G 1/4	G 1/4	58	20	52	21.4
125	10	G 1 1/4	27	35	G 1 1/4	3.5	45	G 1/4	20	72	G 1/4	62.0	72	G 1/4	G 1/4	55	25	60	21.4
	16	G 1 1/4	57	35	G 1 1/4	15	34	G 1/4	76.5	80	G 1/4	86.0	85	G 1/4	G 1/4	86	45	50	40
	25	G 1 1/4	77	42	G 1 1/4	30	34	G 1/4	19	90	G 1/4	109.0	90	G 1/4	G 1/4	103	50	50	52.1
140	10	G 1 1/4	27	35	G 1 1/4	3.5	45	G 1/4	20	72	G 1/4	62.0	72	G 1/4	G 1/4	55	25	60	21.4
	16	G1 1/4	57	35	G 1 1/4	15	34	G 1/4	76.5	80	G 1/4	86.0	85	G 1/4	G 1/4	86	45	50	40
	25	G 1 1/4	77	42	G 1 1/4	30	34	G 1/4	19	90	G 1/4	109.0	90	G 1/4	G 1/4	103	50	50	52.1
160	10	G 1 1/2	27	35	G 1 1/2	3.5	45	G 1/4	19	72	G 1/4	62.0	72	G 1/4	G 1/4	50	25	72	21.4
	16	G 1 1/2	57	35	G 1 1/2	15	34	G 1/4	76.5	80	G 1/4	86.0	85	G 1/4	G 1/4	86	45	60	40
	25	G 1 1/2	77	42	G 1 1/2	30	34	G 1/4	19	90	G 1/4	109.0	90	G 1/4	G 1/4	103	50	60	52.1
180	10	G 1 1/2	27	35	G 1 1/2	3.5	45	G 1/4	19	72	G 1/4	62.0	72	G 1/4	G 1/4	50	25	72	21.4
	16	G 1 1/2	57	35	G 1 1/2	15	34	G 1/4	76.5	80	G 1/4	86.0	85	G 1/4	G 1/4	86	45	60	40
	25	G 1 1/2	77	42	G 1 1/2	30	34	G 1/4	19	90	G 1/4	109.0	90	G 1/4	G 1/4	103	50	60	52.1
200	10	G 1 1/2	27	35	G 1 1/2	3.5	45	G 1/4	19	72	G 1/4	62.0	72	G 1/4	G 1/4	50	25	72	21.4
	16	G 1 1/2	57	35	G 1 1/2	15	34	G 1/4	76.5	80	G 1/4	86.0	85	G 1/4	G 1/4	86	45	60	40
	25	G 1 1/2	77	42	G 1 1/2	30	34	G 1/4	19	90	G 1/4	109.0	90	G 1/4	G 1/4	103	50	60	52.1

Dimensions: Throttle valve
(Dimensions in mm)

ØAL	40	50	63	80	100	125	140	160	180	200	220	250	280	320
Nominal width	4	4	4	5	5	8	8	8	8	8	20	20	20	20

ØAL = piston Ø

- 1
- Throttle valve only with end position damping "E" (180° for bleeding)

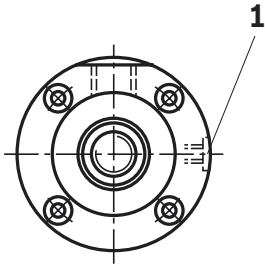


Bleeding / measuring coupling

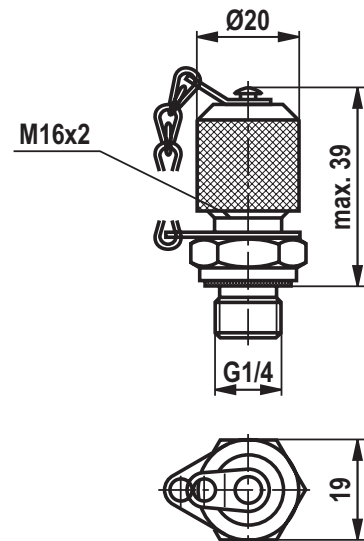
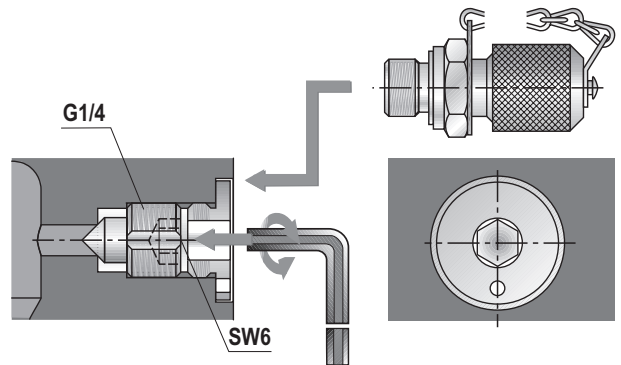
(Dimensions in mm)

By default, a patented safety vent against unintended screwing out in head and base is delivered for all cylinders.

The port allows for the installation of a measuring coupling with check valve for pressure measurement or contamination-free bleeding. Measuring coupling with check valve function, i.e. the measuring or bleeding hose can also be connected when the system is pressurized.



- 1 Bleeding: With view to the piston rod, the position is offset by 90° in relation to the line connection (clockwise)



Scope of delivery: Measuring coupling **G1/4**

MEASURING COUPLING AB 20-11/K1 G1/4 with NBR seal ring

Material no. **R900009090**

MEASURING COUPLING AB 20-11/K1V G1/4 with FKM seal ring

Material no. **R900001264**

Buckling

For the admissible stroke length with flexibly guided load and a factor of 3.5 for safety against buckling, please refer to the relevant table. For other installation positions of the cylinder, the admissible stroke length must be interpolated. Admissible stroke length for non-guided load on request.

Buckling calculations are carried out according to the following formulas:

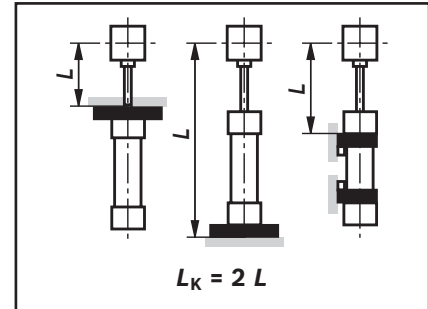
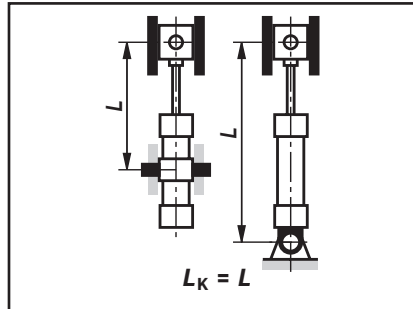
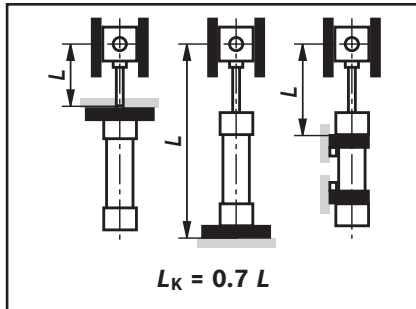
1. Calculation according to Euler

$$F = \frac{\pi^2 \cdot E \cdot I}{\nu \cdot L_K^2} \quad \text{if } \lambda > \lambda_g$$

2. Calculation according to Tetmajer

$$F = \frac{d^2 \cdot \pi (335 - 0.62 \cdot \lambda)}{4 \cdot \nu} \quad \text{if } \lambda \leq \lambda_g$$

Influence of the type of mounting on the buckling length:



Explanation:

E = module of elasticity in N/mm²

= 2.1 x 10⁵ for steel

I = geometrical moment of inertia in mm⁴ for circular cross-section = $\frac{d^4 \cdot \pi}{64} = 0.0491 \cdot d^4$

ν = 3.5 (safety factor)

L_K = free buckling length in mm (depending on the type of mounting, see sketches A, B, C)

d = piston rod Ø in mm

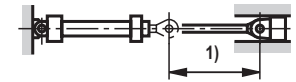
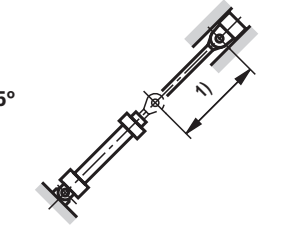
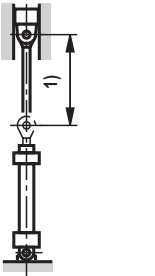
λ = slenderness ratio

$$= \frac{4 \cdot L_K}{d} \quad \lambda_g = \pi \sqrt{\frac{E}{0.8 \cdot R_e}}$$

R_e = yield strength of the piston rod material

Admissible stroke length: CDH3, CSH3 (Dimensions in mm)

Type of mounting: "MP3" and "MP5"

ØAL	ØMM	admissible stroke length with									Installation position
		100 bar			210 bar			350 bar			
		0°	45°	90°	0°	45°	90°	0°	45°	90°	
40	28	360	375	420	225	230	240	140	145	150	0° 
50	36	505	525	351	335	340	355	230	235	240	
63	45	625	650	755	425	430	455	295	300	305	
80	56	765	800	945	530	545	575	375	380	390	
100	70	950	995	1200	680	695	745	495	500	515	
125	90	1200	1270	1610	895	925	1010	665	680	705	45° 
140	100	1335	1405	1785	995	1025	1125	745	755	790	
160	110	1380	1406	1865	1025	1055	1160	755	770	805	
180	125	1580	1670	2150	1180	1220	1350	880	895	940	
200	140	1780	1890	2470	1355	1400	1565	1035	1055	1110	
220	160	1985	2110	2970	1575	1640	1900	1230	1260	1360	90° 
250	180	2190	2340	3310	1740	1820	2120	1370	1400	1510	
280	200	2360	2520	3640	1890	1970	2330	1490	1530	1660	
320	220	2530	2700	3830	2010	2100	2450	1320	1460	1740	

1) Admissible stroke length

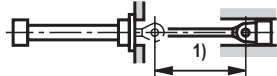
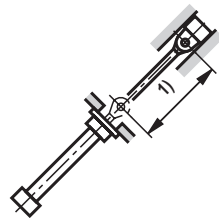


Notice:

- For CSH3, observe the maximum stroke length "X*_{max}", see page 40 ... 51.
- With longer strokes, an extended guide and/or the use of guide rings may make sense to increase the service life, depending on the respective case of application and installation position. Recommendation on request.

Admissible stroke length: CDH3, CGH3, CSH3
(Dimensions in mm)

Type of mounting "MF3"

ØAL	ØMM	admissible stroke length with									Installation position
		100 bar			210 bar			350 bar			
		0°	45°	90°	0°	45°	90°	0°	45°	90°	
40	28	1370	1415	1600	1020	1035	1075	795	800	810	0°
50	36	1755	1825	2135	1345	1370	1440	1060	1070	1090	
63	45	2000	2000	2000	1660	1695	1800	1320	1330	1365	
80	56	2000	2000	2000	2000	2000	2000	1600	1620	1665	
100	70	3000	3000	3000	2470	2530	2740	1900	2010	2085	45°
125	90	3000	3000	3000	3000	3000	3000	2615	2660	2785	
140	100	3000	3000	3000	3000	3000	3000	2875	2920	3000	
160	110	3000	3000	3000	3000	3000	3000	2775	3000	3000	
180	125	3000	3000	3000	3000	3000	3000	3000	3000	3000	90°
200	140	3000	3000	3000	3000	3000	3000	3000	3000	3000	
220	160	6000	6000	6000	5410	5630	6000	4575	4675	5055	
250	180	6000	6000	6000	5950	6000	6000	4815	5160	5605	
280	200	6000	6000	6000	6000	6000	6000	5005	5565	6000	
320	220	6000	6000	6000	6000	6000	6000	4560	5060	6000	

1) Admissible stroke length

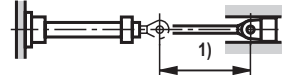
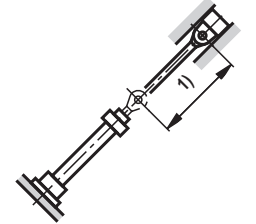
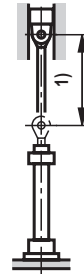


Notice:

- For CSH3, observe the maximum stroke length "X*_{max}", see page 40 ... 51.
- With longer strokes, an extended guide and/or the use of guide rings may make sense to increase the service life, depending on the respective case of application and installation position. Recommendation on request.

Admissible stroke length: CDH3, CSH3 (Dimensions in mm)

Type of mounting "MF4"

ØAL	ØMM	admissible stroke length with									Installation position
		100 bar			210 bar			350 bar			
		0°	45°	90°	0°	45°	90°	0°	45°	90°	
40	28	540	565	675	380	385	410	270	275	280	0° 
50	36	735	770	940	540	550	590	400	405	415	
63	45	900	945	1175	670	690	745	505	510	530	
80	56	1080	1140	1450	825	845	930	630	635	665	
100	70	1330	1400	1840	1030	1070	1190	805	820	860	45° 
125	90	1655	1760	2450	1330	1380	1590	1060	1080	1160	
140	100	1830	1940	2700	1470	1530	1760	1175	1200	1285	
160	110	1905	2030	2830	1530	1590	1835	1035	1160	1300	
180	125	2210	2355	3310	1795	1870	2170	1285	1435	1585	90° 
200	140	2400	2565	3000	1965	2050	2420	1410	1590	1765	
220	160	2655	2850	4445	2245	2360	2935	1735	1930	2160	
250	180	2945	3160	4950	2490	2620	3275	1840	2095	2410	
280	200	3170	3410	5455	2705	2850	3615	1870	2140	2665	
320	220	3425	3680	5775	2905	3055	3820	1675	1925	2815	

1) Admissible stroke length

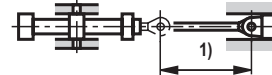
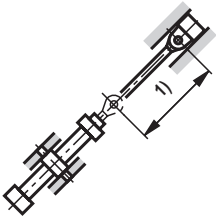
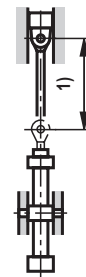


Notice:

- For CSH3, observe the maximum stroke length " X^*_{max} ", see page 40 ... 51.
- With longer strokes, an extended guide and/or the use of guide rings may make sense to increase the service life, depending on the respective case of application and installation position. Recommendation on request.

Admissible stroke length: CDH3, CGH3, CSH3
(Dimensions in mm)

Type of mounting "MT4" trunnion in cylinder center

ØAL	ØMM	admissible stroke length with									Installation position
		100 bar			210 bar			350 bar			
		0°	45°	90°	0°	45°	90°	0°	45°	90°	
40	28	560	580	640	380	385	395	265	270	275	0° 
50	36	760	790	890	353	545	565	390	395	400	
63	45	930	965	1105	665	675	705	490	495	505	
80	56	1125	1170	1365	815	830	875	610	615	625	45° 
100	70	1390	1450	1730	1030	1050	1120	785	790	810	
125	90	1755	1845	2300	1345	1380	1500	1040	1050	1090	
140	100	1935	2030	2545	1485	1525	1660	1150	1165	1210	
160	110	2020	2125	2660	1545	1585	1725	1190	1205	1250	90° 
180	125	2300	2420	3000	1770	1820	1990	1370	1390	1445	
200	140	2555	2695	3000	1990	2050	2270	1555	1580	1655	
220	160	2870	3045	4185	2320	2410	2760	1865	1905	2035	
250	180	3180	3380	4665	2580	2680	3080	2080	2125	2270	
280	200	3430	3645	5130	2800	2915	3390	2270	2325	2500	
320	220	3700	3925	5435	3000	3115	3585	2065	2295	2640	

1) Admissible stroke length

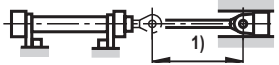
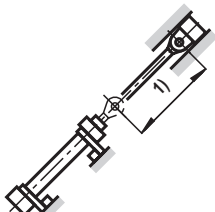
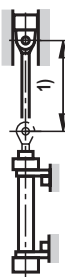


Notice:

- For CSH3, observe the maximum stroke length "X*_{max}", see page 40 ... 51.
- With longer strokes, an extended guide and/or the use of guide rings may make sense to increase the service life, depending on the respective case of application and installation position. Recommendation on request.

Admissible stroke length: CDH3, CGH3, CSH3 (Dimensions in mm)

Type of mounting "MS2"

ØAL	ØMM	admissible stroke length with									Installation position
		100 bar			210 bar			350 bar			
		0°	45°	90°	0°	45°	90°	0°	45°	90°	
40	28	1265	1310	1500	920	935	970	690	695	710	0° 
50	36	1650	1715	2000	1235	1260	1330	950	960	980	
63	45	1995	2000	2000	1520	1550	1655	1180	1190	1220	
80	56	2000	2000	2000	1850	1895	2000	1445	1460	1510	
100	70	2940	3000	3000	2310	2370	2585	1830	1855	1925	
125	90	3000	3000	3000	3000	3000	3000	2640	2685	2810	45° 
140	100	3000	3000	3000	3000	3000	3000	2640	2690	2840	
160	110	3000	3000	3000	3000	3000	3000	2510	2760	2955	
180	125	3000	3000	3000	3000	3000	3000	2900	3000	3000	
200	140	3000	3000	3000	3000	3000	3000	3000	3000	3000	
220	160	6000	6000	6000	5065	5280	6000	4225	4330	4705	90° 
250	180	6000	6000	6000	5590	5835	6000	4455	4805	5250	
280	200	6000	6000	6000	6000	6000	6000	4645	5205	5790	
320	220	6000	6000	6000	6000	6000	6000	4175	4680	6000	

1) Admissible stroke length



Notice:

- For CSH3, observe the maximum stroke length " X^*_{max} ", see page 40 ... 51.
- With longer strokes, an extended guide and/or the use of guide rings may make sense to increase the service life, depending on the respective case of application and installation position. Recommendation on request.

End position damping

End position damping

The objective is to reduce the velocity of a moved mass, the center of gravity of which lies on the cylinder axis, to a level at which neither the cylinder nor the machine into which the cylinder is installed is damaged. For velocities above 20 mm/s, we recommend the use of an end position damping feature, which absorbs energy without requiring the use of additional equipment. It must, however, always be checked whether end position damping is also required for lower velocities with large masses.

Damping capacity

When decelerating masses via end position damping, the structural-inherent damping capacity must not be exceeded. Cylinders with end position cushioning can achieve their full damping capacity only over the entire stroke length. With the adjustable end position cushioning version "E", a throttle valve is additionally provided when compared with version "D". End position damping version "E" allows cycle times

to be optimized. The max. damping capacity can only be achieved when the throttle valve is closed. The calculation depends on the factors mass, velocity, system pressure and installation position. For this reason, mass and velocity are used to determine the characteristic D_m and system pressure and installation position to determine the characteristic D_p . These two characteristics are used for verifying the admissible damping capacity in the "damping capacity" diagram. The intersection point of the characteristics D_m and D_p must always be below the damping capacity curve of the selected cylinder. The values in the diagrams refer to an average oil temperature of +45 ... +65 °C with the throttle valve being closed. For special applications with very short stroke times, high velocities or large masses, cylinders with special end position damping versions can be offered on request. When fixed or adjustable stops are used, special measures must be taken.

Formulas:

$$D_m = \frac{m}{10K} ; K = kv (0.5-v)$$

- m = moved mass in kg
 v = stroke velocity in m/s
 kv = see table page 69

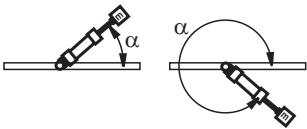
Extension for CDH3 and CSH3

$$D_p = p_s - \frac{m \cdot 9.81 \cdot \sin \alpha}{A_1 \cdot 10}$$

Retraction for CDH3, CGH3 and CSH3; extension for CGH3

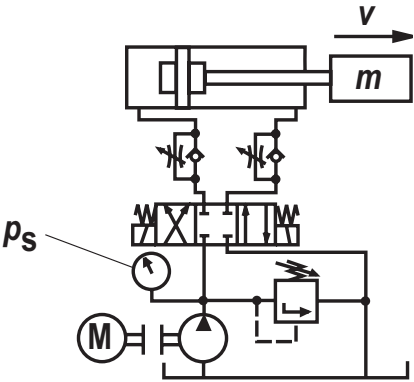
$$D_p = p_s + \frac{m \cdot 9.81 \cdot \sin \alpha}{A_3 \cdot 10}$$

- p_s = system pressure in bar
 A_1 = piston area in cm² (see page 14)
 A_3 = annulus area in cm² (see page 14)
 α = angle to the horizontal in degrees



Damping length

With cylinder strokes within the damping lengths, restrictions with regard to speed / cycle time are possible.



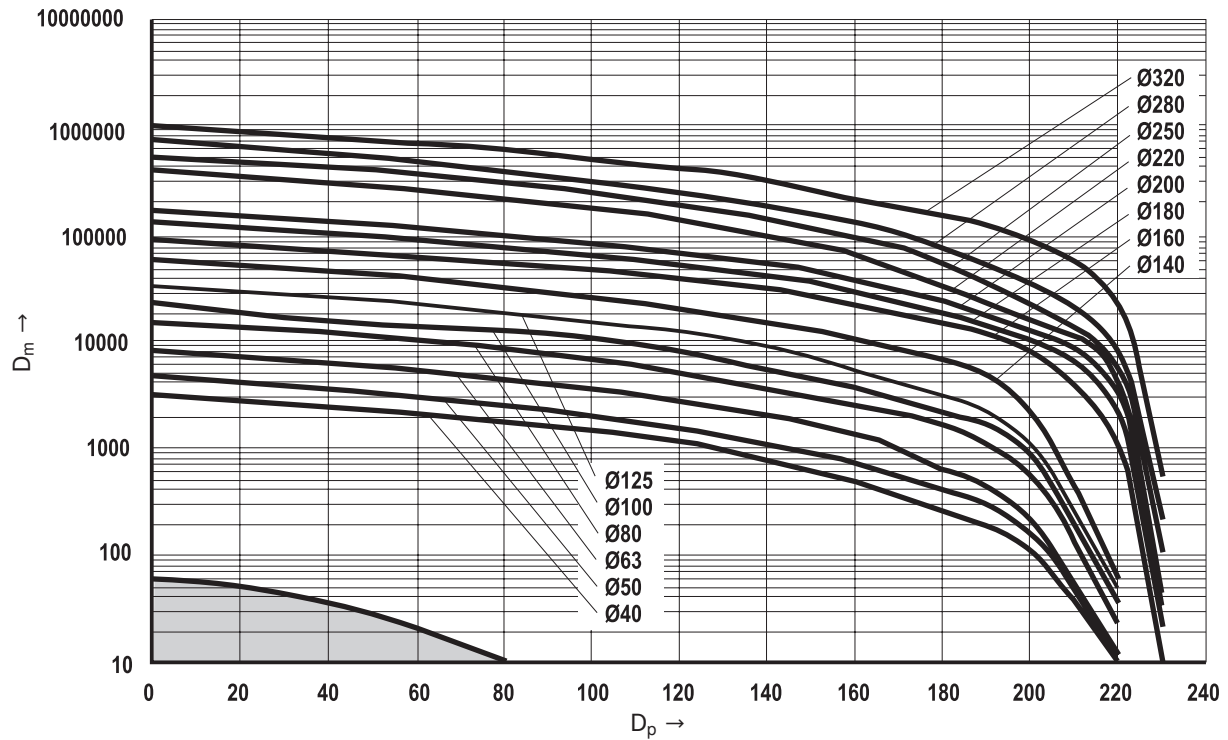
We recommend minimum stroke greater than the damping length plus 10 mm.

ØAL mm	40	50	63	80	100	125	140	160	180	200	220	250	280	320
Head side	21	20	23	25	25	25	33	33	37	37	76	81	86	90
Base side	21	20	23	25	25	25	33	33	37	37	76	81	86	90

End position damping

ØAL mm	40	50	63	80	100	125	140	160	180	200	220	250	280	320
k_v ①	1.72	1.85	1.51	1.85	2.34	2.02	1.85	1.93	1.84	1.65	1.41	1.45	1.58	1.68
k_v ②	2.31	1.85	1.95	1.86	2.25	1.97	1.94	1.92	2.05	1.97	1.64	1.61	1.82	1.94

Damping capacity: Extension for CDH3 and CSH3, with k_v ①

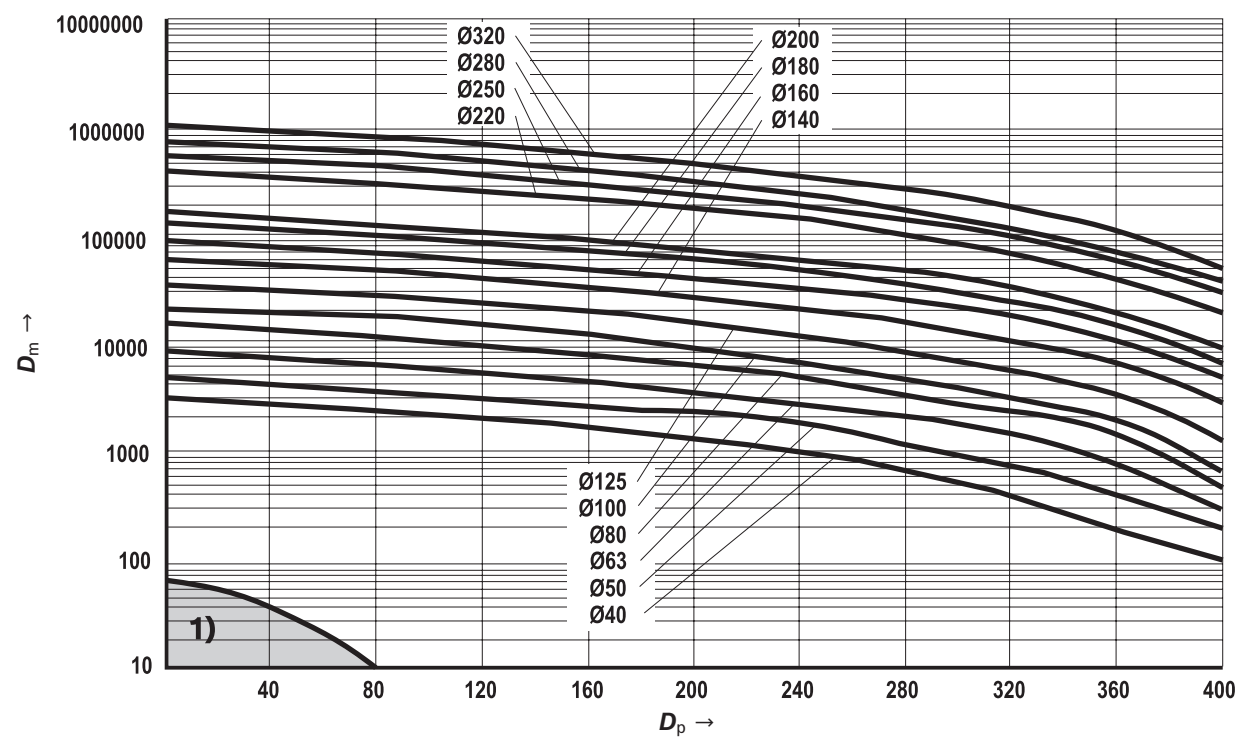


ØAL = piston Ø

1) If with standard applications the calculated intersection point of D_m and D_p is within the marked area, we recommend designing the cylinder without end position damping

End position damping

Damping capacity: Retraction for CDH3, CGH3 and CSH3; extension for CGH3 with k_v ②



1) If with standard applications the calculated intersection point of D_m and D_p is within the marked area, we recommend designing the cylinder without end position damping

Selection criteria for seals

Hydraulic fluids

Hydraulic fluid	Hydraulic fluid temperature range	Seal versions								
		"M"	"G"	"V"	"L"	"A"	"B"	"T"	"R"	"S"
HL, HLP	-20 °C ... +80 °C	++	++	++	++	++	++	++	++	++
HFA	+5 °C ... +55 °C	+/-	+/-	+/-	+/-	+	+/-	++	+/-	+/-
HFC	-20 °C ... +60 °C	-	++	-	-	+/-	-	++	-	-
HFDR	-15 °C ... +80 °C	-	-	++	-	-	++	-	-	++
HFUD	-15 °C ... +80 °C	-	-	++	-	-	++	-	-	++

Environmental conditions

Ambient and piston rod temperature range in the area of the piston rod		Seal versions								
		"M"	"G"	"V"	"L"	"A"	"B"	"T"	"R"	"S"
Standard	-20 °C ... +80 °C ¹⁾	++	+	+ ²⁾	++	++	+ ²⁾	+	++	++ ²⁾
Extended	+80 °C ... +120 °C ¹⁾	-	-	++	-	-	+	-	-	++

Working conditions

Function and velocity	Seal versions								
	"M"	"G"	"V"	"L"	"A"	"B"	"T"	"R"	"S"
Static hold function more than 10 minutes: Application and temperature-dependent	++	+	+	+	++	++	+	+	+
Static holding function short-time < 1 minute	++	++	++	++	++	++	++	++	++
Robust application conditions: Steel works, mining, thin ice	++	++	++	++	++	++	-	++	-
Zero point control, hardly any amplitude, frequency max. 5 Hz, not longer than 5 minutes	-	-	-	+/-	-	-	++	+	++
Cylinder velocity min. 0.001 m/s Stick-slip behavior	++	+	+	++	-	-	++	++	++
Cylinder velocity from 0.01 m/s to 0.5 m/s ³⁾	++	+	+	++	+	+	++	++	++
Cylinder velocity > 0.5 m/s to 0.8 m/s max. ³⁾	-	+/-	+/-	++	-	-	++	+	++
Stroke > 1.0 m	+/-	++	++	++	++	++	++	++	++
Standstill period (wear)	++	+/-	+/-	++	+/-	-	++	++	++
Undissolved air in the oil ⁴⁾	-	+	+	+	-	-	+	+	+

- ++ = very good
+ = good
+/- = conditional, depending on the application parameters
- = inappropriate

- ¹⁾ Moreover, observe the corresponding hydraulic fluid temperature range.
²⁾ Lower temperature limit -15 °C.
³⁾ Standard line connections not designed for that velocity.
⁴⁾ - Seal is destroyed / + Seal is not directly destroyed, leakage may occur.



Notice:

- General technical data in corresponding data sheets will remain valid, see page 11.
- Generally, a hydraulic fluid temperature of approx. 40 °C is recommended. The specified values should be regarded as guidelines; depending on the case of application, it may be necessary to check the suitability of the seal system.
- For detailed information, please contact your local Bosch Rexroth distribution organization under www.boschrexroth.com.

Seal kits

CDH3 – Standard

ØAL	ØMM	Material no. for seal design								
		"M"	"G"	"V"	"L"	"A"	"B"	"T"	"R"	"S"
40	28	R900851087	R961006002	R961006037	R961006072	R900859445	R900859770	R900858841	R961006107	R900861001
50	36	R900849392	R961006005	R961006040	R961006075	R900851515	R900860940	R900860277	R961006110	R900861004
63	45	R900847956	R961006008	R961006043	R961006078	R900851638	R900859678	R900847855	R961006113	R900861007
80	56	R900850905	R961006011	R961006046	R961006081	R900854718	R900851205	R900856180	R961006116	R900861010
100	70	R900853382	R961006014	R961006049	R961006084	R900856094	R900860946	R900860285	R961006119	R900861013
125	90	R900857949	R961006017	R961006052	R961006087	R900856095	R900855464	R900856102	R961006122	R900861016
140	100	R900853965	R961006019	R961006054	R961006089	R900856096	R900860952	R900860290	R961006124	R900849080
160	110	R900851146	R961006021	R961006056	R961006091	R900860933	R900860954	R900857536	R961006126	R900861019
180	125	R900848603	R961006024	R961006059	R961006094	R900860935	R900860956	R900860292	R961006129	R900861021
200	140	R900856431	R961006026	R961006061	R961006096	R900860937	R900860958	R900860293	R961006131	R900861023
220	160	R900888101	R961006028	R961006063	R961006098	R900888117	R900888141	R900888109	R961006133	R900888133
250	180	R900888103	R961006030	R961006065	R961006100	R900888119	R900888143	R900888111	R961006135	R900888135
280	200	R900888105	R961006032	R961006067	R961006102	R900888121	R900888145	R900888113	R961006137	R900888137
320	220	R900888107	–	–	R961006104	R900888123	R900888147	R900888115	R961006139	R900888139

CGH3 – Standard

ØAL	ØMM	Material no. for seal design								
		"M"	"G"	"V"	"L"	"A"	"B"	"T"	"R"	"S"
40	28	R900867252	R961006223	R961006258	R961006293	R900866747	R900867133	R900868889	R961006328	R900868943
50	36	R900864930	R961006226	R961006261	R961006296	R900866750	R900867136	R900868892	R961006331	R900868946
63	45	R900867262	R961006229	R961006264	R961006299	R900866753	R900867139	R900868895	R961006334	R900868949
80	56	R900867265	R961006232	R961006267	R961006302	R900866756	R900867142	R900868898	R961006337	R900868952
100	70	R900867268	R961006235	R961006270	R961006305	R900866759	R900867146	R900868901	R961006340	R900868955
125	90	R900867270	R961006238	R961006273	R961006308	R900866762	R900867149	R900868904	R961006343	R900868957
140	100	R900867272	R961006240	R961006275	R961006310	R900866764	R900867151	R900868906	R961006345	R900868959
160	110	R900867274	R961006242	R961006277	R961006312	R900866766	R900867153	R900868908	R961006347	R900868961
180	125	R900867276	R961006245	R961006280	R961006315	R900866768	R900867155	R900868910	R961006350	R900868963
200	140	R900867278	R961006247	R961006282	R961006317	R900866770	R900867157	R900868912	R961006352	R900868965
220	160	R900888021	R961006249	R961006284	R961006319	R900888037	R900888061	R900888029	R961006354	R900888053
250	180	R900888023	R961006251	R961006286	R961006321	R900888039	R900888063	R900888031	R961006356	R900888055
280	200	R900888025	R961006253	R961006288	R961006323	R900888041	R900888065	R900888033	R961006358	R900888057
320	220	R900888027	–	–	R961006325	R900888043	R900888067	R900888035	R961006360	R900888059

ØAL = piston Ø
 ØMM = piston rod Ø



Notice:

Seal kits for proximity switches and subplate mounting separate material no.

Seal kits

CDH3 – Standard + additional option "F"

ØAL	ØMM	Material no. for seal design					
		"M" + "F"	"G" + "F"	"V" + "F"	"T" + "F"	"R" + "F"	"S" + "F"
40	28	R900861025	R961006142	R961006169	R900861050	R961006196	R900861100
50	36	R900861028	R961006145	R961006172	R900861053	R961006199	R900861103
63	45	R900861031	R961006148	R961006175	R900861056	R961006202	R900861106
80	56	R900861034	R961006151	R961006178	R900861059	R961006205	R900861109
100	70	R900861037	R961006154	R961006181	R900861062	R961006208	R900861115
125	90	R900861040	R961006157	R961006184	R900861065	R961006211	R900861122
140	100	R900861042	R961006159	R961006186	R900861067	R961006213	R900861126
160	110	R900861044	R961006161	R961006188	R900861069	R961006215	R900861130
180	125	R900861046	R961006164	R961006191	R900861071	R961006218	R900861135
200	140	R900861048	R961006166	R961006193	R900861073	R961006220	R900861143

CGH3 – Standard + additional option "F"

ØAL	ØMM	Material no. for seal design					
		"M" + "F"	"G" + "F"	"V" + "F"	"T" + "F"	"R" + "F"	"S" + "F"
40	28	R900868999	R961006363	R961006390	R900869026	R961006417	R900869093
50	36	R900869002	R961006366	R961006393	R900869029	R961006420	R900869096
63	45	R900869005	R961006369	R961006396	R900869032	R961006423	R900869099
80	56	R900869008	R961006372	R961006399	R900869035	R961006426	R900869102
100	70	R900869013	R961006375	R961006402	R900869038	R961006429	R900869105
125	90	R900869016	R961006378	R961006405	R900869041	R961006432	R900869108
140	100	R900869018	R961006380	R961006407	R900869043	R961006434	R900869110
160	110	R900869020	R961006382	R961006409	R900869045	R961006436	R900869112
180	125	R900869022	R961006385	R961006412	R900869047	R961006439	R900869114
200	140	R900869024	R961006387	R961006414	R900869049	R961006441	R900869116

ØAL = piston Ø
ØMM = piston rod Ø



Notice:

Seal kits for proximity switches and subplate mounting separate material no.

Seal kits

CSH3 – Standard

ØAL	ØMM	Material no. for seal design						
		"M"	"G"	"V"	"L"	"T"	"R"	"S"
40	28	R900861025	R961006142	R961006169	R961006072	R900861050	R961006196	R900861100
50	36	R900861028	R961006145	R961006172	R961006075	R900861053	R961006199	R900861103
63	45	R900861031	R961006148	R961006175	R961006078	R900861056	R961006202	R900861106
80	56	R900861034	R961006151	R961006178	R961006081	R900861059	R961006205	R900861109
100	70	R900861037	R961006154	R961006181	R961006084	R900861062	R961006208	R900861115
125	90	R900861040	R961006157	R961006184	R961006087	R900861065	R961006211	R900861122
140	100	R900861042	R961006159	R961006186	R961006089	R900861067	R961006213	R900861126
160	110	R900861044	R961006161	R961006188	R961006091	R900861069	R961006215	R900861130
180	125	R900861046	R961006164	R961006191	R961006094	R900861071	R961006218	R900861135
200	140	R900861048	R961006166	R961006193	R961006096	R900861073	R961006220	R900861143
220	160	R900888101	R961006028	R961006063	R961006098	R900888109	R961006133	R900888133
250	180	R900888103	R961006030	R961006065	R961006100	R900888111	R961006135	R900888135
280	200	R900888105	R961006032	R961006067	R961006102	R900888113	R961006137	R900888137
320	220	R900888107	–	–	R961006104	R900888115	R961006139	R900888139

ØAL = piston Ø
 ØMM = piston rod Ø



Notice:

Seal kits for position measurement system and subplate mounting separate material no.

Seal kits

Only for proximity switches

ØAL	Material no. for seal design								
	"M", "M" + "F"	"T", "T" + "F"	"G", "G" + "F"	"L"	"R", "R" + "F"	"A"	"S", "S" + "F"	"V", "V" + "F"	"B"
40 ... 200	R900885938						R900885939		
220 ... 320	R900894997						R900894998		

Only for subplate mounting

ØAL	Material no. for seal design	
	"M", "T", "G", "L", "R", "A"	"S", "B", "V"
40	R961006022	R961006243
50	R961006022	R961006243
63	R961006092	R961006313
80	R961006092	R961006313
100	R961006092	R961006313
125	R961006162	R961006383
140	R961006162	R961006383
160	R961006189	R961006410
180	R961006189	R961006410
200	R961006189	R961006410

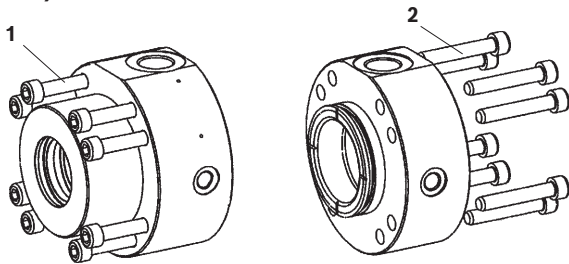
Only for position measurement system

ØAL	Material no. for seal design	
	"M", "T", "G", "L", "R"	"S", "V"
40	R900885935	R900885937
50	R900894958	R900894979
63	R900894959	R900894980
80	R900894960	R900894981
100	R900894961	R900894982
125	R900894962	R900894983
140	R900894963	R900894985
160	R900894964	R900894986
180	R900894973	R900894987
200	R900894974	R900894988
220	R900894975	R900894989
250	R900894976	R900894991
280	R900894977	R900894993
320	R900894978	R900894994

ØAL = piston Ø

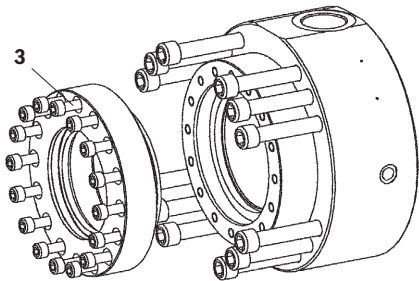
Tightening torques

Screws: Head and base (item 1 and 2)



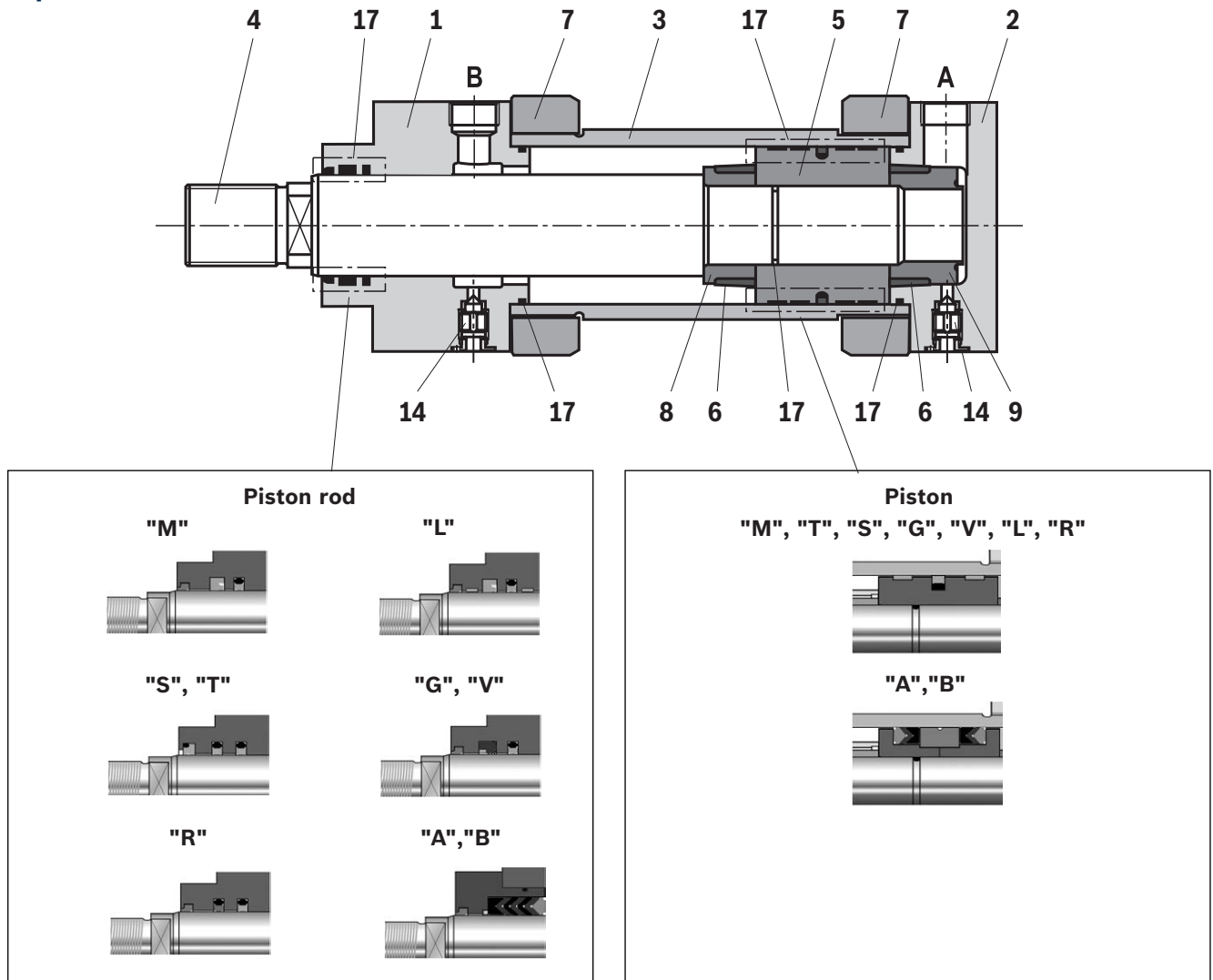
Series	Piston Ø	Screw	Quantity	Quality class	Tightening torque
CDH3; CGH3; CSH3	40	M10	4	10.9	40 Nm
CDH3; CGH3; CSH3	50	M8	8	10.9	25 Nm
CDH3; CGH3; CSH3	63	M10	8	10.9	50 Nm
CDH3; CGH3; CSH3	80	M12	8	10.9	90 Nm
CDH3; CGH3; CSH3	100	M16	8	10.9	175 Nm
CDH3; CGH3; CSH3	125	M20	8	10.9	350 Nm
CDH3; CGH3; CSH3	140	M20	8	10.9	450 Nm
CDH3; CGH3; CSH3	160	M24	8	10.9	670 Nm
CDH3; CGH3; CSH3	180	M24	12	10.9	580 Nm
CDH3; CGH3; CSH3	200	M24	12	10.9	720 Nm
CDH3; CGH3; CSH3	220	M24	16	10.9	750 Nm
CDH3; CGH3; CSH3	250	M30	16	10.9	1400 Nm
CDH3; CGH3; CSH3	280	M30	16	10.9	1600 Nm
CDH3; CGH3; CSH3	320	M42	12	10.9	4200 Nm

Screws: Seal cover (item 3)

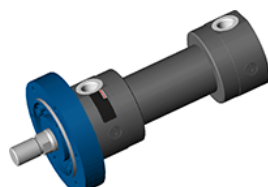


Only with seal design "A" and "B"

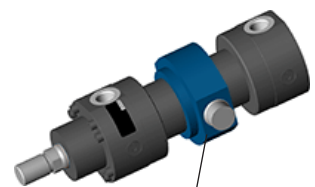
Series	Piston Ø	Piston rod Ø	Screw	Quantity	Quality class	Tightening torque
CDH3; CGH3	160	110	M10	16	10.9	60 Nm
CDH3; CGH3	180	125	M12	16	10.9	80 Nm
CDH3; CGH3	200	140	M12	16	10.9	90 Nm
CDH3; CGH3	220	160	M12	24	10.9	90 Nm
CDH3; CGH3	250	180	M16	16	10.9	90 Nm
CDH3; CGH3	280	200	M16	16	10.9	230 Nm
CDH3; CGH3	320	220	M16	24	10.9	230 Nm

Components: Series CDH3

10



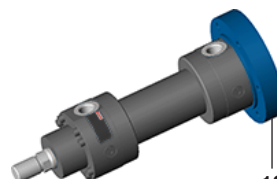
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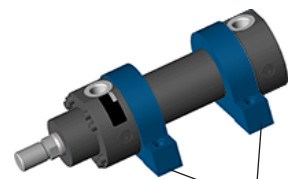
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11



13



16

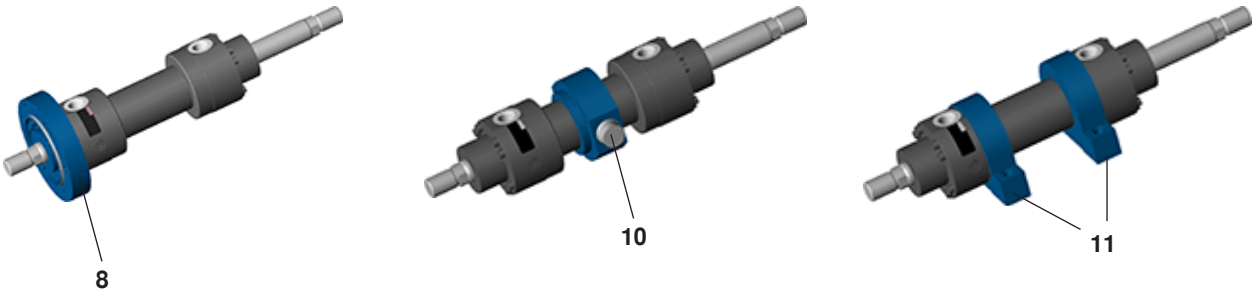
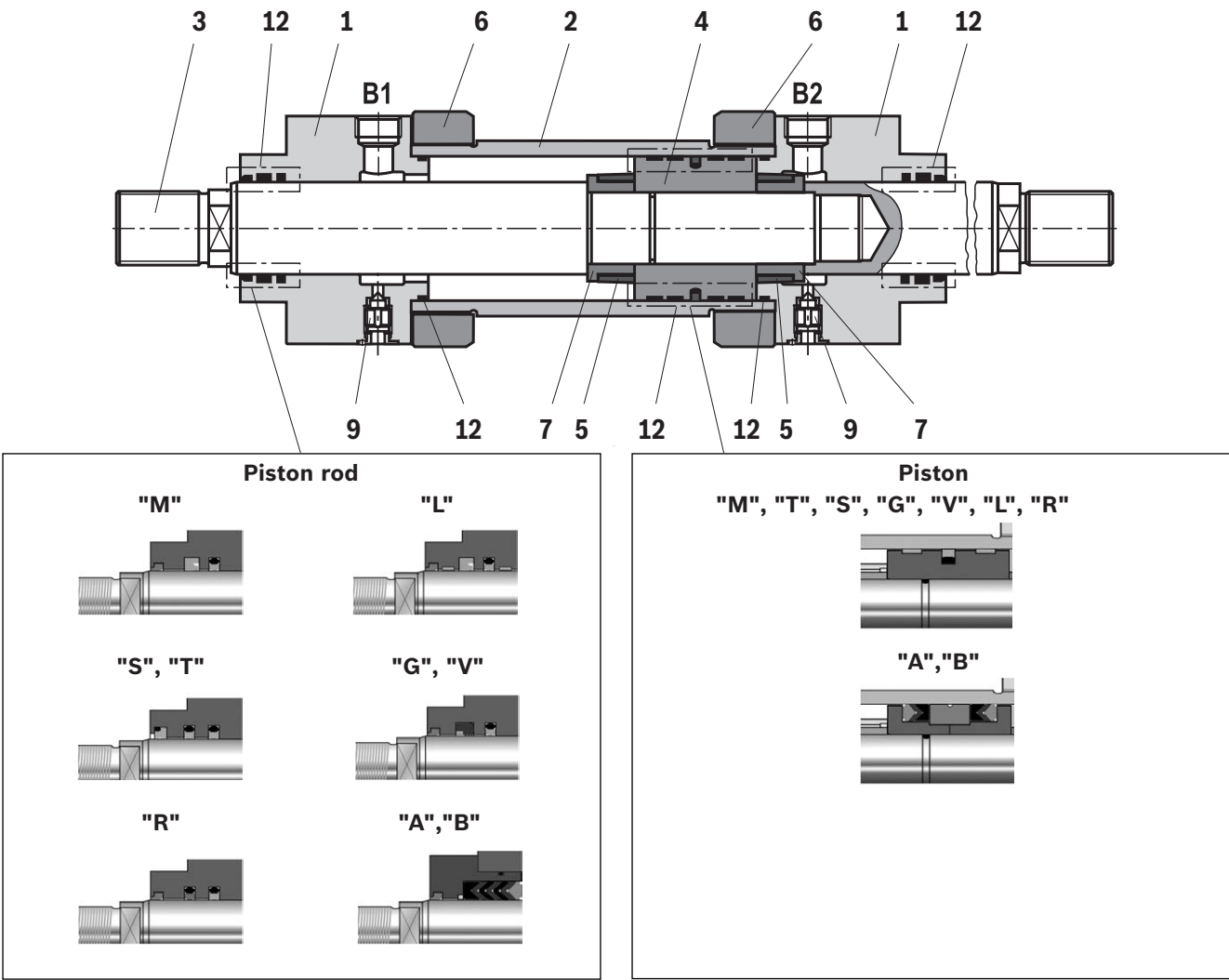
- A** Piston chamber
B Annulus area
1 Head
2 Base
3 Pipe
4 Piston rod

- 5** Piston
6 Damping bush
7 Flange
8 Socket
9 Socket
10 Base MP3

- 11** Base "MP5"
12 Round flange "MF3"
13 Round flange "MF4"
14 Bleeding
15 Trunnion mounting "MT4"
16 Foot "MS2"

- 17** Seal kit:
 Wiper
 Rod seal
 Piston seal
 O-ring
 Guide ring

Components: Series CGH3



- B1

Annulus area 1
- B2

Annulus area 2
- 1

Head
- 2

Pipe
- 3

Piston rod
- 4

Piston
- 5

Damping bush
- 6

Flange
- 7

Socket
- 8

Round flange "MF3"
- 9

Bleeding
- 10

Trunnion mounting "MT4"
- 11

Foot "MS2"
- 12

Seal kit:

Wiper

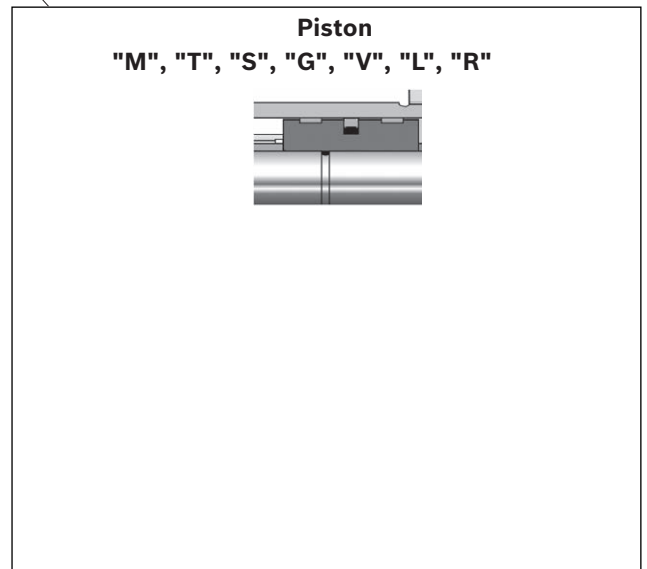
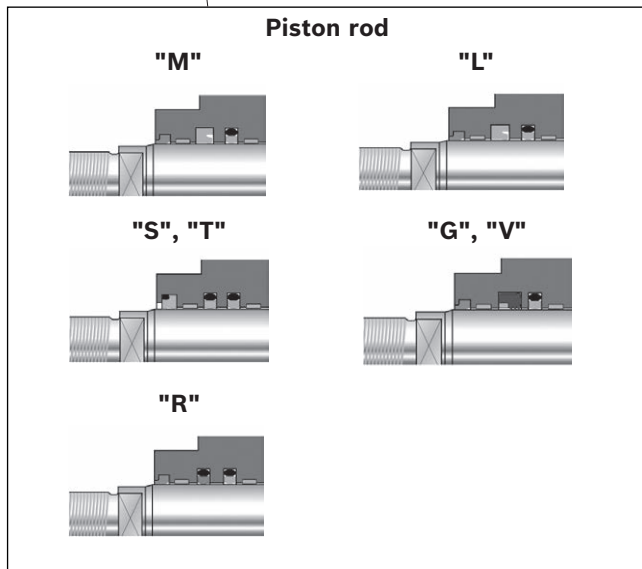
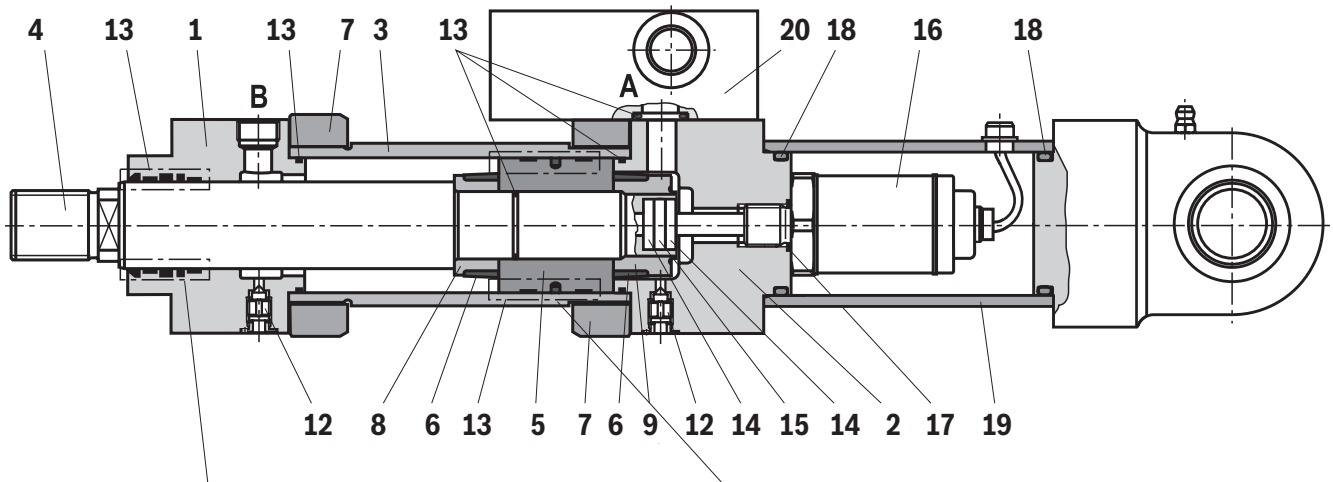
Rod seal

Piston seal

O-ring

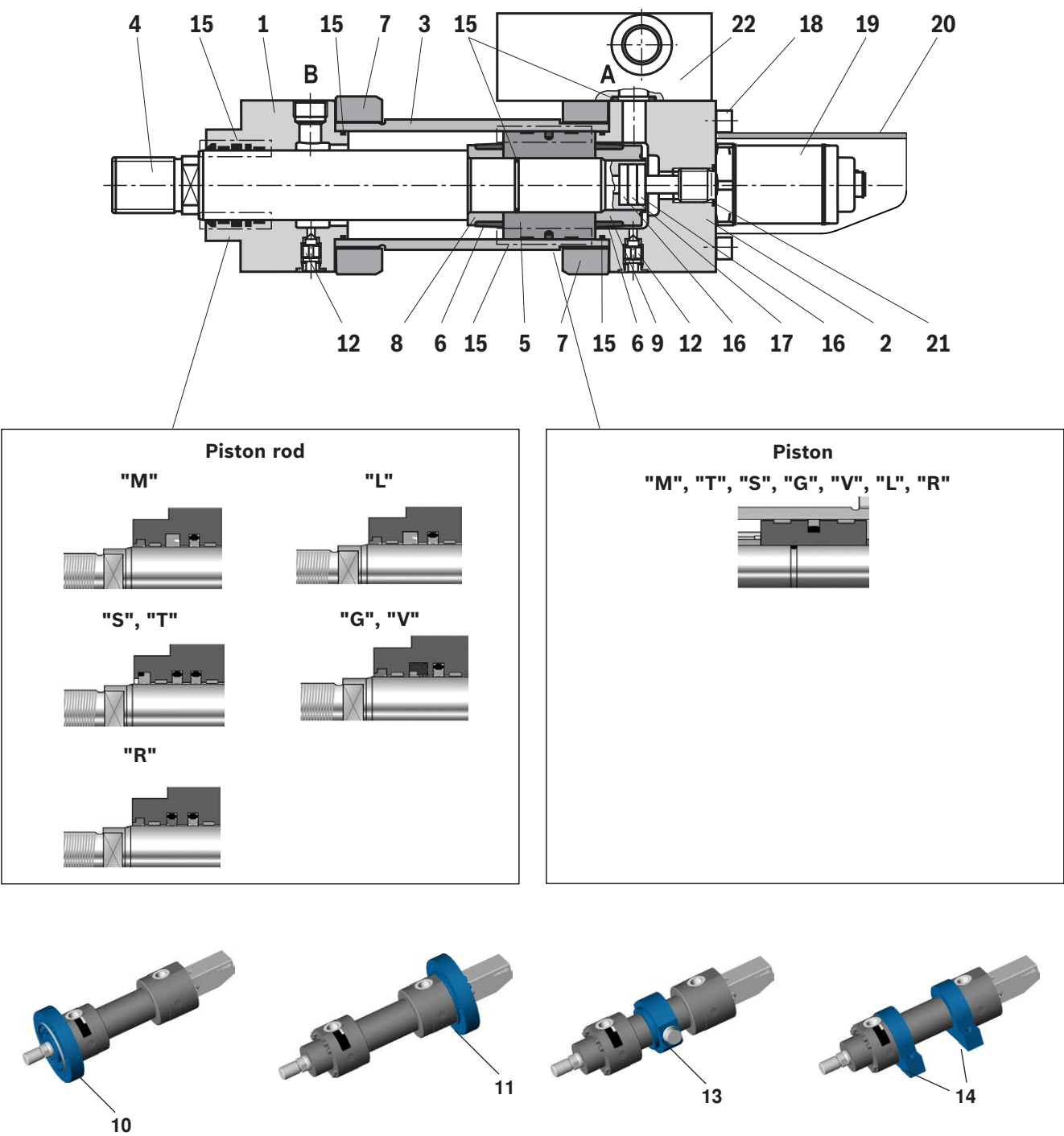
Guide ring

Components: Series CSH3 – "MP3" and "MP5"



A	Piston chamber	8	Socket	13	Seal kit:	16	Position measurement system
B	Annulus area	9	Socket		Wiper	17	Seal
1	Head	10	Base "MP3"		Rod seal	18	Seal
2	Base	11	Base "MP5"		Piston seal	19	Protective pipe
3	Pipe	12	Bleeding		O-ring	20	Subplate
4	Piston rod			14	Insulating socket		
5	Piston			15	Solenoid		
6	Damping bush						
7	Flange						

Components: Series CSH3 – "MF3", "MF4", "MT4" and "MS2"



A	Piston chamber	8	Socket	15	Seal kit:	18	Hexagon socket head cap screw
B	Annulus area	9	Socket		Wiper	19	Position measurement system
1	Head	10	Round flange "MF3"		Rod seal	20	Protective housing
2	Base	11	Round flange "MF4"		Piston seal	21	Seal
3	Pipe	12	Bleeding		O-ring	22	Subplate
4	Piston rod	13	Trunnion mounting "MT4"	16	Guide ring		
5	Piston	14	Foot "MS2"	17	Insulating socket		
6	Damping bush				Solenoid		
7	Flange						

Priming / painting

Priming | By default, hydraulic cylinders are primed with a coating (color gentian blue RAL 5010) of min. 40 µm. Other colors on request.

With hydraulic cylinders and attachment parts, the following surfaces are not primed:

- ▶ All fit diameters to the customer side
- ▶ Sealing surfaces for line connection
- ▶ Sealing surfaces for flange connection
- ▶ Connection surface for valve mounting
- ▶ Inductive proximity switches
- ▶ Position measurement system
- ▶ Measuring coupling
- ▶ Spherical/plain bearings
- ▶ Grease nipples

Painting | By default, hydraulic cylinders can be ordered in the CP4 to CP7 corrosivity category in the RAL colors.

With hydraulic cylinders and attachment parts, the following surfaces are not painted:

- ▶ All fit diameters to the customer side
- ▶ All mounting surfaces to the customer side
- ▶ Sealing surfaces for line connection
- ▶ Sealing surfaces for flange connection
- ▶ Connection surface for valve mounting
- ▶ Inductive proximity switches
- ▶ Position measurement system
- ▶ Measuring coupling
- ▶ Spherical/plain bearings
- ▶ Grease nipples

Notice:

Surfaces not primed or painted are protected with solvent-free corrosion protection agent. Accessories ordered as a separate order item are not primed or painted by default. Corresponding priming and/or painting on request.

Corrosivity categories

	Class	Certification by	Applications Inside	Outside
Priming	CP3	240 h salt spray test SST (EN ISO 9227) 240 h condensation water test KKT (EN ISO 6270-2) Layer thickness: at least 40 µm	Field of application e. g. hall atmosphere, air humidity ≤ 60%, no thermal load.	Not suitable for outdoor exposure.
Painting	CP4	480 h salt spray test SST (EN ISO 9227) 480 h condensation water test KKT (EN ISO 6270-2) Nominal layer thickness: 120 µm	Unheated buildings in which there may be condensation (production rooms, storage and sport halls).	Urban and industrial atmosphere with little salt or sulfur dioxide load.
	CP5	720 h salt spray test SST (EN ISO 9227) 480 h condensation water test KKT (EN ISO 6270-2) Nominal layer thickness: 140 µm	Silo and debris facilities, chemical plants, boathouses above sea water, laundries, breweries with high humidity and medium contamination.	Industrial and coastal areas with medium salt load.
	CP6	1000 h salt spray test NSS (EN ISO 9227) 480 h condensation water test KKT (EN ISO 6270-2) Nominal layer thickness: 220 µm	Buildings or areas with almost permanent condensation and serious contamination.	Industrial areas with large humidity and aggressive atmosphere.
	CP7	1440 h salt spray test NSS (EN ISO 9227) 720 h condensation water test KKT (EN ISO 6270-2) Nominal layer thickness: 320 µm	Buildings or areas with almost permanent condensation and with serious contamination.	Coastal and offshore areas with high salt load.

Notice:

Please refer to the information on priming/painting on page 81. The specified resistances of the individual Bosch Rexroth classes only refer to the primed / painted cylinder areas, not, for example, to piston rods, trunnion mountings, etc. In this connection, special measures may be necessary.

Accessories (separate order)**Mating connectors and cable sets**

Designation	Version	Short designation	Material number	Data sheet
Cable sets; for cylinders with proximity switch, 5-pole	Straight, 5 m	M12 x 1, A-coded	R913016852	08006
	Angled, 5 m	M12 x 1, A-coded	R988064311	
Mating connectors; for cylinders with position measurement system – analog output "F" and "C" or digital output "D" and "S"	Straight, PG7	6-pole	R901511626	08006
	Straight, PG7	7-pole	R900079551	
Mating connectors; for cylinders with position measurement system – IO-Link interface, 5-pole	Straight	M12, A-coded	R913045873	08006
	Angled	M12, A-coded	R901500328	
Mating connectors; for cylinders with position measurement system – Profibus "D53"	Straight	M8, 4-pole	R901132799	08006
	Straight	M12, 2-pole, B-coded	R900773386	
	Straight	M12, 2-pole, B-coded	R901091655	
	Straight	M12, 4-pole, B-coded	R901070126	
Mating connectors; for cylinders with position measurement system – Profibus "D63", 6-pole	Straight, PG9	M16, signal input	R900705950	08006
	Straight, PG9	M16, end plug	R900722518	
	Straight, PG9	M16, signal output	R900705951	
Cable sets; for cylinders with position measurement system – Profibus "D53", 4-pole	Straight, 5 m	M8 x 1	R901213191	08006
	Straight, 10 m	M8 x 1	R913008737	
	Straight, 15 m	M8 x 1	R913008738	
Mating connectors; for cylinders with position measurement system – Profinet	Straight	M12; 4-pole, D-coded	R913045872	08006
	Straight	M12; 5-pole, A-coded	R913045873	

Project planning information

Standards

Bosch Rexroth standard; main dimensions like piston Ø and piston rod Ø correspond to ISO 3320.

Acceptance

Every hydraulic cylinders is tested according to Bosch Rexroth standards and following ISO 10100:2020 with module L.

Safety instructions

For assembly, commissioning and maintenance of hydraulic cylinders, observe the operating instructions 07100-B. Service and repair work has to be performed by Bosch Rexroth AG or by personnel especially trained for this purpose. No warranty is accepted for damage as a consequence of assembly, maintenance or repair work not performed by Bosch Rexroth AG.

Check lists for hydraulic cylinders

Cylinders the characteristics and/or application parameters of which deviate from the values specified in the data sheet can only be offered as a special version upon request. For offers, the deviations of the characteristics and / or application parameters must be described in the check lists for hydraulic cylinders (07200).

Minimum strokes

When using end position damping, the minimum stroke must also be observed, see page 68 "End position damping". Observe the minimum stroke when selecting subplates for on/off and proportional directional control valves or isolation valves type SL and SV, see page 57 and 59 "Dimensions: Subplates for valve mounting".

Boundary and application conditions

- ▶ The mechanical alignment of the movement axis and thus the mounting points of hydraulic cylinder and piston rod must be ensured. Lateral forces on the guides of piston rod and piston are to be avoided. It may be necessary to consider the own weight of the hydraulic cylinder ("MP3", "MP5" or "MT4") or the piston rod.
- ▶ The buckling length / buckling load of the piston rod and / or the hydraulic cylinder must be observed (see page 62).
- ▶ The maximum admissible stroke velocities with regard to the suitability / load of seals must be observed as must their compatibility with the properties of the hydraulic fluid (see page 71).

- ▶ The maximum admissible velocities / kinetic energies when moving into the end positions, also considering external loads, must be observed (excess pressure).
- ▶ The maximum admissible operating pressure must be complied with in any operating state of the hydraulic cylinder. Possible pressure intensification resulting from the area ratio of annulus area and piston area and possible throttling points are to be observed.
- ▶ Detrimental environmental influences, like e.g. aggressive finest particles, vapors, high temperatures, etc. as well as contaminations and deterioration of the hydraulic fluid are to be avoided.

Project planning software ICS (Interactive Catalog System)

The ICS (Interactive Catalog System) is a selection and project planning aid for hydraulic cylinders. The ICS allows designers for machines and systems to quickly and reliably find the perfect hydraulic cylinder solution through logic-guided type key enquiries. This software helps to solve design and project planning tasks more quickly and efficiently. After having been guided through the product selection, the user quickly and reliably gets the exact technical data of the selected component as well as 3D CAD data in the correct file format for all common CAD systems.

This allows users to reduce costs while increasing their competitiveness.

Notice:

- ▶ This list does not claim to be complete. In case of questions regarding the compatibility with the medium or exceedance of the boundary or application conditions, please contact us.
- ▶ All graphical representations in the data sheet are examples. The product supplied may therefore differ from the figure shown.

More information

▶ Mineral oil-based hydraulic fluids	Data sheet 90220
▶ Environmentally compatible hydraulic fluids	Data sheet 90221
▶ Flame-resistant, water-free hydraulic fluids	Data sheet 90222
▶ Flame-resistant hydraulic fluids – containing water (HFAE, HFAS, HFB, HFC)	Data sheet 90223
▶ Mounting elements for hydraulic cylinders	Data sheet 17042
▶ Mating connectors and cable sets for valves, cylinders and sensors in hydraulics	Data sheet 08006
▶ Hydraulic cylinders	Data sheet 07100-B
▶ Information on available spare parts	www.boschrexroth.com

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